E-Bank

Technical specification and API contract

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1 Introduction

1.1 Overview

E-Bank gives a bank one platform for onboarding customers, running their accounts, cards and loans, and keeping every operation compliant, so the bank runs one coherent system instead of several products that were never designed to work together. The buyer is the bank. The users are its customers (individuals and small companies, called legal entities), who work in a mobile and web app, and its employees (support, compliance and AML officers, loan officers, accountants, administrators), who work in a back-office console.

The platform consists of eight modules that talk to each other through synchronous APIs (application programming interfaces) and a shared event bus: core banking (CBS) with the payment system, KYC (know your customer), CRM (customer relationship management), card management, loans, accounting, AML (anti-money laundering) and notifications.

1.2 The problem

A bank usually runs customer data, accounts, cards, payments, compliance and accounting on separate systems that were bought or built at different times by different teams. They were not designed to work together, so integrating them is slow and error-prone, and every change risks breaking something else.

1.3 Goals

#	Goal	Measurable target	Defined in
G1	Fast onboarding	A low-risk individual is registered and approved in **under 10 minutes**; automated checks finish within 60 s for 95% of cases	KYC and CRM
G2	Fast, safe payments	95% of internal transfers complete within **2 s**; every outgoing payment is screened by AML (decision within 500 ms for 95%, never later than 2 s)	CBS and Payment system, AML
G3	One version of the customer	A status change (restrict, block) reaches all modules within 5 s (p95, the value that 95% of measurements do not exceed)	KYC and CRM
G4	Reliable books	End-of-day posting and reconciliation of 100,000 transactions in **under 1 hour** with 0 unbalanced entries	Loans and Accounting
G5	Always available	**99.9%** yearly availability for core services	Non-functional requirements

1.4 Scope

In scope: individuals and legal entities as customers; EUR, USD (US dollar) and GBP (British pound) accounts; internal, SEPA (Single Euro Payments Area) and SWIFT (Society for Worldwide Interbank Financial Telecommunication) payments; debit and credit cards (EUR purchases); consumer loans; general ledger and reports; AML monitoring with reporting to the Lithuanian financial intelligence unit (FNTT); notifications by push, e-mail and SMS (short message service).

Out of scope in this version: interest calculation for credit cards and monthly card bills, cash withdrawals, card purchases in other currencies, and real connections to external providers. The identity verification provider, company registry, SEPA and SWIFT networks, card network, FNTT and the notification providers are described as interfaces and can be mocks during implementation.

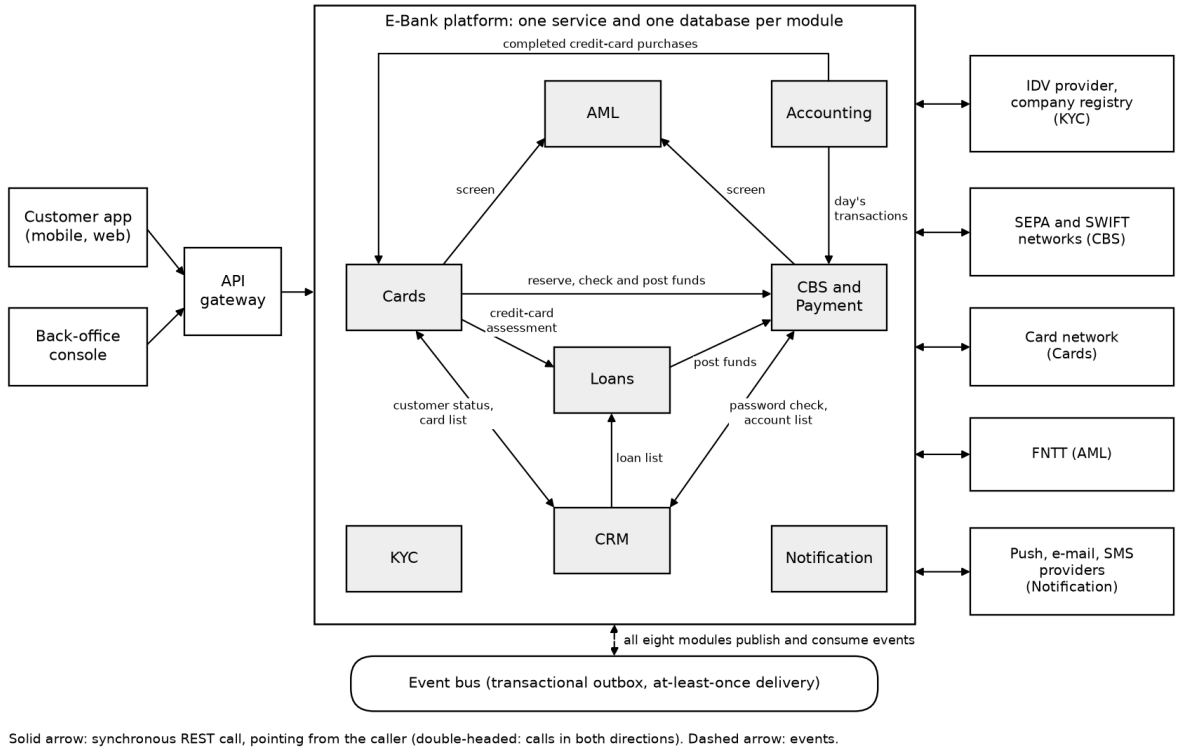
2 Feature overview

The platform has 69 features. Each module chapter contains the full feature table with actors and priorities; the table below only counts them. The identifiers are used in the flows and wireframes.

Module	Feature IDs	Must	Should	Could	Total
KYC	KYC-01 to KYC-09	7	2	0	9
CRM	CRM-01 to CRM-07	5	1	1	7
CBS and Payment system	CBS-01 to CBS-06, PAY-01 to PAY-06	9	2	1	12
Cards	CARD-01 to CARD-06	3	3	0	6
Loans	LOAN-01 to LOAN-06	3	2	1	6
Accounting	ACC-01 to ACC-06	3	2	1	6
AML	AML-01 to AML-14	12	2	0	14
Notification	NTF-01 to NTF-09	7	1	1	9
Total		**49**	**15**	**5**	**69**

3 System architecture

3.1 Context



System context: modules, synchronous calls, event bus and external systems

The platform has eight modules: KYC (know your customer), CRM (customer relationship management), CBS (core banking system) and Payment, Cards, Loans, Accounting, AML (anti-money laundering) and Notification. Customers and employees reach them through the API (application programming interface) gateway. External systems are the IDV (identity verification) provider, the company registry, the SEPA (Single Euro Payments Area) and SWIFT (Society for Worldwide Interbank Financial Telecommunication) networks, the card network, FNTT (the Lithuanian financial intelligence unit) and the push, e-mail and SMS (short message service) providers.

Synchronous calls between modules: Cards and CBS call AML to screen a payment; Cards and Loans call CBS to reserve, post or check funds; Cards calls Loans for a credit-card assessment and CRM for a customer status when it has no replica; CBS calls CRM to check the password at sign-in; Accounting pulls the day's transactions from CBS and the day's completed credit-card purchases from Cards; CRM reads accounts and products from CBS, Cards and Loans (read-only).

3.2 Communication rules

Synchronous calls (REST, Representational State Transfer, with JSON, JavaScript Object Notation) are used only when the caller must wait for an answer, as listed in section 3.1. Customer and employee calls go through the API gateway under /api/v1; calls between services use /internal/<module>.

Events are used for everything else. A module saves its change and the event in one database transaction (transactional outbox) and the relay publishes it at least once. Consumers ignore an event whose eventId they have already processed. Events about one customer are keyed by customerId, so their order is kept.

Independence: every module owns its database; no module reads another module's tables. Other modules keep only ids and small replicas fed by events (for example the customer status in CBS and Cards).

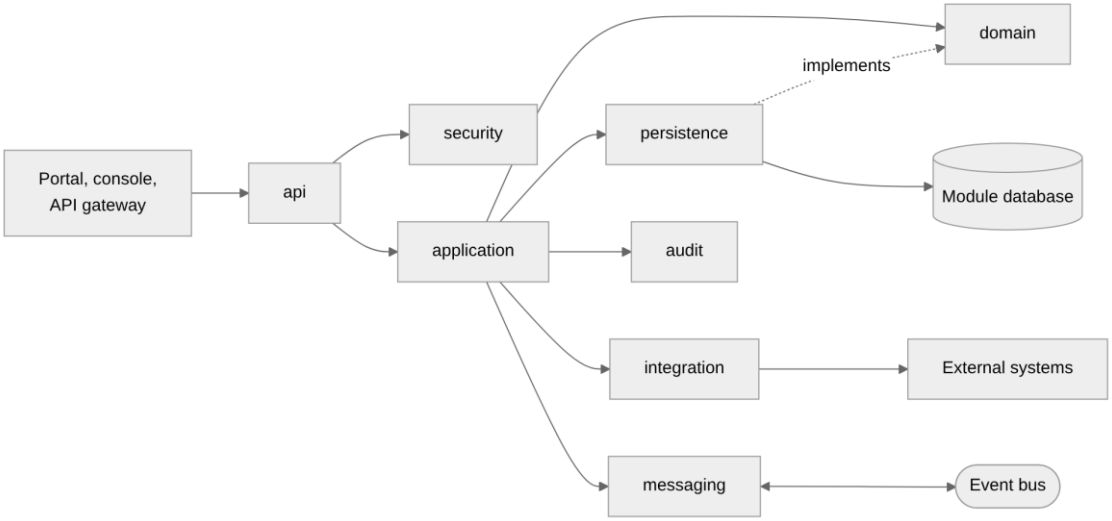
Envelope of every event: eventId (UUID), eventType, occurredAt (ISO 8601 UTC), source, payload. Event names use UPPER_SNAKE_CASE; schemas change only by adding fields.

Errors of every API use the body {"error": {"code": "...", "message": "..."}}, for client errors 400, 401, 403, 404, 409 and 422 and for server errors 500, 502, 503 and 504. A caller treats a 5xx response, or no answer within its timeout, as a temporary failure. It retries only idempotent calls, meaning GET or calls with an Idempotency-Key or another unique id, with backoff, and then follows the fallback of the flow. For AML screening the fallback is section 11.3.1, A4 for payments and A6 for cards.

Conventions: identifiers are UUID (universally unique identifier); money is Decimal(18,2), in the API a string with two decimals; times are ISO 8601 (the international date and time format) in UTC (Coordinated Universal Time); screens show Europe/Vilnius time.

3.3 Module structure

Every module is one deployable service with its own database. The packages are layered; dependencies point downwards and the domain package has no dependency on infrastructure.



Common package layout of a module (prefix kyc., crm., and so on)

3.4 Shared entities

Other modules reference these entities by id and never redefine them. Each entity has one owning module.

Entity	Owner	Fields
User	CRM	id, email, phone, passwordHash, status (PENDING, ACTIVE, BLOCKED), language (LT, EN), createdAt
Customer	CRM	id, userId, type (INDIVIDUAL, LEGAL_ENTITY), kycStatus, createdAt
Account	CBS	id, customerId, iban, currency, balance, type (CURRENT, SAVINGS), status (OPEN, FROZEN, CLOSED), openedAt
Transaction	CBS	id, accountId, toAccountId (internal only), paymentId, referenceId, amount (negative = debit), currency, type, status (PENDING, COMPLETED, FAILED, REJECTED, ON_HOLD), counterpartyName, counterpartyIban, counterpartyCountry, createdAt, completedAt
LedgerEntry	Accounting	id, transactionId, ledgerAccountId, debit, credit, postedAt
Employee	AML	id, userId, role (SUPPORT, COMPLIANCE, LOAN_OFFICER, ACCOUNTANT, ADMIN)
Notification	Notification	id, eventId, userId, templateId, channel, title, body, status, createdAt, sentAt

Only CRM reads passwordHash. CBS runs the sign-in flow and sends the e-mail and password to POST /internal/crm/users/authenticate (section 7.5.3), so it does not read CRM tables. Customer.kycStatus is CRM's read-only copy of the latest KYC outcome; the KYC module makes that decision.

3.5 Event catalogue

Event	Published by	Consumed by
CUSTOMER_REGISTERED, CUSTOMER_PROFILE_UPDATED, SENSITIVE_DATA_CHANGED	CRM	KYC, registration and profile updates also Notification
CUSTOMER_ACTIVATED, CUSTOMER_STATUS_CHANGED	CRM	CBS, Cards, Loans, Accounting, KYC, CUSTOMER_ACTIVATED also AML
CONSENT_CHANGED	CRM	Notification
KYC_CASE_OPENED, KYC_REJECTED, KYC_EXPIRED	KYC	CRM (rejection also AML)
KYC_APPROVED, KYC_INFO_REQUESTED	KYC	CRM, Notification, KYC_APPROVED also AML
SCREENING_REQUESTED / SCREENING_COMPLETED	KYC / AML	AML / KYC

Event	Published by	Consumed by
ACCOUNT_OPENED, ACCOUNT_STATUS_CHANGED, MANDATE_CHANGED	CBS	Notification
PAYMENT_PENDING_APPROVAL, PAYMENT_REJECTED, TRANSACTION_FAILED	CBS	Notification
TRANSACTION_COMPLETED	CBS	AML, Notification
AML_SCREENING_FAILED	CBS	AML
OTP_REQUESTED, NEW_DEVICE_LOGIN	CBS	Notification
TRANSACTION_RELEASED	AML	CBS
TRANSACTION_REJECTED	AML	CBS, Notification
CARD_TRANSACTION_AUTHORIZED	Cards	AML, Notification
CARD_BLOCKED	Cards	Notification
LOAN_INSTALLMENT_DUE, LOAN_INSTALLMENT_MISSED	Loans	Notification
RECONCILIATION_DISCREPANCY	Accounting	Accounting back-office and audit log
NOTIFICATION_FAILED	Notification	Audit log

Money movements and their accounting: every transfer, debit-card purchase, credit-card repayment, loan payout and installment is a completed CBS transaction, and the nightly Accounting batch reads all completed transactions from CBS. A credit-card purchase moves no money on a bank account, so it has no CBS transaction; the batch reads the completed credit-card purchases from Cards (section 9.9.3). This way each money movement receives a ledger entry.

4 Non-functional requirements

These requirements are the baseline for all modules.

ID	Area	Requirement
NFR-1	Security	**Session timeout:** a session ends after 15 minutes of inactivity. The access token is valid 5 minutes and is renewed with a refresh token that CBS (core banking system) checks against the server-side Session (chapter 8, BR-15). Services stay stateless. Encryption**:** TLS (Transport Layer Security) 1.3 for all communication; AES-256 (Advanced Encryption Standard) for sensitive data at rest, keys in a key management service. **SCA (PSD2):** a one-time code at login, for payments above **EUR 500**, for every payment approval for mandate changes, for card activation and for credit-card repayment. **Access control:** role-based and enforced on the server. **Audit:** payment, mandate, account-status and back-office changes are written to an append-only audit log (user, time, old and new value).
NFR-2	Availability and reliability	**99.9%** availability per year (at most about 8.8 hours of downtime). Baseline recovery **RPO (recovery point objective) 5 minutes, RTO (recovery time objective) 1 hour**; the CBS ledger and payments are stricter: **RPO 0** (synchronous replication across two availability zones) and **RTO 15 minutes**. An outage of AML (anti-money laundering), Notification or a payment network never loses a payment: calls are retried with backoff, and a payment not settled within 60 s after it is sent to the payment network becomes FAILED, with its hold released. The 60 s limit does not apply while a transaction is ON_HOLD because of AML, as in section 11.3.1, A2 to A4: it stays ON_HOLD, with its hold active, until the AML officer releases or rejects it.
NFR-3	Performance and scalability	First screen loads in under **1.0 s** (p95 (the value that 95% of measurements do not exceed)). Lists use server-side paging (50 rows) and render in under **200 ms**. Payment creation and validation respond in under **2.0 s** (p95) at up to **1,000 requests per second**; reads in under **500 ms** (p95). Export of 10,000 rows in under **5 s**. Services are stateless and scale horizontally.
NFR-4	Supported platforms	**iOS 16** and newer, **Android 10 (API 29)** and newer, and the latest two versions of **Chrome, Firefox, Safari and Edge**. Minimum screen width 360 px. Native desktop clients are out of scope.
NFR-5	Localization	Customer and back-office screens in **Lithuanian (LT)** and **English (EN)**, default LT. Timestamps stored in UTC (ISO 8601 (the international date and time format)) and shown in Europe/Vilnius time; amounts with 2 decimals; IBANs (International Bank Account Numbers) in groups of four characters.
NFR-6	Compliance	**GDPR (General Data Protection Regulation):** data minimization, export of personal data within the legal limit of one month, which can be extended by two more months for complex or numerous requests, access audit. **Retention:** customer, account, payment, case and audit records are kept **8 years** after the end of the relationship or the closing of the case (anti-money-laundering rules), then deleted or anonymized. **PSD2 (Payment Services Directive 2):** SCA as in NFR-1. **AML:** reporting to FNTT (the Lithuanian financial intelligence unit) as in the AML chapter.

ID	Area	Requirement
NFR-7	Data integrity	Money is stored as Decimal(18,2), never as floating point. A balance cannot become negative. Payments and postings are idempotent (Idempotency-Key). Balance and hold changes run in one database transaction with optimistic locking. Events use the transactional outbox and at-least-once delivery with idempotent consumers.

5 UI principles

5.1 Navigation

Customer app (mobile, same structure on the web): a bottom navigation bar with five items: **Home** (account list), **Payments** (new transfer), **Cards**, **Loans** and **More** (profile, notification settings, pending approvals, mandates, saved recipients, KYC (know your customer) status and support, sign out). The active item is filled. Back is always a “<” at the top left; the app never uses the bottom bar for actions.

Back-office console (web): a left menu that shows only the items allowed for the employee role: KYC review, Customers, Accounts, AML (anti-money laundering) queue, Loan review, General ledger and reports, Audit log.

5.2 Shared components

Component	Rule
Buttons	One **primary** button per screen (dark fill); **secondary** (outline) for alternatives; **destructive** (double border, underlined) for reject, delete, report; disabled buttons are grey and have a hint.
Form errors	A field error is shown directly under the field in text, never by colour alone. Errors from the server use the API (application programming interface) message in a dashed banner above the main button. The main button stays enabled and shows the errors on press.
Confirmation dialog	Used for irreversible actions: one sentence, **Cancel** and the confirm button. Decisions that need a reason also show a required comment field and keep the confirm button disabled until it is filled.
One-time code dialog	Six boxes, a countdown of the 5 minute validity, **Resend** after 30 s, and the message “Code is wrong” after each failed attempt (3 attempts).
Status labels	Rounded labels with the status text (for example FROZEN, VERIFIED); the text is always shown.
Empty and loading states	A list without rows shows one sentence (“No payments waiting for approval.”); loading shows a placeholder of the list rows.
Touch targets	At least 44 px high; one column on mobile.

5.3 Wireframe notation

Rectangles with a thin border are fields; a dark filled rectangle is the primary button, an outlined one the secondary, a double-bordered underlined one the destructive button; dashed boxes are banners, notices or upload areas; “v” marks a dropdown and “<” a back button. A numbered circle marks an element that is described in the table below the pictures; the table reference is the screen id and the number (for example S4.7). Screens of other modules use the same notation.

6 KYC

6.1 Scope and ownership

Know Your Customer (KYC) is the module that decides whether the bank may trust the identity of a customer. It runs verification cases, keeps the identity evidence, records the result of every check, calculates the customer risk level and stores the review decisions. KYC is one deployable service with its own database and an encrypted document vault. It is operated by compliance officers (employee role COMPLIANCE), because KYC data has stricter access and retention rules than other customer data.

Owned by KYC	Only referenced (owned by other modules)
KYC cases, uploaded documents, verification checks (identity document, liveness, company registry, UBO match), the customer risk score and risk level, review decisions, information requests, the KYC audit trail	Customer, user and legal-entity structure (CRM; KYC keeps a read-only snapshot), matching against sanctions and PEP lists (AML), message delivery (Notification), employees (Employee.id)

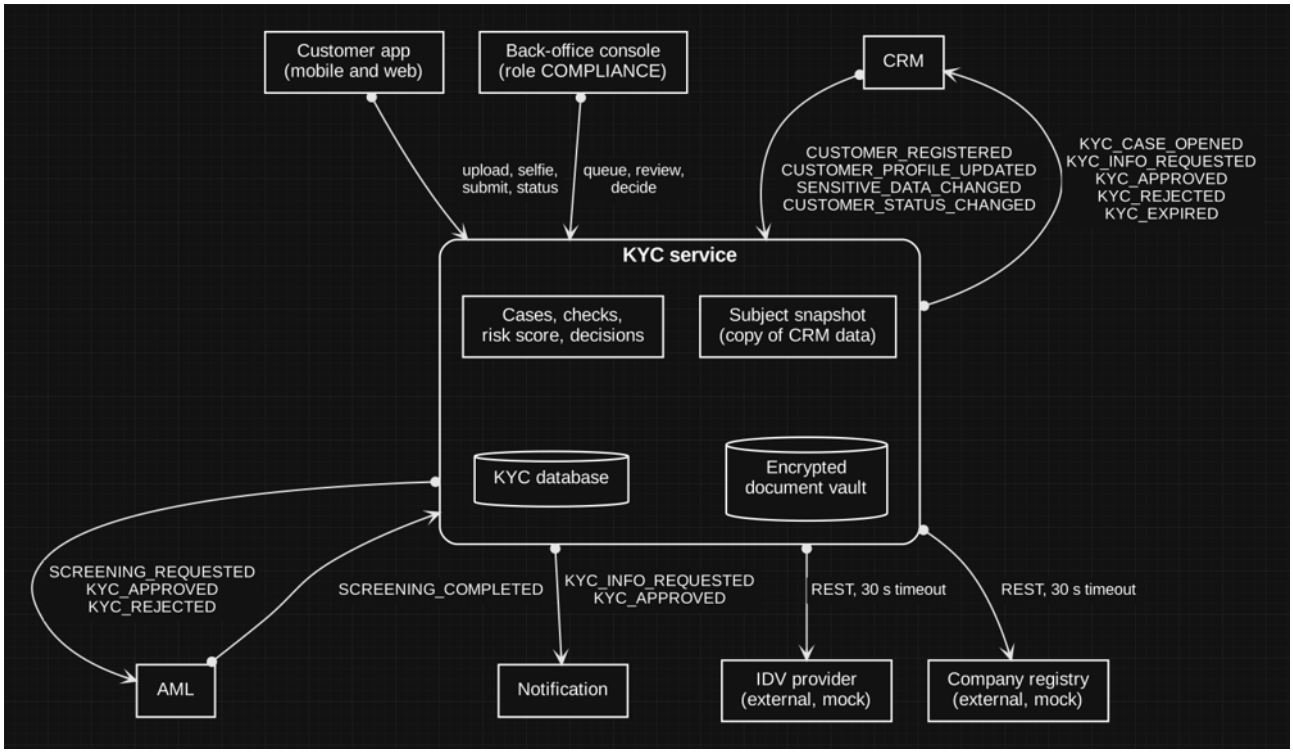
KYC works with CRM, the Anti-Money Laundering (AML) module, Notification, the customer app and the back-office console. It also calls two external providers: an identity verification (IDV) provider that checks document authenticity and liveness, and the company registry. Both providers are described as interfaces in 6.10.5 and can be mocks during implementation.

Terms used in this chapter: a **UBO** (ultimate beneficial owner) is a natural person who owns more than 25 % of a company; a **PEP** (politically exposed person) is a person who holds or held a prominent public function and needs enhanced checks; the **four-eyes principle** means that a decision needs two different officers; a **company representative** is a natural person who acts for a legal entity (see 7.1).

6.2 Feature table

ID	Feature	Actor	Priority
KYC-01	Document submission: identity document, selfie and proof of address; for a legal entity also the registry extract and the UBO declaration	Customer, company representative	Must
KYC-02	Automated identity verification (document authenticity and liveness) through the IDV provider	System	Must
KYC-03	Sanctions and PEP screening of the customer, its representatives and UBOs; AML matches the lists, KYC records the result and decides	System	Must
KYC-04	Risk scoring LOW / MEDIUM / HIGH with configurable rules (6.4.2)	System	Must
KYC-05	Manual review queue; four-eyes approval for HIGH risk	Compliance officer	Must
KYC-06	Information request to the customer and resubmission	Compliance officer, customer	Should
KYC-07	Periodic review, document-expiry handling and review after a sensitive data change	System, compliance officer	Should
KYC-08	Legal-entity verification: company registry check and UBO identification	System, compliance officer	Must
KYC-09	Append-only KYC audit trail and retention control	Compliance officer, administrator	Must

6.3 Integration with other modules



KYC integration with other modules

All exchanges with other modules are asynchronous events (common envelope, transactional outbox, idempotent consumers). The only synchronous calls go to the two external providers. Payloads are defined in 6.10.3 (published events) and 6.10.4 (consumed events).

6.4 Business rules

6.4.1 General rules

ID	Rule
K-BR-01	**Required documents.** INDIVIDUAL: one identity document (PASSPORT, ID_CARD or RESIDENCE_PERMIT), one SELFIE and one PROOF_OF_ADDRESS issued in the last 3 months. LEGAL_ENTITY: a REGISTRY_EXTRACT issued in the last 3 months and a UBO_DECLARATION; every active representative must be an INDIVIDUAL customer whose own onboarding case is APPROVED.
K-BR-02	**Files.** PDF (Portable Document Format), JPG or PNG, at most 10 MB per file and 10 files per case. The file goes to the encrypted vault; the database keeps storageRef and the SHA-256 hash. A stored file is never changed: a new upload of the same type replaces the old document, which stays in the case with status REJECTED and reason REPLACED.
K-BR-03	**IDV attempts.** At most 3 IDV attempts per case. After the third FAIL the case goes to MANUAL_REVIEW.
K-BR-04	**Screening.** KYC publishes one SCREENING_REQUESTED per screened party: the subject, every active representative and every UBO. A HIT is never cleared automatically; only a compliance officer can clear it as a false positive, with a comment.
K-BR-05	**Risk score.** The score is the sum of the points of all matching active risk rules (6.4.2). Level: LOW 0–29, MEDIUM 30–59, HIGH 60 or more.
K-BR-06	**Automatic approval.** Only when the level is LOW, every check is PASS and there is no uncleared HIT. In every other case the case goes to MANUAL_REVIEW.
K-BR-07	**Four-eyes principle.** A HIGH risk case needs two APPROVE decisions by two different officers (approvalOrder 1 and 2). The officer who gave approval 1 cannot give approval 2. A REJECT or REQUEST_INFO by either officer cancels approval 1. The rule is enforced on the server.
K-BR-08	**Comments.** REJECT, REQUEST_INFO and clearing a HIT need a comment of 1–1000 characters; for APPROVE the comment is optional. A case with an uncleared sanctions HIT cannot be approved.
K-BR-09	**Review dates.** nextReviewDate = approval date + 1 year (HIGH), 3 years (MEDIUM) or 5 years (LOW). The periods are configuration values.
K-BR-10	**Periodic review.** A daily job at 02:00 Europe/Vilnius time opens a PERIODIC_REVIEW case when the due date is 30 days away. The due date is the earlier of nextReviewDate and the expiry date of the identity document. The current approval stays valid until the due date. A review case that is not APPROVED 30 days after the due date becomes EXPIRED.
K-BR-11	**Information requests.** The due date is 30 days after the request. KYC sends one reminder 7 days before the due date. A case still in PENDING_CUSTOMER after the due date becomes EXPIRED.
K-BR-12	**Change review.** SENSITIVE_DATA_CHANGED opens a CHANGE_REVIEW case; if a case is already open, the new snapshot is attached to it instead. Required documents: name, personal code or date of birth changed → new identity document; address changed → proof of address; company structure changed → new UBO declaration and a new registry check.
K-BR-13	**Customer status.** KYC never changes the customer status. It publishes its outcome and CRM applies it (7.5.4).
K-BR-14	**Visibility.** A customer sees only the cases of their own INDIVIDUAL customer and of companies where they are an active representative. A customer never sees the risk score, the screening details or the reason SANCTIONS_MATCH.

6.4.2 Risk scoring rules

The rules are stored in RiskRule and loaded with these default values at deployment. Points are added up for the subject and all screened parties of the case. There is no screen for editing the rules; an administrator changes them through configuration.

Code	Factor	Condition	Points
RF-01	COUNTRY	Residence or registration country is on the high-risk country list. The list is configuration; default KP (North Korea) and IR (Iran), the same list as AML rule R3.	60

RF-02	COUNTRY	Nationality of the subject, a representative or a UBO is on the high-risk country list	60
RF-03	COUNTRY	Residence or registration country is outside the EU (European Union) and EEA (European Economic Area)	20
RF-04	SUBJECT_TYPE	Subject is a LEGAL_ENTITY	10
RF-05	STRUCTURE	A UBO or representative lives outside the EU and EEA, or an owner is itself a legal entity	20
RF-06	PEP	PEP match confirmed by an officer (the officer did not clear the HIT)	60
RF-07	CHECK_RESULT	IDV passed only after a failed attempt	10
RF-08	CHECK_RESULT	Registry data differs from the declared structure and the officer accepted the declaration	15

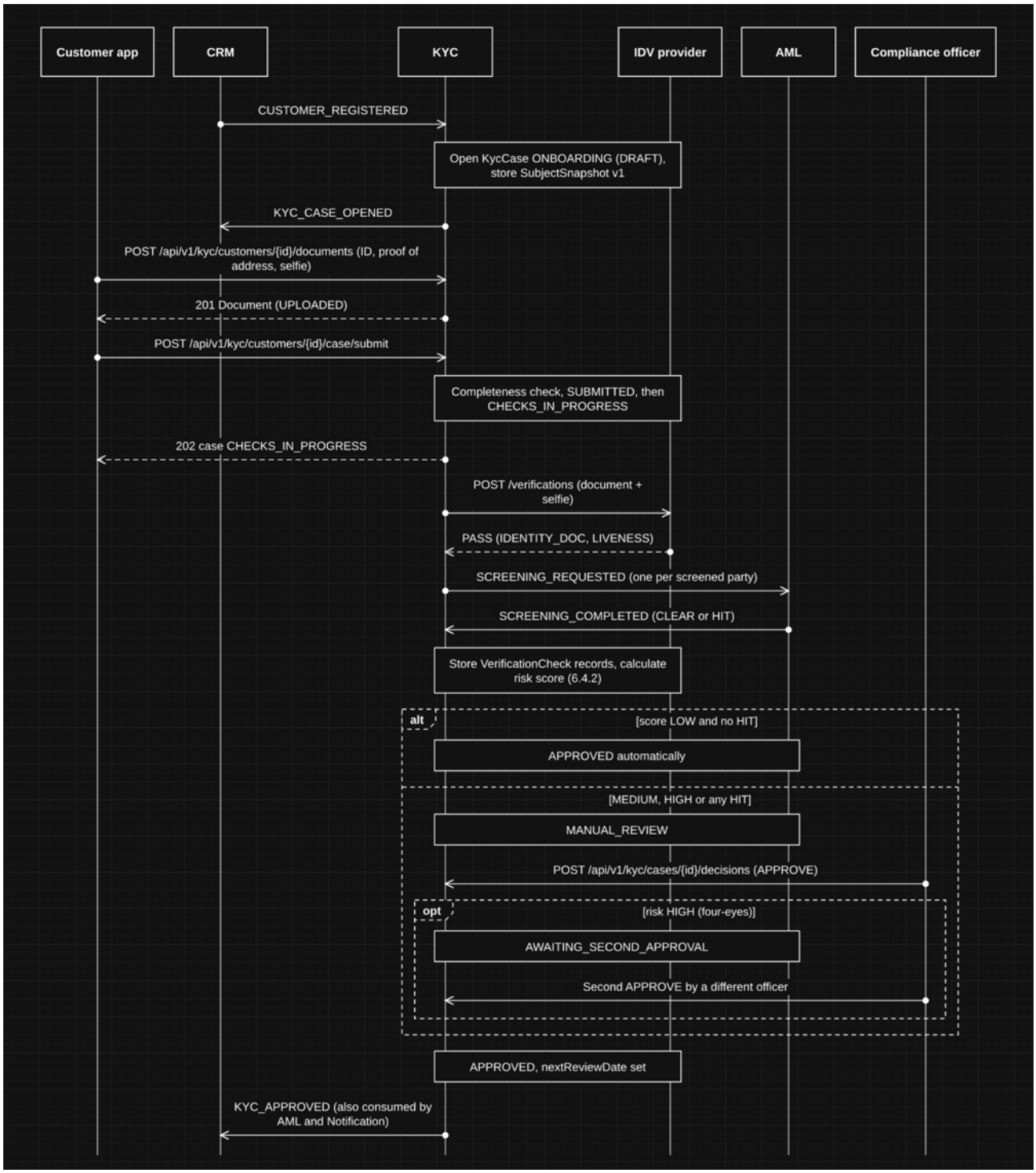
Example: an individual living in Lithuania with a confirmed PEP match whose IDV passed on the second attempt scores RF-06 + RF-07 = 70, so the level is HIGH and two approvals are needed (wireframe K-W3).

6.5 Business flows

6.5.1 Flow 1: Onboarding of an individual

Actors: customer, CRM, KYC, IDV provider, AML, compliance officer. **Trigger:** CUSTOMER_REGISTERED with type INDIVIDUAL. **Result:** KYC_APPROVED, after which CRM activates the customer.

1. KYC consumes CUSTOMER_REGISTERED, opens a KycCase of type ONBOARDING in status DRAFT, stores the first SubjectSnapshot and publishes KYC_CASE_OPENED. Later CUSTOMER_PROFILE_UPDATED events add new snapshot versions.
2. The customer opens **More > Verification** (K-W1), uploads an identity document and proof of address and takes a selfie (K-BR-01, K-BR-02).
3. The customer presses **Submit**. KYC checks that all required documents exist, sets SUBMITTED and then CHECKS_IN_PROGRESS, and answers at once; the customer does not wait for the checks.
4. KYC sends the identity document and the selfie to the IDV provider and stores the result as two VerificationCheck records (IDENTITY_DOC, LIVENESS).
5. KYC publishes SCREENING_REQUESTED for the subject. AML answers with SCREENING_COMPLETED (CLEAR or HIT), stored as SANCTIONS and PEP checks.
6. KYC calculates the risk score (6.4.2). A LOW case without a HIT is approved automatically (K-BR-06); every other case goes to MANUAL_REVIEW (Flow 3).
7. KYC sets APPROVED, calculates nextReviewDate (K-BR-09) and publishes KYC_APPROVED. CRM, AML and Notification consume it. KYC's part ends here.



Alternative paths

- **1A Required document missing (step 3).** KYC answers 422 DOCUMENTS_MISSING and the case stays DRAFT. In K-W1 the **Submit** button is disabled until the list is complete.
- **1B IDV fails (step 4).** The document gets status REJECTED with a reason (for example DOCUMENT_UNREADABLE, FACE_MISMATCH), the case goes to PENDING_CUSTOMER and KYC publishes KYC_INFO_REQUESTED so that the customer is notified. After a new upload and **Submit** the flow continues at step 3. The third failure sends the case to MANUAL_REVIEW (K-BR-03).
- **1C Sanctions or PEP hit (step 5).** The case goes to MANUAL_REVIEW. A confirmed sanctions match ends in REJECT with reason SANCTIONS_MATCH; the customer sees only the neutral status “NOT APPROVED”. A false positive is cleared by the officer with a comment. A confirmed PEP adds RF-06, so the case becomes HIGH and needs two approvals.
- **1D Provider or AML outage (steps 4–5).** No answer within 30 s or a 5xx answer stores the check as ERROR and the call is retried after 1, 5 and 15 minutes and then every hour. After 24 hours without a result the case goes to

MANUAL_REVIEW, where the officer decides with the evidence available. The customer is never blocked by the outage.

- **1E Customer does not answer.** A case that stays in PENDING_CUSTOMER after the due date becomes EXPIRED and KYC publishes KYC_EXPIRED (K-BR-11). For an onboarding case the customer can start a new case in K-W1.

6.5.2 Flow 2: Onboarding of a legal entity

The flow is the same as Flow 1, with these differences. The trigger is CUSTOMER_REGISTERED with type LEGAL_ENTITY, published when a company representative adds a company in CRM (7.5.2).

1. The representative selects the company in the **For** field of K-W1 and uploads the registry extract and the UBO declaration.
2. On **Submit**, KYC checks that every active representative in the latest snapshot has an APPROVED onboarding case of their own. If not, it answers 422 REPRESENTATIVE_NOT_VERIFIED and names the persons.
3. KYC queries the company registry with the registration number and country. The REGISTRY check is PASS when the company is ACTIVE and the legal name matches. The UBO_MATCH check is PASS when every registry owner with more than 25 % is declared and each declared share differs from the registry by at most 1 percentage point.
4. KYC publishes SCREENING_REQUESTED for the company, every UBO and every representative (K-BR-04).
5. Scoring and decision as in Flow 1. A UBO_MATCH FAIL always goes to manual review; the officer may accept the declaration (RF-08) or request information.

6.5.3 Flow 3: Manual review and four-eyes approval

Actors: compliance officer (role COMPLIANCE), a second compliance officer for HIGH risk. **Precondition:** the case is in MANUAL_REVIEW or AWAITING_SECOND_APPROVAL.

1. The officer opens **KYC review** (K-W2). The queue shows cases in MANUAL_REVIEW, the oldest first. The tab **Awaiting second approval** shows HIGH cases with one approval.
2. The officer opens a case (K-W3); KYC assigns it to the officer (assignedOfficerId). The officer reviews the snapshot, the checks, the risk breakdown and the documents. Every document view is written to the audit trail.
3. The officer may clear a HIT as a false positive with a comment; KYC then recalculates the score.
4. The officer chooses **Approve**, **Request information** or **Reject** (K-BR-08). KYC stores a ReviewDecision and writes a KycAuditEvent.
5. Approve on a LOW or MEDIUM case: APPROVED and KYC_APPROVED. Approve on a HIGH case: AWAITING_SECOND_APPROVAL; the case appears in the second tab for all other officers. When a different officer approves, the case becomes APPROVED.
6. Request information: KYC creates an InformationRequest with the requested document types and the due date, sets PENDING_CUSTOMER and publishes KYC_INFO_REQUESTED. Reject: REJECTED and KYC_REJECTED with a reason code.

Alternative paths

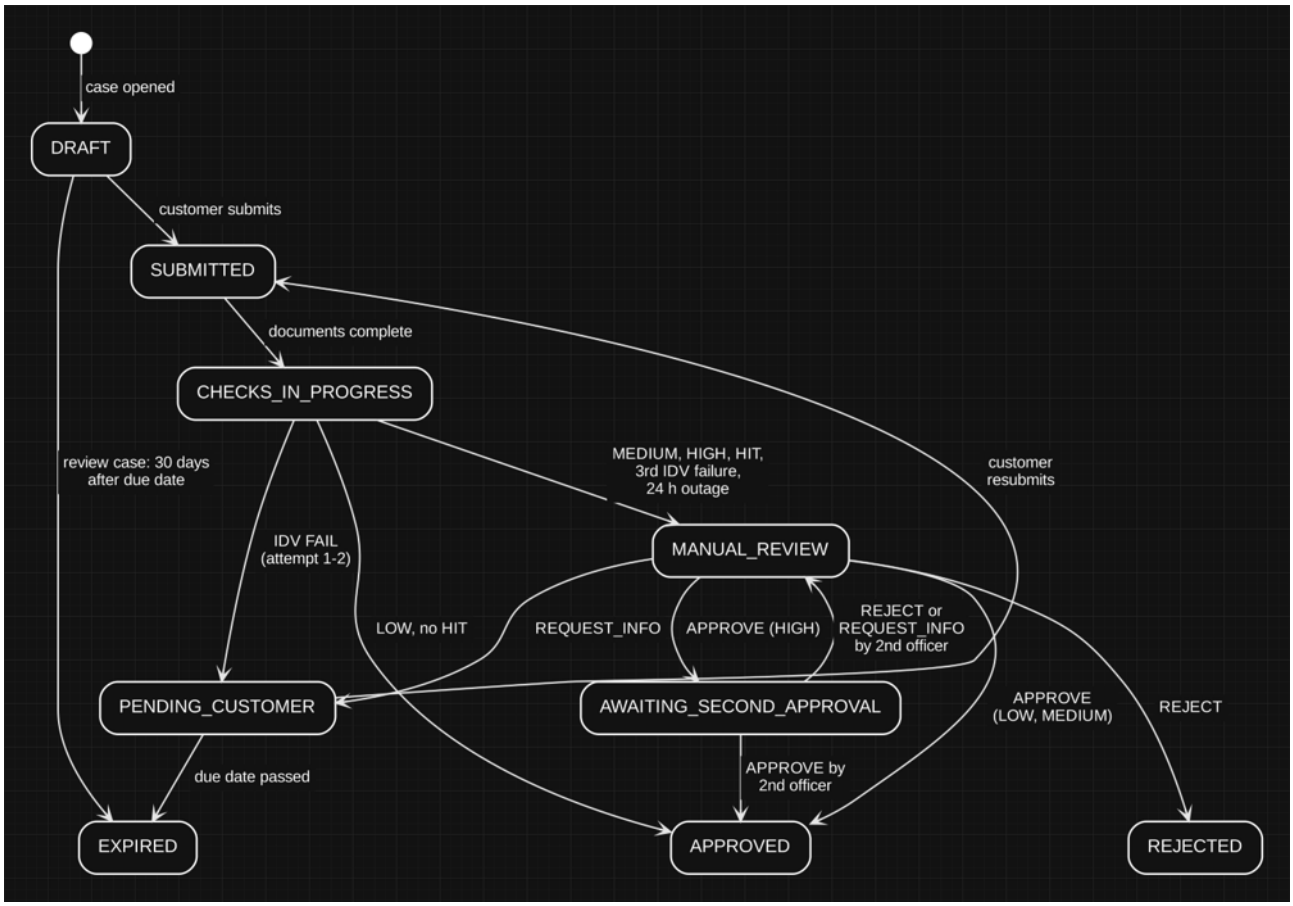
- **3A Same officer gives the second approval.** KYC answers 403 SAME_OFFICER_APPROVAL; in K-W3 the **Approve** button is disabled for that officer.
- **3B Two officers decide at the same time.** Optimistic locking (KycCase.version) accepts the first decision; the second officer gets 409 VERSION_CONFLICT and the screen reloads.

6.5.4 Flow 4: Periodic review and change review

1. The daily job opens a PERIODIC_REVIEW case 30 days before the due date (K-BR-10), or CRM's SENSITIVE_DATA_CHANGED opens a CHANGE_REVIEW case (K-BR-12). KYC publishes KYC_CASE_OPENED (with the due date) and KYC_INFO_REQUESTED with the documents the customer must provide, so Notification informs the customer.
2. The customer confirms the data or uploads the requested documents in K-W1 and presses **Submit**.
3. KYC re-runs IDV for new identity documents, publishes SCREENING_REQUESTED for all parties again and recalculates the score.
4. If the level is LOW or MEDIUM, is not higher than before and there is no HIT, the case is approved automatically and KYC publishes KYC_APPROVED with the new nextReviewDate. Otherwise the case goes to manual review (Flow 3).
5. If the case is not APPROVED 30 days after the due date, it becomes EXPIRED and KYC publishes KYC_EXPIRED; CRM restricts the customer (7.5.4). A later KYC_APPROVED lifts the restriction.

Adverse findings in a review case (a new HIT or a higher level) always go to manual review; a HIGH result needs two approvals. KYC never blocks a customer itself; if an officer decides to block, they do it in Customer 360 (C-W1).

6.5.5 Case status model

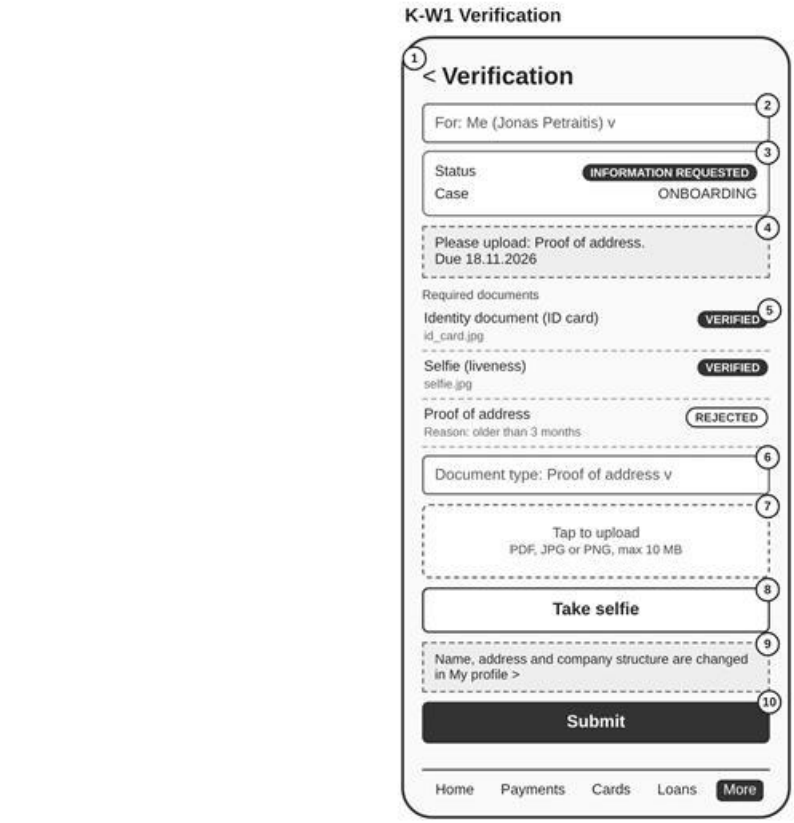


The customer app shows a simpler label for each status, and CRM keeps its own KYC status in KycReplica. For PERIODIC_REVIEW and CHANGE_REVIEW cases the CRM value stays APPROVED until KYC publishes KYC_EXPIRED or KYC_REJECTED.

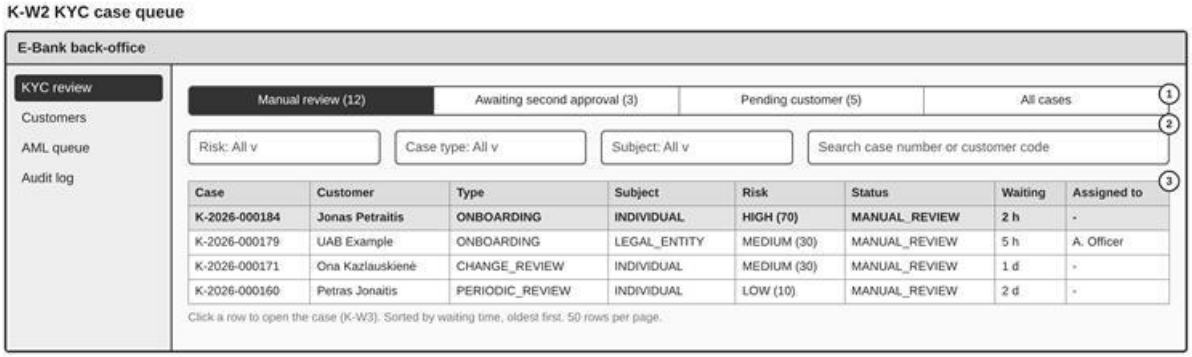
KycCase.status	Label in K-W1	CRM KycReplica.kycStatus (onboarding case)
DRAFT	NOT SUBMITTED	IN_PROGRESS
SUBMITTED, CHECKS_IN_PROGRESS	CHECKING	IN_PROGRESS
PENDING_CUSTOMER	INFORMATION REQUESTED	IN_PROGRESS
MANUAL_REVIEW, AWAITING_SECOND_APPROVAL	IN REVIEW	IN_PROGRESS
APPROVED	VERIFIED	APPROVED
REJECTED	NOT APPROVED	REJECTED
EXPIRED	EXPIRED	EXPIRED

6.6 Wireframes

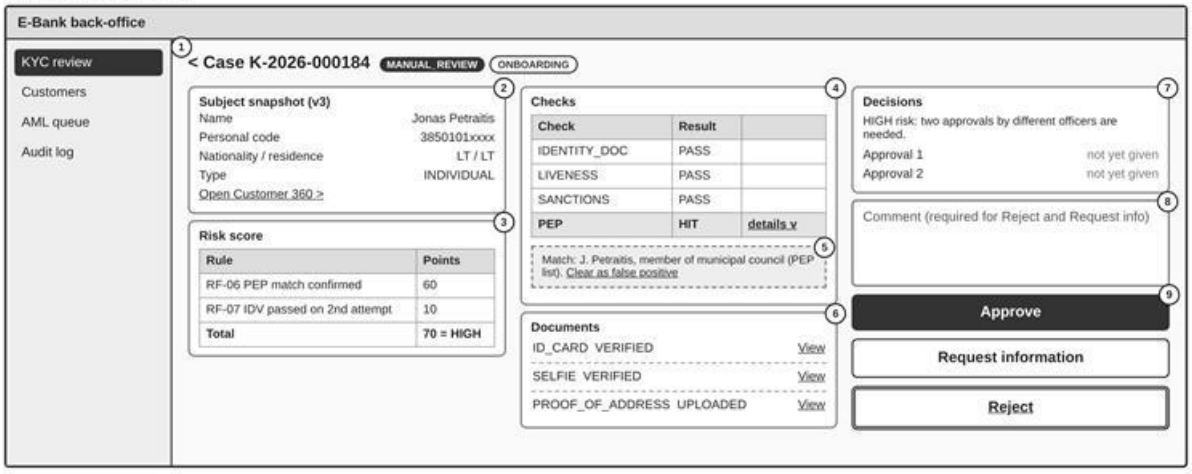
K-W1 is a screen of the customer app (mobile, same structure on the web) reached from **More > Verification**. K-W2 and K-W3 are back-office screens for employees with the role COMPLIANCE. The notation is described in 5.3.



Wireframe K-W1 (verification, customer app)



Wireframe K-W2 (KYC case queue, back-office)



Wireframe K-W3 (KYC case review, back-office)

Transitions: More > Verification opens K-W1; **Submit** stays on K-W1 and shows the label CHECKING; K-W1.9 opens My profile (C-W2) or, for a company, Company structure (C-W5). The back-office menu **KYC review** opens K-W2; a row opens K-W3; a decision returns to K-W2; the link in K-W3.2 opens Customer 360 (C-W1).

Ref	Element	What happens when it is used
K-W1.1	< back	Returns to **More**.
K-W1.2	For	Selects whose verification is shown: the user's own INDIVIDUAL customer or a company where the user is an active representative.
K-W1.3	Status card	Shows the status label (6.5.5) and the case type. For an EXPIRED onboarding case it shows the button **Start again**, which opens a new case.
K-W1.4	Request banner	Shown while an InformationRequest is OPEN: requested documents, officer message and due date.
K-W1.5	Document list	One row per required document with its status (UPLOADED, VERIFIED, REJECTED, EXPIRED or REQUIRED when missing); a rejected document shows the reason.
K-W1.6	Document type	Lists the document types required for the customer type (K-BR-01).
K-W1.7	Upload area	Uploads the file (K-BR-02). A wrong type or a file over 10 MB shows the error under the area.
K-W1.8	Take selfie	Opens the camera and uploads the picture as SELFIE. Not shown for a company.
K-W1.9	Profile notice	Personal data and company structure are not edited here; the link leaves KYC.
K-W1.10	Submit	Enabled when all required documents exist and the case is DRAFT or PENDING_CUSTOMER; calls submit (Flow 1, step 3).
K-W2.1	Tabs	Manual review, Awaiting second approval, Pending customer, All cases; the number of cases in brackets.
K-W2.2	Filters and search	Filter by risk level, case type and subject type; search by case number, customer name, personal code or registration number.
K-W2.3	Case table	Paged, 50 rows, oldest first. A HIGH case is shown in bold. Clicking a row opens K-W3.
K-W3.1	Case header	Case number, status and type; < returns to K-W2.
K-W3.2	Subject snapshot	Read-only data of the latest snapshot. **Open Customer 360** opens C-W1.
K-W3.3	Risk score	Matching rules with points, the total and the level.
K-W3.4	Checks	Each VerificationCheck with its result; a HIT is shown in bold and expands to the match details.
K-W3.5	Match details	List entry that matched; **Clear as false positive** opens a dialog with a required comment.
K-W3.6	Documents	**View** opens the document in a preview; every view is audited.
K-W3.7	Decisions	Approvals given so far (officer and time). For HIGH risk two approvals are needed.
K-W3.8	Comment	Required for Reject and Request information (K-BR-08).
K-W3.9	Decision buttons	Approve, Request information (opens a dialog with document types and due date) and Reject (opens a confirmation dialog with a reason code). Approve is disabled for the officer who gave approval 1 and while a sanctions HIT is open.

6.7 Non-functional requirements

The shared non-functional requirements (chapter 4) apply. Module-specific targets:

- **Security and privacy:** documents and personal data are AES-256 (Advanced Encryption Standard) encrypted at rest; every document view is audited; the four-eyes rule is enforced on the server; only the role COMPLIANCE can decide and clear hits; the role ADMIN can read and export the audit trail; a customer sees only their own cases (K-BR-14).
- **Resilience:** an IDV, registry or AML outage never stops case intake: retries with backoff, a circuit breaker and escalation to MANUAL_REVIEW after 24 hours (alternative 1D).
- **Performance:** API (application programming interface) p95 (the value that 95% of measurements do not exceed) below 500 ms (uploads excluded); a 10 MB upload below 5 s; automated checks finished within 60 s for 95 % of cases (goal G1); 100 new cases per minute at peak.
- **Integration and retention:** events keyed by customerId, idempotent consumers (processed eventIds are stored), transactional outbox, additive schema changes. Evidence, cases and audit events are kept 8 years after the end of the customer relationship and then deleted; KycAuditEvent is append-only.

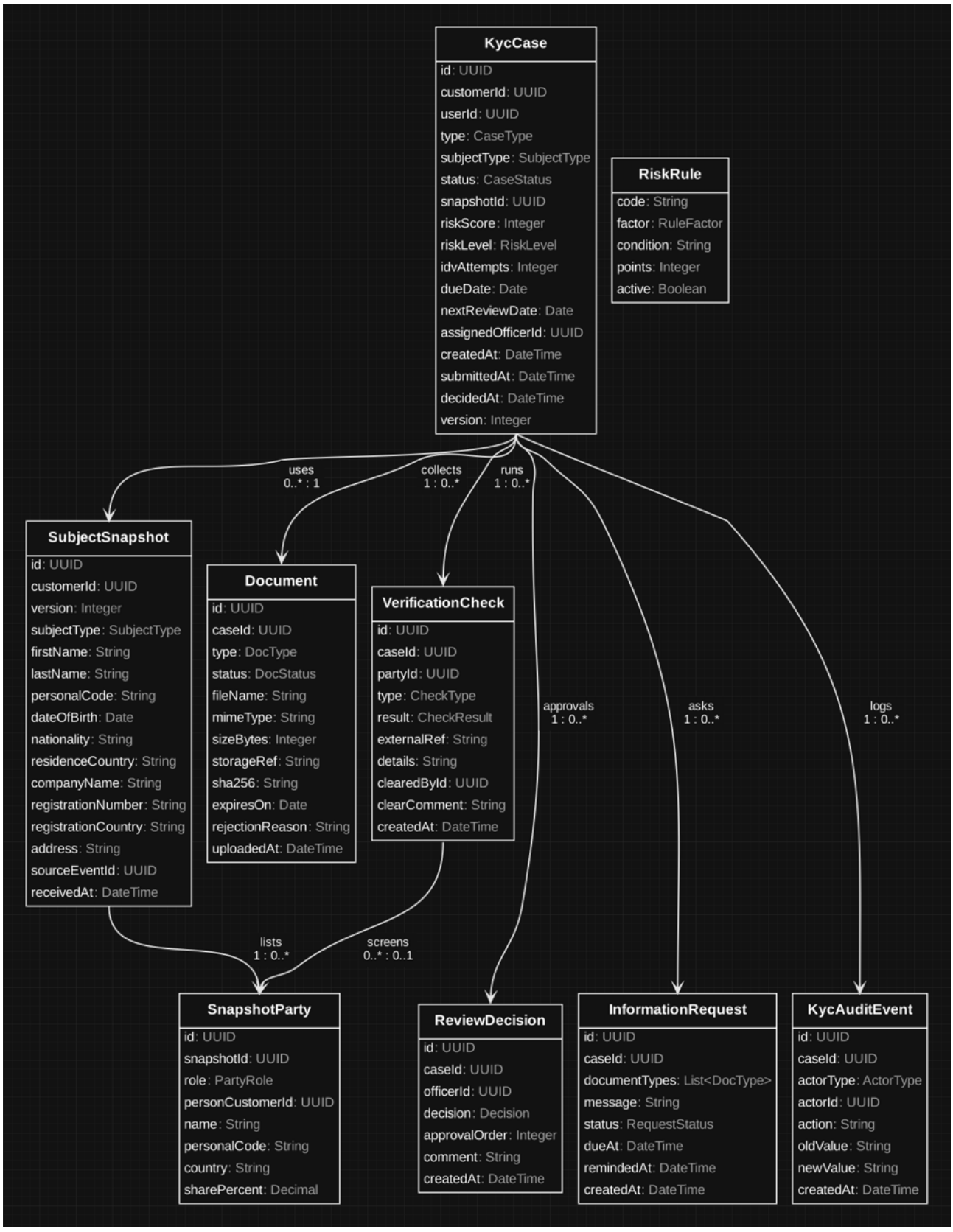
6.8 High-level architecture

KYC uses the common package layout of the System architecture chapter (3.3). It is one deployable service with its own database and document vault. Dependencies point downwards (api, application, domain); the domain package has no dependency on infrastructure. Other modules are reached only through the event bus.

Package	Responsibility
kyc.api	REST (Representational State Transfer) controllers for the customer app and the back-office console; validation and mapping to data transfer objects (DTOs), no business logic.
kyc.security	Token and role check on every call: COMPLIANCE decides, ADMIN reads the audit trail, a customer reaches only cases allowed by K-BR-14.
kyc.application	Use cases: case lifecycle, check orchestration, risk scoring, review queue, information requests, daily review job.
kyc.domain	Entities and rules without framework code: case state machine, four-eyes policy, risk rules, required-document rules.
kyc.persistence	Repositories for the KYC database and the client of the encrypted document vault.
kyc.integration	Adapters for the IDV provider and the company registry: timeouts, retries, circuit breaker, mock switch.
kyc.audit	Append-only KycAuditEvent for every state change, decision and document view; PDF export.
kyc.messaging	Idempotent consumers of CRM and AML events; outbox relay that publishes KYC events.

6.9 Data model

The KYC database is used only by KYC. customerId and userId are references without foreign keys. SubjectSnapshot is KYC's own copy of CRM data (a new version for every CUSTOMER_PROFILE_UPDATED), so KYC never reads CRM tables. Documents live in the encrypted vault; the database keeps storageRef and sha256.



KYC data model (fields and types)

Entity	Meaning and rules
KycCase	One verification case of one customer. userId is the user who is notified (for a company the main contact). dueDate is set for review cases. riskScore and riskLevel are set after the checks. version is used for optimistic locking. At most one open case (not APPROVED, REJECTED or EXPIRED) per customer.
SubjectSnapshot	Versioned copy of the CRM profile (person or company fields, address). sourceEventId is the eventId of the CRM event. A case always points to the latest snapshot.

SnapshotParty	Representative or UBO from the CRM structure. personCustomerId is set when the party is a customer; sharePercent only for UBOs.
Document	Uploaded file. expiresOn is filled from the IDV result for identity documents. rejectionReason is set when the status is REJECTED.
VerificationCheck	Result of one check. partyId is empty for the subject itself and set for a representative or UBO. externalRef is the IDV verification id or the screening id. details holds the provider answer as JSON (JavaScript Object Notation). clearedById and clearComment are set when an officer clears a HIT.
ReviewDecision	Decision of one officer (officerId = Employee.id). approvalOrder is 1 or 2 for APPROVE, empty otherwise.
InformationRequest	Documents asked from the customer, with message and due date; remindedAt is set when the reminder was sent.
RiskRule	Configurable scoring rule (6.4.2). condition is a rule expression evaluated by kyc.domain.
KycAuditEvent	Append-only record of every state change, decision and document view (actor, time, old and new value).

Enumerations

Enumeration	Values
CaseType	ONBOARDING, PERIODIC_REVIEW, CHANGE_REVIEW
CaseStatus	DRAFT, SUBMITTED, CHECKS_IN_PROGRESS, PENDING_CUSTOMER, MANUAL_REVIEW, AWAITING_SECOND_APPROVAL, APPROVED, REJECTED, EXPIRED
RiskLevel	LOW, MEDIUM, HIGH
SubjectType	INDIVIDUAL, LEGAL_ENTITY
PartyRole	REPRESENTATIVE, UBO
DocType	PASSPORT, ID_CARD, RESIDENCE_PERMIT, SELFIE, PROOF_OF_ADDRESS, REGISTRY_EXTRACT, UBO_DECLARATION
DocStatus	UPLOADED, VERIFIED, REJECTED, EXPIRED
CheckType	IDENTITY_DOC, LIVENESS, SANCTIONS, PEP, REGISTRY, UBO_MATCH
CheckResult	PASS, FAIL, HIT, ERROR
Decision	APPROVE, REJECT, REQUEST_INFO
RequestStatus	OPEN, FULFILLED, EXPIRED
RuleFactor	COUNTRY, SUBJECT_TYPE, STRUCTURE, PEP, CHECK_RESULT
ActorType	CUSTOMER, EMPLOYEE, SYSTEM
RejectReason	SANCTIONS_MATCH, FRAUD_SUSPECTED, DOCUMENTS_INVALID, OTHER

6.10 API contract

Conventions are in the System architecture chapter (JSON, /api/v1, error body, event envelope, UUID (universally unique identifier) identifiers, ISO 8601 (the international date and time format) UTC (Coordinated Universal Time) times). Customer endpoints need a customer token; {customerId} must be the caller's own INDIVIDUAL customer or a company where the caller is an active representative, otherwise the answer is 404 CASE_NOT_FOUND. Back-office endpoints need an employee token with the role COMPLIANCE (audit endpoints also ADMIN).

6.10.1 Customer endpoints

Method and path	Purpose	Request (main fields)	Success
GET /api/v1/kyc/customers/{customerId}/case	Current case (K-W1)	none	200 case: id, type, status, label, requiredDocuments, openRequest
POST /api/v1/kyc/customers/{customerId}/cases	Start a new onboarding case after EXPIRED	none	201 case (DRAFT)
POST /api/v1/kyc/customers/{customerId}/documents	Upload a document	multipart: type, file	201 Document (UPLOADED)
GET /api/v1/kyc/customers/{customerId}/documents	Documents of the open case	none	200 list (no file content)

POST /api/v1/kyc/customers/{customerId}/case/submit	Submit the case	none	202 case (CHECKS_IN_PROGRESS)

6.10.2 Back-office endpoints

Method and path	Purpose	Request (main fields)	Success
GET /api/v1/kyc/cases	Queue (K-W2)	query: status, riskLevel, type, subjectType, q, page, size (max 50)	200 page of cases
GET /api/v1/kyc/cases/{caseId}	Case details (K-W3); assigns the case to the caller if unassigned	none	200 case with snapshot, parties, checks, documents, risk breakdown, decisions, requests
GET /api/v1/kyc/cases/{caseId}/documents/{documentId}/content	View a document (audited)	none	200 file
POST /api/v1/kyc/cases/{caseId}/checks/{checkId}/clear	Clear a HIT as false positive	comment, version	200 check
POST /api/v1/kyc/cases/{caseId}/decisions	Approve, reject or request information	decision, comment, reasonCode (reject), documentTypes and dueAt (request info), version	201 ReviewDecision and the new case status
GET /api/v1/kyc/cases/{caseId}/audit	Audit trail (COMPLIANCE, ADMIN)	query: format (json, pdf)	200 list or PDF file

6.10.3 Events published by KYC

All events use the common envelope and are keyed by customerId. userId is the user to notify.

Event	Published when	Payload fields
KYC_CASE_OPENED	A case is opened	caseId, customerId, userId, caseType, dueDate (review cases)
KYC_INFO_REQUESTED	IDV failed, an officer requests information, a review case needs documents, or the reminder is due	caseId, customerId, userId, requestId, documentTypes, dueAt, reminder (true or false)
KYC_APPROVED	A case is approved	caseId, customerId, userId, caseType, riskLevel, nextReviewDate
KYC_REJECTED	A case is rejected	caseId, customerId, userId, caseType, reasonCode (RejectReason); the reason is never shown to the customer
KYC_EXPIRED	A case expired	caseId, customerId, userId, caseType, expiredAt
SCREENING_REQUESTED	Screening of one party is needed (K-BR-04)	screeningId, caseId, customerId, partyRole (SUBJECT, REPRESENTATIVE, UBO), subjectType, name, birthDate (optional), personalCode or registrationNumber, countryCode

6.10.4 Events consumed by KYC

Event	From	Used for
CUSTOMER_REGISTERED	CRM	Opens the onboarding case; stores the mapping customerId–userId
CUSTOMER_PROFILE_UPDATED	CRM	New SubjectSnapshot version and the list of parties
SENSITIVE_DATA_CHANGED	CRM	Opens a CHANGE_REVIEW case (K-BR-12)
CUSTOMER_STATUS_CHANGED	CRM	CLOSED: open cases become EXPIRED without an event and no new review cases are opened
SCREENING_COMPLETED	AML	Stores the SANCTIONS and PEP checks; matched by screeningId

6.10.5 External interfaces

Both providers can be mocks during implementation. Calls use a 30 s timeout and the retry policy of alternative 1D.

IDV provider. POST {IDV_URL}/verifications with referenceId (the check id), documentType, documentFileName, documentImage and selfieImage (both Base64). The answer contains verificationId, documentResult and livenessResult (PASS or FAIL), reasonCode and the extracted data (firstName, lastName, dateOfBirth, documentNumber, expiryDate, nationality). The mock returns FAIL with FACE_MISMATCH when the file name contains “fail”, otherwise PASS with expiryDate five years from today.

{ "verificationId": "idv-7f3a", "documentResult": "PASS", "livenessResult": "PASS",
"reasonCode": null, "extracted": { "firstName": "Jonas", "lastName": "Petraitis",
"dateOfBirth": "1985-01-01", "documentNumber": "LT1234567", "expiryDate": "2031-11-04", "nationality": "LT" } }

Company registry. GET {REGISTRY_URL}/companies/{country}/{registrationNumber} returns registrationNumber, legalName, status (ACTIVE or LIQUIDATED), registeredAddress, representatives (personalCode, name, role) and owners (name, personalCode or registrationNumber, sharePercent), or 404 when the company is unknown. The mock reads the answers from a fixture file.

6.10.6 Error codes

Code	HTTP	Meaning
VALIDATION_ERROR, UNSUPPORTED_FILE_TYPE, FILE_TOO_LARGE	400	Field or file rules failed (K-BR-02).
PERMISSION_DENIED, SAME_OFFICER_APPROVAL	403	Missing role; the second approval is given by the first officer (K-BR-07).
CASE_NOT_FOUND	404	No such case or no access (K-BR-14).
CASE_NOT_EDITABLE, CASE_ALREADY_OPEN, VERSION_CONFLICT	409	Upload or submit in a wrong status; a new case while one is open; concurrent change.
DOCUMENTS_MISSING, REPRESENTATIVE_NOT_VERIFIED, COMMENT_REQUIRED, SANCTIONS_HIT_OPEN	422	Business rule failed (K-BR-01, K-BR-08).

7 CRM

7.1 Scope and ownership

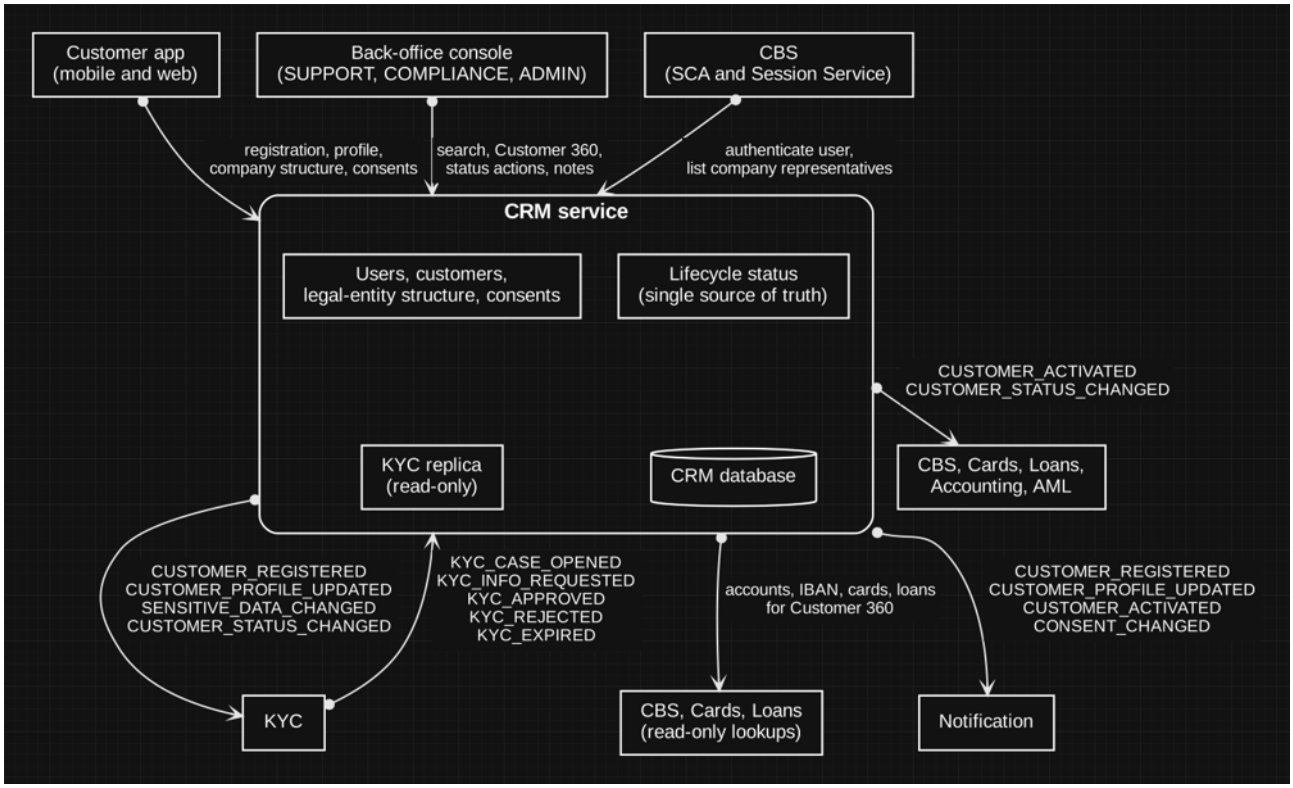
CRM (customer relationship management) is the system of record for who the customer is: master data, legal-entity structure (representatives, beneficial owners), consents and lifecycle status. It has its own service and database. Every other module (CBS, Payments, Cards, Loans, Accounting, AML (anti-money laundering), Notification) treats a customer as an ID plus a small status replica fed by CRM events, so CRM is the single place where status is decided and published. KYC is one of its neighbours: CRM keeps only a read-only copy of the KYC outcome and applies its own rules to it

7.2 Feature table

ID	Feature	Actor	Priority
CRM-01	Registration of an individual (two steps); an active individual adds a legal entity	Customer, company representative	Must
CRM-02	Profile maintenance; sensitive changes (name, personal code, address, company structure) are flagged for KYC	Customer, support agent	Must
CRM-03	Legal-entity structure: representatives with roles and validity dates, beneficial owners (more than 25 %)	Company representative, support agent	Must
CRM-04	Lifecycle and status control (restrict, lift restriction, block, unblock, close) with reason	Compliance officer, administrator	Must
CRM-05	Consents, data export and erasure (GDPR)	Customer	Must
CRM-06	Customer 360 and search (profile, structure, KYC outcome, products through module APIs (application programming interfaces))	Support agent, compliance officer	Should

ID	Feature	Actor	Priority
CRM-07	Internal customer notes	Support agent	Could

7.3 Integration with other modules



CRM integration with other modules

7.4 Business rules

ID	Rule
C-BR-01	Uniqueness. E-mail is unique across users (case-insensitive). Personal code is unique across INDIVIDUAL customers. Registration number plus country is unique across LEGAL_ENTITY customers.
C-BR-02	Password. 10–64 characters with at least one letter and one digit, stored as an Argon2id hash. Five wrong passwords in a row lock the user for 15 minutes (lockedUntil).
C-BR-03	User status. PENDING after registration step 1, ACTIVE after step 2 (terms and DATA_PROCESSING consent accepted). A PENDING user that is not completed within 24 hours is deleted. A user is BLOCKED while their INDIVIDUAL customer is BLOCKED. Only an ACTIVE user can sign in. The customer status, not the user status, decides which products can be used, so a PROSPECT or ONBOARDING customer can sign in and complete KYC.
C-BR-04	Customer status. The statuses and the allowed transitions are in 7.5.4. Every transition is written to CustomerChangeLog and published: the first activation as CUSTOMER_ACTIVATED, every other transition as CUSTOMER_STATUS_CHANGED.
C-BR-05	Status actions. Restrict, Lift restriction, Block and Unblock need the role COMPLIANCE or ADMIN; Close needs ADMIN or COMPLIANCE. A reason code and a comment are mandatory.
C-BR-06	Closing. A customer can be closed only when CBS (core banking system) has no OPEN or FROZEN account of the customer, Cards has no card that is not PERM_BLOCKED, EXPIRED or CANCELLED, and Loans has no open loan. Otherwise the answer is 409 PRODUCTS_OPEN with the list of products. A person who is an active representative of a company that is not CLOSED cannot be closed.
C-BR-07	Sensitive fields. Person: first name, last name, personal code, date of birth, nationality, residence address. Company: legal name, registration number, registered address, representatives, beneficial owners. Saving any of them publishes SENSITIVE_DATA_CHANGED with the changed field groups. Personal code and date of birth can be edited by the customer only while the customer is PROSPECT or ONBOARDING; later only SUPPORT can change them, with a reason.
C-BR-08	Contacts. A change of e-mail or phone needs the current password, because the phone number receives the one-time codes. The new e-mail must be unique (C-BR-01). The change is written to the history and published in CUSTOMER_PROFILE_UPDATED, so Notification uses the new contact details.

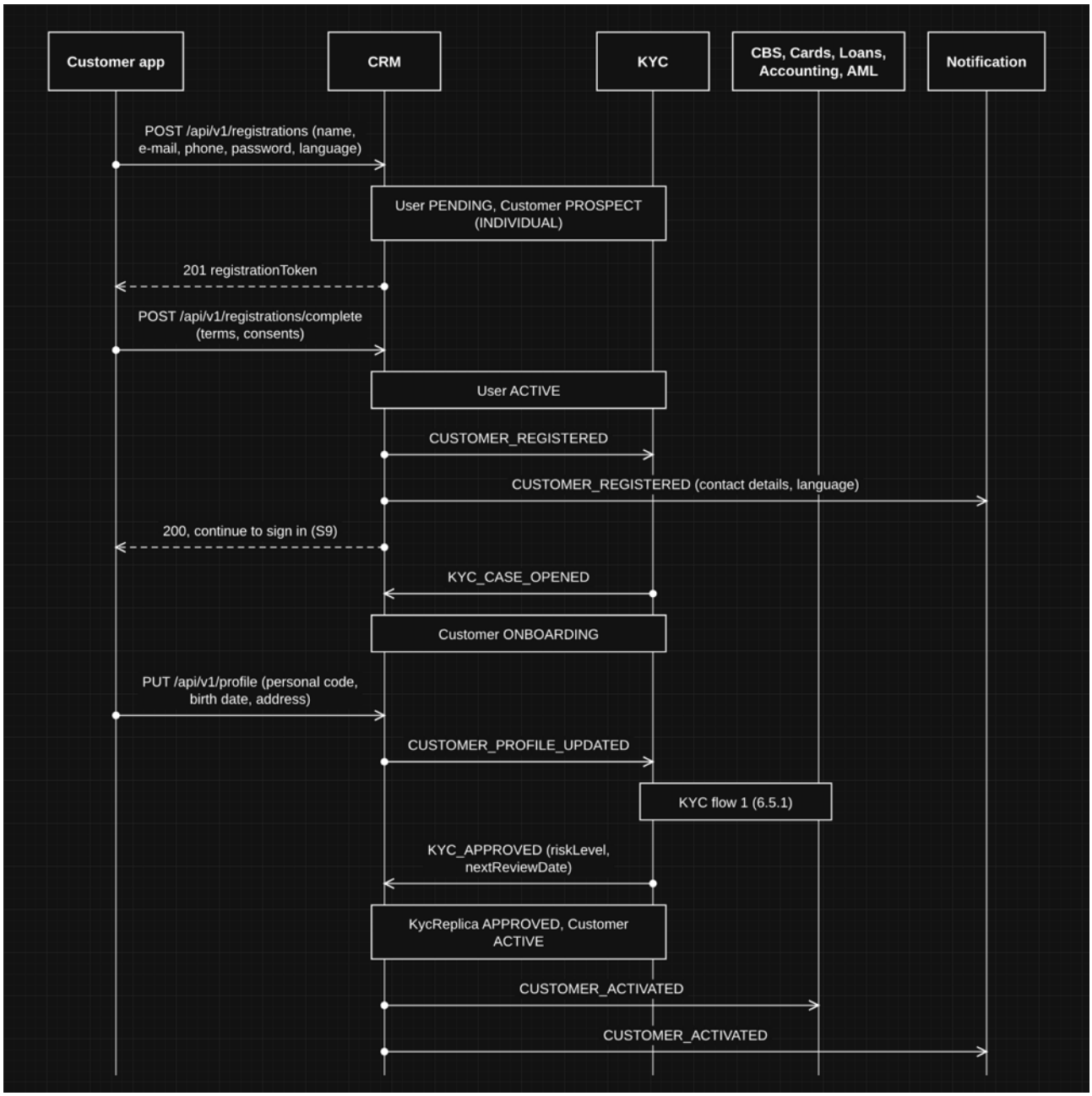
C-BR-09	Representatives. A representative must be an existing INDIVIDUAL customer, found by personal code, with status ONBOARDING or ACTIVE. Roles: DIRECTOR, ACCOUNTANT, PROCURATOR. validTo is empty or not before validFrom. A company always has at least one active DIRECTOR. Only an active DIRECTOR (or SUPPORT on request) can change the structure.
C-BR-10	Beneficial owners. A person who owns more than 25 % of the company: sharePercent from 25.01 to 100.00 with two decimals; the sum of active shares is at most 100.00. A beneficial owner does not have to be a customer (name, personal code and country are stored).
C-BR-11	Consents. DATA_PROCESSING is required and cannot be switched off while the customer has products; switching it off means a request to close the customer. MARKETING and E_STATEMENTS are optional. Every change is stored with time and user and published as CONSENT_CHANGED. Marketing messages may be sent only while MARKETING is granted; the channel choice per message category stays in Notification (12.6).
C-BR-12	Language. User.language (LT or EN, default LT) is owned by CRM, changed in My profile and published in CUSTOMER_PROFILE_UPDATED.
C-BR-13	Data export. The export contains all CRM data of the customer as JSON (JavaScript Object Notation) and PDF (Portable Document Format). It is ready within 24 hours (the legal limit is one month, GDPR (General Data Protection Regulation) Art. 12(3)) and can be downloaded for 7 days. Other modules export their own data.
C-BR-14	Erasure. CRM data is erased only when the customer is CLOSED and the 8-year retention period (NFR-6) has passed. An earlier request is answered REFUSED with the reason RETENTION_OBLIGATION (GDPR Art. 17(3)(b)) and the date from which the data will be erased. After the retention period a nightly job anonymizes the customer automatically.
C-BR-15	KYC replica. CRM never decides about KYC. It stores the latest outcome received from KYC events in KycReplica and shows it read-only.

7.5 Business flows

7.5.1 Flow 1: Registration and activation of an individual

Actors: customer, CRM, KYC, Notification, other modules. **Precondition:** the e-mail is not used (C-BR-01).

1. The customer opens **Register** on the sign-in screen and fills in name, e-mail, phone, password and language (C-W3, step 1). CRM validates the data, creates the User (PENDING) and the Customer (INDIVIDUAL, PROSPECT) and returns a registration token valid for 24 hours.
2. The customer accepts the terms and the DATA_PROCESSING consent and chooses the optional consents (step 2). CRM stores the consents, sets the user ACTIVE and publishes CUSTOMER_REGISTERED. KYC opens the onboarding case; Notification stores the contact details and the language.
3. The app opens the sign-in screen (S9). The customer signs in with e-mail, password and the SMS (short message service) code (CBS F6; CRM checks the password, 7.5.3).
4. CRM consumes KYC_CASE_OPENED, sets the customer ONBOARDING and the replica IN_PROGRESS.
5. The customer completes the profile in My profile (C-W2): personal code, date of birth, nationality and address. Each save publishes CUSTOMER_PROFILE_UPDATED; KYC stores a new snapshot. The customer then completes KYC (K-W1).
6. CRM consumes KYC_APPROVED, stores the risk level and the next review date, sets the customer ACTIVE and publishes CUSTOMER_ACTIVATED.
7. CBS, Cards and Loans now accept the customer, Accounting and AML register it, and Notification informs the customer.



CRM F1: registration and activation of an individual

Alternative paths

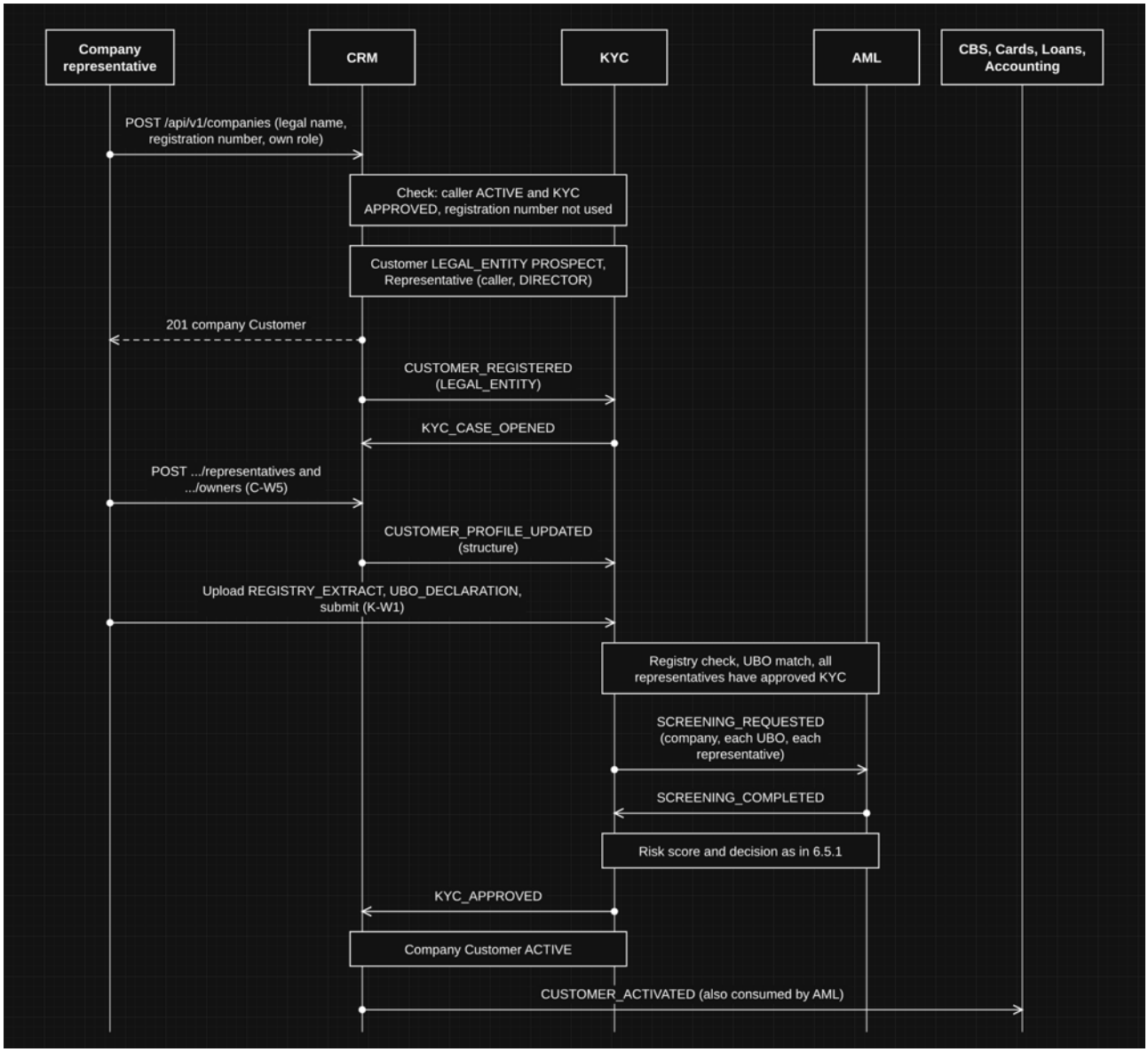
- **1A E-mail or personal code already used (step 1 or 5).** CRM answers 409 EMAIL_ALREADY_USED or 409 PERSONAL_CODE_ALREADY_USED; the error is shown under the field; nothing is published.
- **1B Step 2 not completed.** After 24 hours the PENDING user and the PROSPECT customer are deleted; nothing was published, so no other module knows them.
- **1C KYC rejected.** CRM consumes KYC_REJECTED, sets the replica REJECTED and the customer back to PROSPECT. The customer can sign in but cannot open products. A compliance officer may block the customer with a reason.
- **1D KYC expired during onboarding.** CRM consumes KYC_EXPIRED, sets the replica EXPIRED and the customer back to PROSPECT. When the customer starts a new case, KYC_CASE_OPENED sets ONBOARDING again.
- **1E KYC unavailable.** CRM keeps working; the KYC status in the app and in Customer 360 shows the last known value.

7.5.2 Flow 2: Registration of a legal entity and its structure

Actors: company representative (an ACTIVE individual customer with APPROVED KYC), CRM, KYC, other modules.

1. The representative opens **More > My companies > Add company** (C-W4) and enters legal name, registration number, country, legal form, registered address and their own role.

2. CRM checks that the caller is ACTIVE with KYC APPROVED and that the company is not registered yet (C-BR-01). It creates the LEGAL_ENTITY customer (PROSPECT, userId = caller), the company profile and the first Representative, and publishes CUSTOMER_REGISTERED.
3. KYC opens the onboarding case of the company; CRM consumes KYC_CASE_OPENED and sets ONBOARDING.
4. In Company structure (C-W5) the representative adds the other representatives by personal code (C-BR-09) and the beneficial owners (C-BR-10). Each change is saved at once, written to the history and published as CUSTOMER_PROFILE_UPDATED; while the company is ACTIVE it is also published as SENSITIVE_DATA_CHANGED (field group STRUCTURE).
5. The representative completes the company KYC in K-W1 (KYC Flow 2). After KYC_APPROVED CRM sets the company ACTIVE and publishes CUSTOMER_ACTIVATED. The representative can now open company accounts and assign mandates in CBS.



CRM F2: registration of a legal entity

Alternative paths

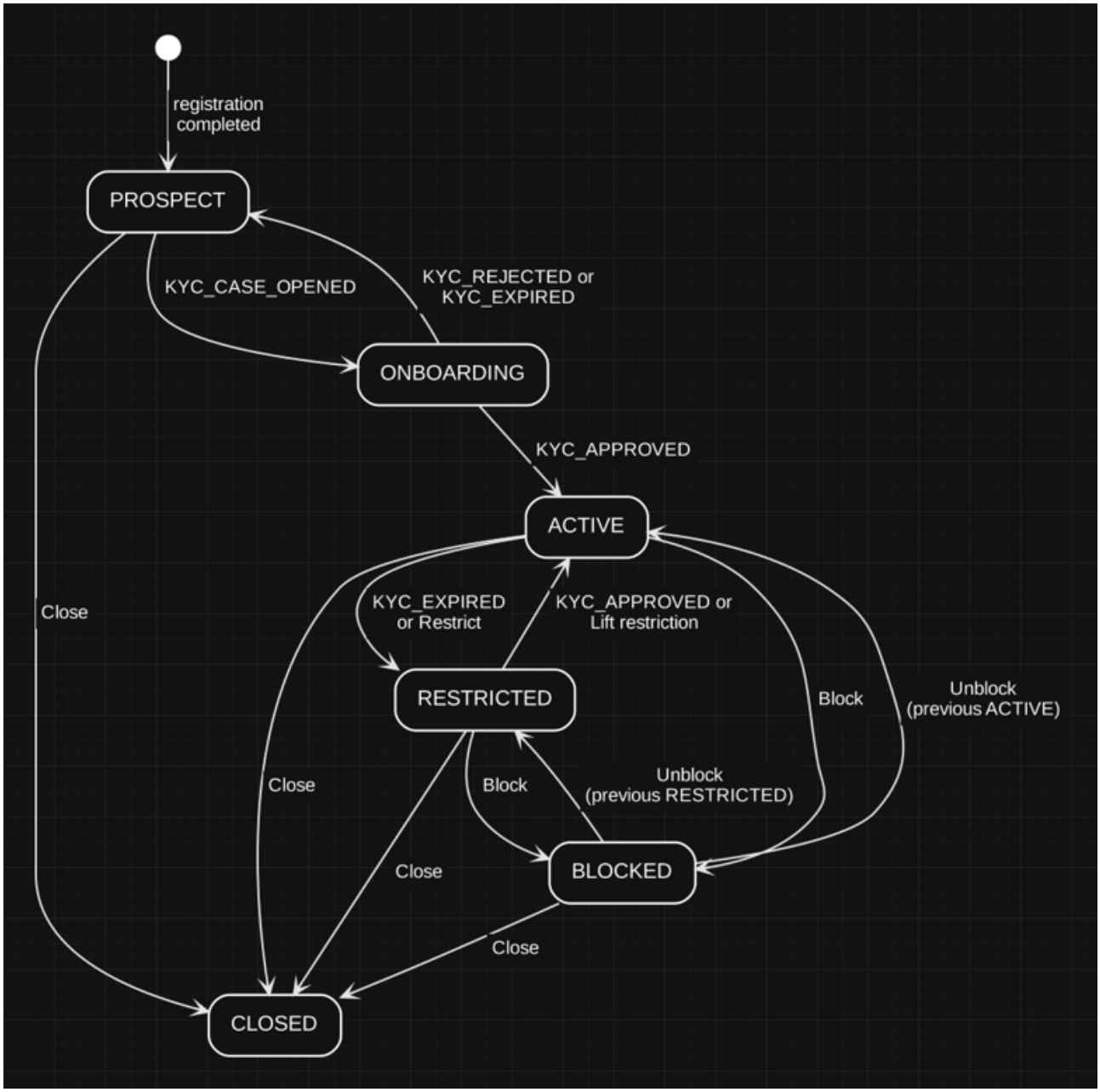
- **2A Caller not verified (step 2).** 422 KYC_NOT_APPROVED: “Complete your own verification before adding a company.”
- **2B Company already registered (step 2).** 409 COMPANY_ALREADY_REGISTERED; the error is shown under the registration number. The existing representatives must add the caller instead.
- **2C Representative not found (step 4).** 422 PERSON_NOT_CUSTOMER: “This person must first register as a customer.”
- **2D Last director removed (step 4).** 409 LAST_DIRECTOR: the change is refused.

7.5.3 Flow 3: Password check for sign-in

CBS runs the sign-in flow (8.5.6 F6) and the one-time code. CRM only checks the password, because it owns the password hash. In step 1 of F6 CBS calls POST /internal/crm/users/authenticate with e-mail and password.

1. CRM finds the user by e-mail. If the user is locked (lockedUntil in the future), it answers 403 USER_LOCKED.
2. CRM compares the password with the hash. A wrong password increases failedLogins; the fifth one in a row sets lockedUntil = now + 15 minutes (C-BR-02). The answer for an unknown e-mail and for a wrong password is the same: 401 INVALID_CREDENTIALS.
3. A correct password resets failedLogins and returns userId, userStatus, customerId (the user's INDIVIDUAL customer) and customerStatus. CBS continues only when the user is ACTIVE and the customer is not BLOCKED or CLOSED; in every other case it shows the neutral message "Sign-in failed."

7.5.4 Flow 4: Status control



Status model of Customer

Transition	Trigger and actor	Effect in other modules
PROSPECT → ONBOARDING	KYC_CASE_OPENED for an onboarding case	None; products are not available yet.
ONBOARDING → ACTIVE	KYC_APPROVED	CUSTOMER_ACTIVATED: CBS, Cards and Loans accept the customer.

ONBOARDING → PROSPECT	KYC_REJECTED or KYC_EXPIRED	None.
ACTIVE → RESTRICTED	KYC_EXPIRED (reason KYC_EXPIRED) or **Restrict** (COMPLIANCE, ADMIN)	CBS and Cards block outgoing payments, card purchases and new products; incoming funds are still accepted.
RESTRICTED → ACTIVE	KYC_APPROVED when the reason was KYC_EXPIRED, or **Lift restriction** when the restriction was manual	Outgoing payments and new products allowed again.
ACTIVE or RESTRICTED → BLOCKED	**Block** (COMPLIANCE, ADMIN); the previous status is stored in statusBeforeBlock	CBS refuses sign-in and all operations; for an individual the user becomes BLOCKED.
BLOCKED → previous status	**Unblock** (COMPLIANCE, ADMIN)	Back to the rules of the previous status; the user becomes ACTIVE.
any except ONBOARDING → CLOSED	**Close** (ADMIN, COMPLIANCE) after C-BR-06	Final; the retention period starts (C-BR-14).

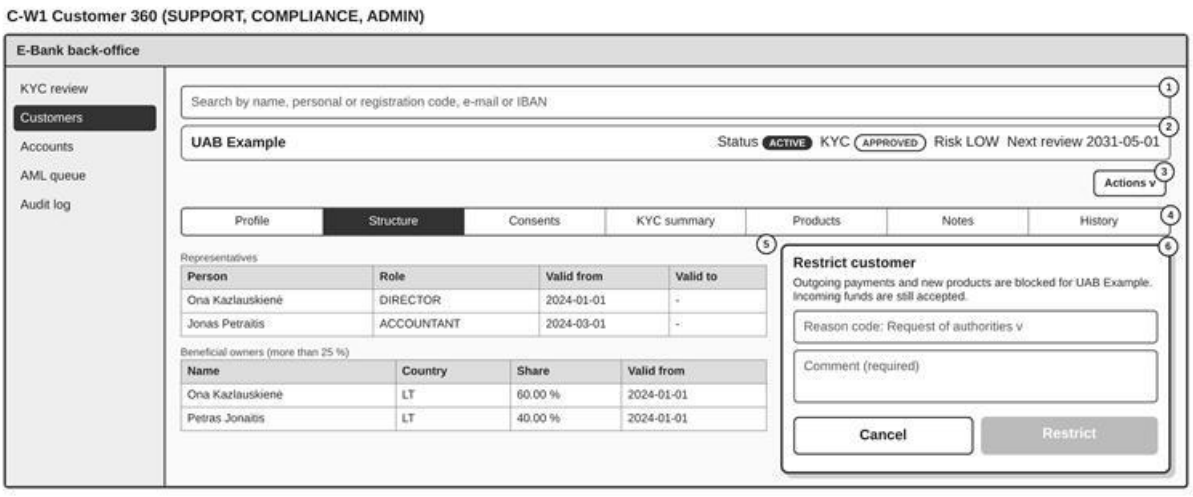
1. An officer opens the customer in Customer 360 (C-W1) and chooses an action in **Actions**. Only the transitions allowed from the current status are enabled.
2. A dialog asks for a reason code and a comment; the confirm button stays disabled until both are filled.
3. CRM checks the role (C-BR-05) and the preconditions, saves the status, the history entry and the event in one transaction and publishes CUSTOMER_STATUS_CHANGED with the old and new status, the KYC status and the reason code.
4. Consumers update their replicas. Goal G3: the change reaches all modules within 5 s (p95).

7.5.5 Flow 5: Profile changes and data rights

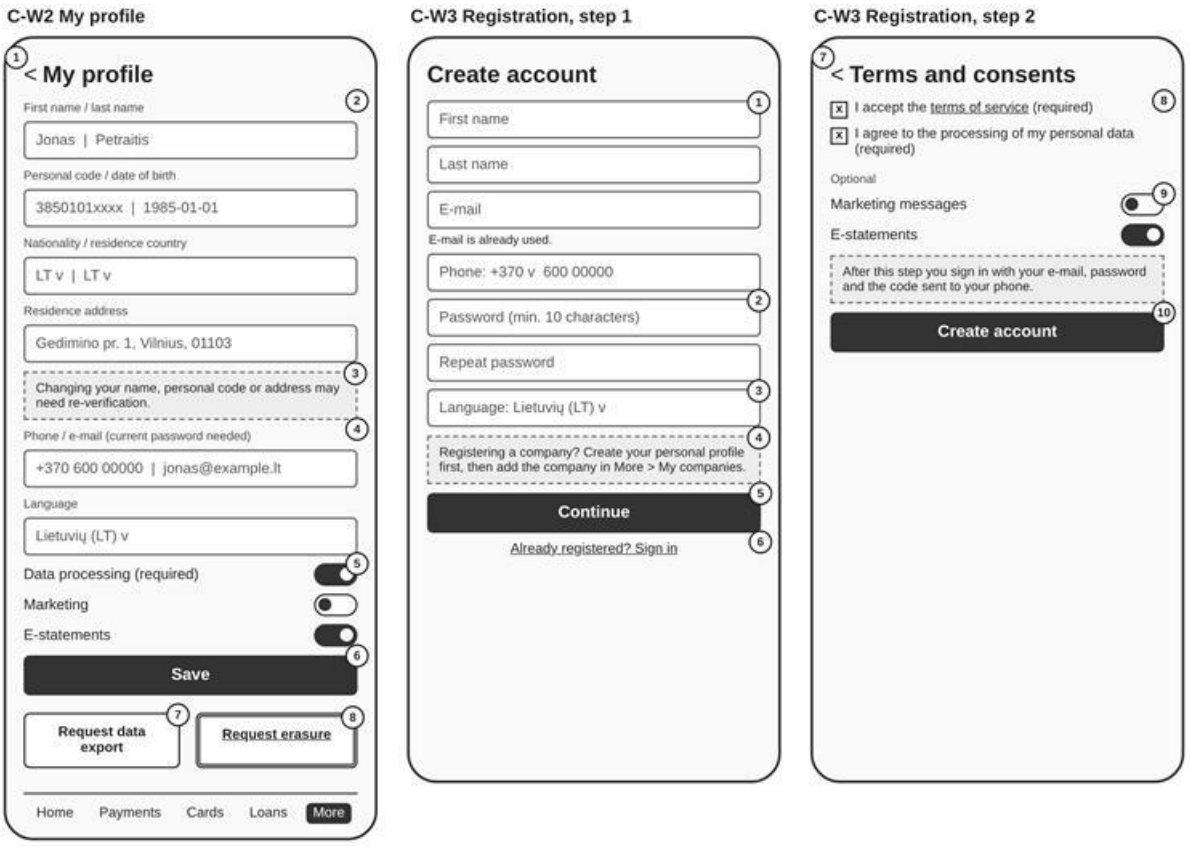
- **Profile change.** The customer edits My profile (C-W2) and presses **Save**. CRM validates the fields, saves them with the history (old and new value) and publishes CUSTOMER_PROFILE_UPDATED. If a sensitive field changed (C-BR-07), it also publishes SENSITIVE_DATA_CHANGED; KYC decides whether a review case is needed (KYC Flow 4).
- **Consent change.** A consent switch saves at once and publishes CONSENT_CHANGED (C-BR-11).
- **Data export. Request data export** creates a DataRequest (EXPORT, OPEN). A background job builds the file and sets COMPLETED; the customer downloads it in My profile (C-BR-13).
- **Erasure. Request erasure** opens a confirmation dialog. CRM creates a DataRequest (ERASURE) and answers it at once: REFUSED with the reason and date (C-BR-14) while the customer has products or the retention period has not passed.

7.6 Wireframes

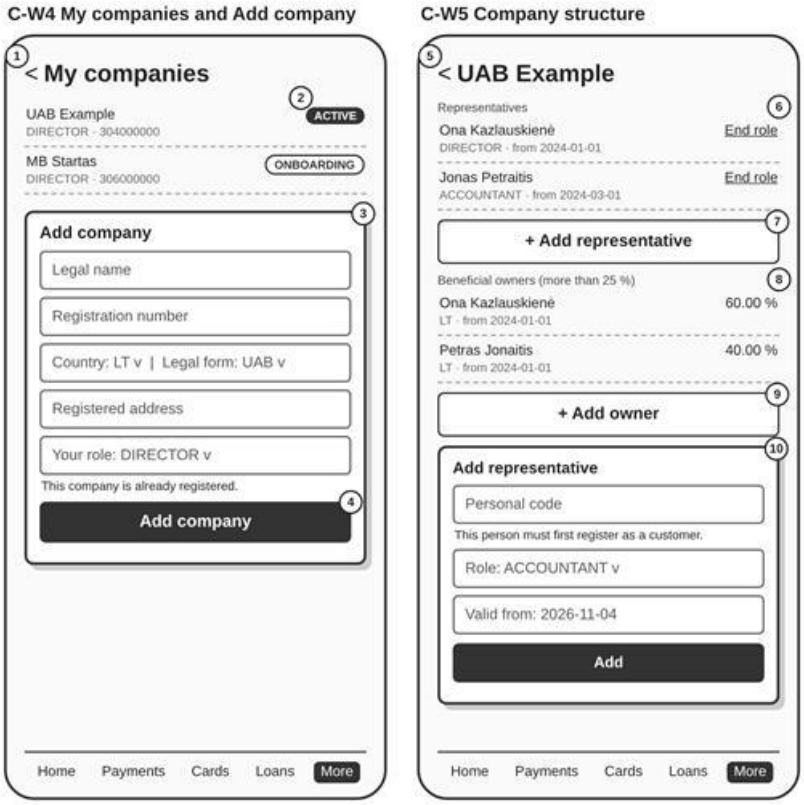
C-W2 to C-W5 are screens of the customer app (mobile, same structure on the web). My profile and My companies are reached from **More**; registration from the sign-in screen (S9). C-W1 is a back-office screen. The notation is described in 5.3.



Wireframe C-W1 (Customer 360, back-office)



Wireframes C-W2 (My profile) and C-W3 (registration, two steps)



Wireframes C-W4 (My companies) and C-W5 (company structure)

Transitions: the back-office search opens C-W1; **Actions** opens the status dialog and returns to C-W1. S9 **Register** opens C-W3 step 1; **Continue** opens step 2; **Create account** returns to S9. **More > My profile** opens C-W2; **Save** stays on C-W2. **More > My companies** opens C-W4; a company row opens C-W5; **Add company** opens C-W5 of the new company.

Ref	Element	What happens when it is used
C-W1.1	Customer search	Finds customers by name, personal or registration code, e-mail or IBAN (the IBAN is resolved through CBS GET /internal/cbs/accounts/by-iban/{iban}). Results are paged, 50 rows.
C-W1.2	Status header	Customer status from CRM; KYC status, risk level and next review date from KycReplica (read-only).
C-W1.3	Actions	Restrict, Lift restriction, Block, Unblock, Close; only actions allowed from the current status and for the employee role are enabled (7.5.4).
C-W1.4	Tabs	Profile, Structure, Consents, KYC summary, Products, Notes, History. Products are loaded from CBS, Cards and Loans; if one module does not answer, only its panel shows “Not available”.
C-W1.5	Structure tab	Representatives and beneficial owners with validity dates (read-only for COMPLIANCE; SUPPORT can edit on the customer’s request with a reason).
C-W1.6	Status dialog	Reason code and comment are required; the confirm button is disabled until both are filled. Confirm runs Flow 4.
C-W2.1	< back	Returns to **More**.
C-W2.2	Personal data	Name, personal code, date of birth, nationality, residence country and address; validated on save. Personal code and date of birth are read-only after activation (C-BR-07).
C-W2.3	Sensitive-data notice	Warns that a change may need re-verification.
C-W2.4	Contacts and language	Phone and e-mail need the current password in a dialog (C-BR-08); the language applies to the app and to notifications.
C-W2.5	Consent switches	Saved at once (C-BR-11). Data processing cannot be switched off here.
C-W2.6	Save	Saves the personal data (Flow 5).
C-W2.7	Request data export	Creates an export request; when it is ready, a download row appears under the button.
C-W2.8	Request erasure	Confirmation dialog; the answer and its reason are shown in a banner (C-BR-14).
C-W3.1	Name, e-mail, phone	Required; errors are shown under the field, for example “E-mail is already used.”
C-W3.2	Password	Rules of C-BR-02; the repeat field must match.
C-W3.3	Language	LT or EN.
C-W3.4	Company notice	Explains that a company is added after the personal registration (Flow 2).
C-W3.5	Continue	Step 1 of Flow 1; opens step 2.
C-W3.6	Sign in link	Opens S9.
C-W3.7	< back	Returns to step 1 with the entered data.
C-W3.8	Required consents	Terms and data processing; **Create account** is disabled until both are checked.
C-W3.9	Optional consents	Marketing and e-statements.
C-W3.10	Create account	Step 2 of Flow 1; opens S9.
C-W4.1	< back	Returns to **More**.
C-W4.2	Company list	Companies where the user is an active representative, with role and status. A row opens C-W5.
C-W4.3	Add company form	Legal name, registration number, country, legal form, registered address and own role; errors under the fields (alternatives 2A, 2B).
C-W4.4	Add company	Runs Flow 2, steps 1–2.
C-W5.5	< back	Returns to C-W4.
C-W5.6	Representatives	Active representatives with role and validity; **End role** asks for the end date (C-BR-09).
C-W5.7	Add representative	Opens the dialog C-W5.10.
C-W5.8	Beneficial owners	Owners with country and share; tapping a row edits the share or ends the ownership (C-BR-10).
C-W5.9	Add owner	Dialog with name, personal code, country, share and valid-from date.
C-W5.10	Add representative dialog	Personal code, role and valid-from date; the person must already be a customer (alternative 2C).

7.7 Non-functional requirements

The shared non-functional requirements (chapter 4) apply. Module-specific targets:

- **Security and privacy:** personal code, date of birth, addresses and the old and new values in the history are encrypted at field level (AES-256 (Advanced Encryption Standard)); passwords are stored only as Argon2id hashes and never published; only COMPLIANCE and ADMIN change a status; every change is written to CustomerChangeLog.
- **Resilience:** other modules keep working from their status replicas when CRM is down; CRM keeps working when KYC is down (last known KYC value). Sign-in depends on the password check, so CRM is a core service (99.9 %, NFR-2) and runs at least two instances.
- **Performance:** password check p95 (the value that 95% of measurements do not exceed) below 200 ms; profile read p95 below 300 ms; search p95 below 1 s on 5 million customers; 200 requests per second sustained; a status change reaches all consumers within 5 s (p95, goal G3).
- **Consistency and privacy rights:** status change, history entry and event are saved in one transaction (outbox); optimistic locking (Customer.version) prevents lost updates; data export within 24 hours; erasure after closure and the 8-year retention period.

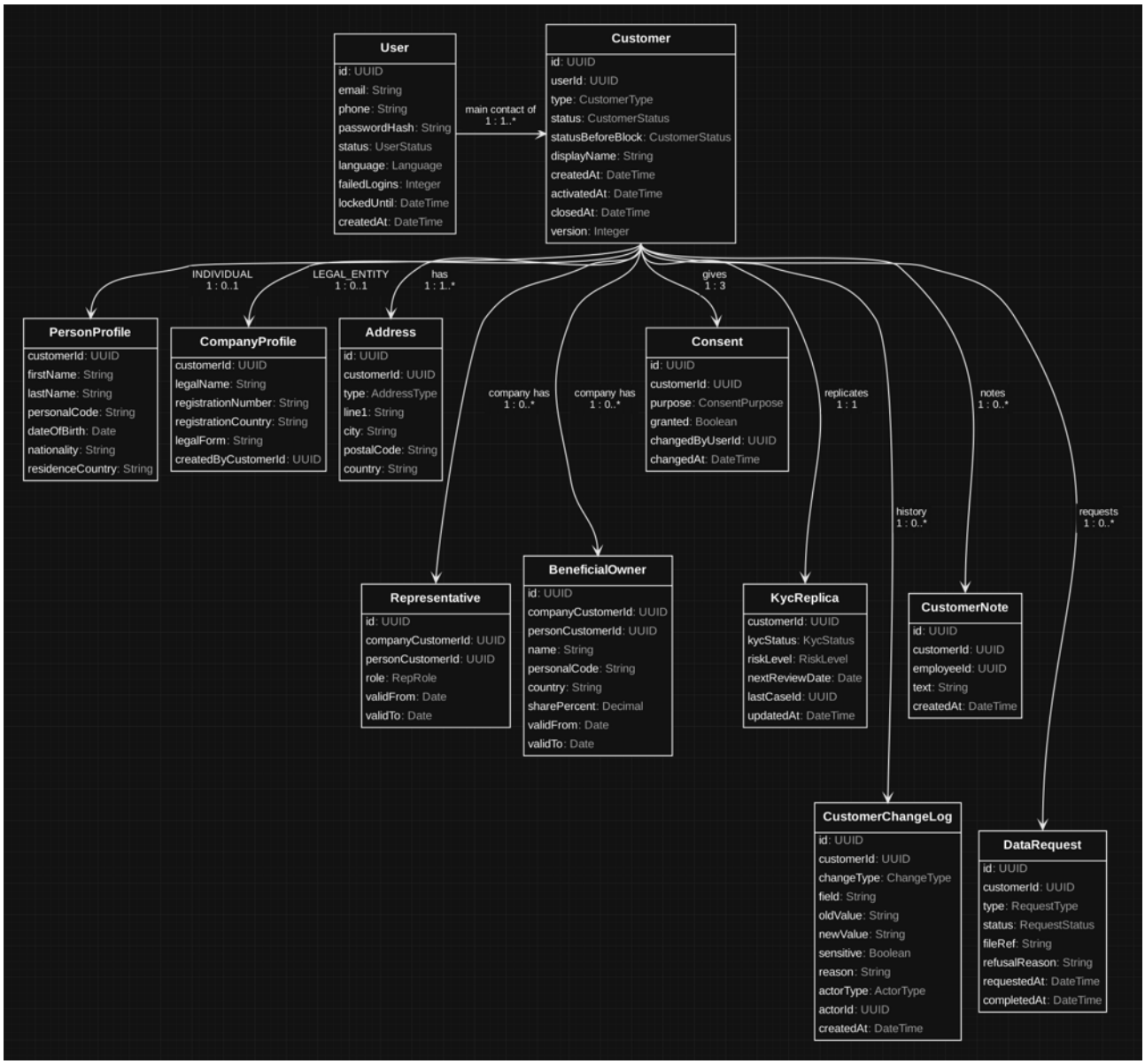
7.8 High-level architecture

CRM uses the common package layout of the System architecture chapter (3.3). It is one deployable service with its own database. Dependencies point downwards; the domain package has no dependency on infrastructure. Other modules are reached through the event bus and read-only lookups.

Package	Responsibility
crm.api	REST (Representational State Transfer) controllers for the customer app, the back-office console and the internal endpoints; validation and mapping to data transfer objects (DTOs), no business logic.
crm.security	Token and role check on every call (C-BR-05); access of a representative to company data; internal endpoints reachable only from other services.
crm.application	Use cases: registration, password check, profile, structure, lifecycle, consents, data rights, search.
crm.domain	Customer aggregate and rules: status transitions, sensitive fields, representatives, beneficial-owner shares, password policy.
crm.persistence	Repositories for the CRM database and a search index over name, codes and e-mail.
crm.integration	Read-only product and IBAN lookups in CBS, Cards and Loans; a failure hides only that panel.
crm.audit	CustomerChangeLog for every status, profile, contact, consent and structure change; serves the History tab.
crm.messaging	Idempotent consumers of KYC events; outbox relay that publishes customer events.
crm.jobs	Deletion of unfinished registrations, data exports, anonymization after the retention period.

7.9 Data model

The CRM database is used only by CRM. Other modules keep only customerId (and userId) and a status replica.



CRM data model (fields and types)

Entity	Meaning and rules
User	Login identity of one natural person (C-BR-02, C-BR-03). email is unique. failedLogins and lockedUntil support the lock.
Customer	Customer of the bank. userId is the person's own user (INDIVIDUAL) or the main contact (LEGAL_ENTITY). statusBeforeBlock is set while BLOCKED. displayName is "first name last name" or the legal name. version is used for optimistic locking.
PersonProfile	Person data of an INDIVIDUAL customer. Countries are ISO 3166 alpha-2 codes. personalCode is unique.
CompanyProfile	Company data of a LEGAL_ENTITY customer. createdByCustomerId is the representative who added it.
Address	One address per type: RESIDENCE (person), REGISTERED (company), optional MAILING.
Representative	Link between a company and an INDIVIDUAL customer (personCustomerId) with a legal role and validity dates (C-BR-09).
BeneficialOwner	Owner with more than 25 % (C-BR-10). personCustomerId is set when the owner is a customer.
Consent	Current value of each consent purpose; earlier values are in the history.
KycReplica	Latest KYC outcome received from KYC events (C-BR-15).
CustomerChangeLog	Append-only history: change type, field, old and new value (encrypted), sensitive flag, reason, actor and time.
CustomerNote	Internal note written by an employee (CRM-07); never shown to the customer.
DataRequest	Export or erasure request (C-BR-13, C-BR-14); fileRef points to the export file.

Enumerations

Enumeration	Values
CustomerType	INDIVIDUAL, LEGAL_ENTITY
CustomerStatus	PROSPECT, ONBOARDING, ACTIVE, RESTRICTED, BLOCKED, CLOSED
UserStatus	PENDING, ACTIVE, BLOCKED
Language	LT, EN
AddressType	RESIDENCE, REGISTERED, MAILING
RepRole	DIRECTOR, ACCOUNTANT, PROCURATOR
ConsentPurpose	DATA_PROCESSING, MARKETING, E_STATEMENTS
ChangeType	STATUS, PROFILE, CONTACT, CONSENT, STRUCTURE
KycStatus	NONE, IN_PROGRESS, APPROVED, REJECTED, EXPIRED
RiskLevel	LOW, MEDIUM, HIGH
RequestType / RequestStatus	EXPORT, ERASURE / OPEN, COMPLETED, REFUSED
ActorType	CUSTOMER, EMPLOYEE, SYSTEM
StatusReason	KYC_EXPIRED, KYC_APPROVED, AUTHORITY_REQUEST, SUSPECTED_FRAUD, CUSTOMER_REQUEST, DECEASED, OTHER

7.10 API contract

Conventions are in the System architecture chapter. Registration endpoints are public and rate-limited (10 requests per minute per IP (Internet Protocol) address). Other customer endpoints need a customer token; company endpoints are allowed only for active representatives. Back-office endpoints need an employee token with the role named in C-BR-05; SUPPORT is enough for reading.

7.10.1 Customer endpoints

Method and path	Purpose	Request (main fields)	Success
POST /api/v1/registrations	Registration step 1	firstName, lastName, email, phone, password, language	201 registrationToken, expiresAt
POST /api/v1/registrations/complete	Registration step 2	header X-Registration-Token; termsAccepted, dataProcessing, marketing, eStatements	200 userId, customerId
GET /api/v1/profile	Own profile	none	200 profile, contacts, language, consents, status, kycStatus, version
PUT /api/v1/profile	Change personal data	firstName, lastName, personalCode, dateOfBirth, nationality, residenceCountry, address, version	200 profile
PUT /api/v1/profile/contacts	Change e-mail or phone	email, phone, currentPassword	200 contacts
PUT /api/v1/profile/language	Change language	language	200
PUT /api/v1/profile/consents/{purpose}	Change a consent	granted	200 consent
POST /api/v1/profile/data-requests	Export or erasure request	type	201 DataRequest
GET /api/v1/profile/data-requests	Own requests	none	200 list
GET /api/v1/profile/data-requests/{id}/file	Download the export	none	200 file (ZIP with JSON and PDF)
GET /api/v1/companies	My companies	none	200 list with role and status

POST /api/v1/companies	Add a company	legalName, registrationNumber, registrationCountry, legalForm, registeredAddress, ownRole	201 company
GET /api/v1/companies/{id}	Company with structure	none	200 company, representatives, owners
POST /api/v1/companies/{id}/representatives	Add a representative	personalCode, role, validFrom	201 Representative
PATCH /api/v1/companies/{id}/representatives/{repId}	End a role	validTo	200 Representative
POST /api/v1/companies/{id}/owners	Add a beneficial owner	name, personalCode, country, sharePercent, validFrom	201 BeneficialOwner
PATCH /api/v1/companies/{id}/owners/{ownerId}	Change share or end ownership	sharePercent or validTo	200 BeneficialOwner

7.10.2 Back-office endpoints

Method and path	Purpose	Request (main fields)	Success
GET /api/v1/admin/customers	Search (C-W1.1)	query: q, status, type, page, size (max 50)	200 page
GET /api/v1/admin/customers/{id}	Customer 360	none	200 profile, structure, consents, kycReplica
GET /api/v1/admin/customers/{id}/products	Products tab	none	200 accounts, cards, loans; a missing module is marked unavailable
GET /api/v1/admin/customers/{id}/history	History tab	query: page, size	200 page of CustomerChangeLog
POST /api/v1/admin/customers/{id}/status-actions	Status action (Flow 4)	action (RESTRICT, LIFT_RESTRICTION, BLOCK, UNBLOCK, CLOSE), reasonCode, comment, version	200 customer
PUT /api/v1/admin/customers/{id}/profile	Profile change on the customer's request (SUPPORT)	changed fields, reason, version	200 profile
GET, POST /api/v1/admin/customers/{id}/notes	Notes tab	text (1–2000 characters)	200 list, 201 note

7.10.3 Internal endpoints

Method and path	Used by	Request / response
POST /internal/crm/users/authenticate	CBS sign-in (F6 step 1)	email, password / 200 userId, userStatus, customerId, customerStatus; errors in 7.10.6
GET /internal/crm/companies/{customerId}/representatives	CBS mandate assignment (S7 user search)	200 list of personCustomerId, userId, name, role, validFrom, validTo
GET /internal/crm/customers/{customerId}	On-demand status check (Cards authorization, 9.3.3); replicas are preferred	200 id, type, status, kycStatus, userId, displayName

7.10.4 Events published by CRM

All events use the common envelope and are keyed by customerId. userId is the user to notify (for a company the main contact).

Event	Published when	Payload fields

CUSTOMER_REGISTERED	Registration step 2 completed, or a company added	customerId, userId, type, displayName, email, phone, language; for a company also registrationNumber, registrationCountry, representativeCustomerId
CUSTOMER_PROFILE_UPDATED	Any profile, contact, language or structure change	customerId, userId, version, type, person or company data, address, representatives (personCustomerId, name, personalCode, role, validFrom, validTo), owners (name, personalCode, country, sharePercent), email, phone, language
SENSITIVE_DATA_CHANGED	A sensitive field changed (C-BR-07)	customerId, userId, version, fieldGroups (NAME, PERSONAL_CODE, DATE_OF_BIRTH, NATIONALITY, ADDRESS, STRUCTURE), changedAt
CUSTOMER_ACTIVATED	First activation	customerId, userId, type, kycStatus, riskLevel, activatedAt
CUSTOMER_STATUS_CHANGED	Every other status transition	customerId, userId, type, oldStatus, newStatus, kycStatus, reasonCode (StatusReason), changedAt
CONSENT_CHANGED	A consent changed	customerId, userId, purpose, granted, changedAt

7.10.5 Events consumed by CRM

Event	From	Used for
KYC_CASE_OPENED	KYC	Onboarding case: PROSPECT → ONBOARDING, replica IN_PROGRESS
KYC_INFO_REQUESTED	KYC	Shows “information requested” in the KYC summary
KYC_APPROVED	KYC	Replica APPROVED with risk level and next review date; activation or lifting a KYC_EXPIRED restriction
KYC_REJECTED	KYC	Replica REJECTED; onboarding customer back to PROSPECT
KYC_EXPIRED	KYC	Replica EXPIRED; ACTIVE → RESTRICTED, ONBOARDING → PROSPECT

7.10.6 Error codes

Code	HTTP	Meaning
VALIDATION_ERROR, WEAK_PASSWORD, INVALID_SHARE	400	Field rules failed (C-BR-02, C-BR-10).
INVALID_CREDENTIALS, REGISTRATION_TOKEN_INVALID	401	Wrong e-mail or password; registration token missing or expired.
PERMISSION_DENIED, USER_LOCKED	403	Missing role or not a representative; user locked (C-BR-02).
CUSTOMER_NOT_FOUND	404	No such customer or no access.
EMAIL_ALREADY_USED, PERSONAL_CODE_ALREADY_USED, COMPANY_ALREADY_REGISTERED, INVALID_STATUS_TRANSITION, PRODUCTS_OPEN, LAST_DIRECTOR, VERSION_CONFLICT	409	Uniqueness or state rule failed.
KYC_NOT_APPROVED, PERSON_NOT_CUSTOMER, CONSENT_REQUIRED, COMMENT_REQUIRED	422	Business rule failed (Flow 2, C-BR-05, C-BR-11).

8 CBS and Payment system

8.1 Scope and ownership

The Core Banking System (CBS) is the system of record for accounts, balances, balance holds and account transactions. The Payment system turns a payment instruction (internal, SEPA (Single Euro Payments Area) or SWIFT (Society for Worldwide Interbank Financial Telecommunication)) into a transaction: it calculates fee and FX (foreign exchange), asks for strong customer authentication (SCA), asks AML (anti-money laundering) to screen the payment, handles dual-signature approval and settles through the payment networks. CBS also contains the identity and access function (the sign-in flow, sessions and one-time codes, called IAM (identity and access management) in the CRM (customer relationship management) chapter), because it protects every payment.

Owned by this module	Only referenced (owned by other modules)
Accounts, IBANs, balances, limits; mandates; balance holds; payments, transactions, beneficiaries; exchange rates; sessions, one-time codes, devices	Customer identity, status and password hash (CRM), identity verification (KYC (know your customer)), screening decisions (AML), general ledger (Accounting), cards (Cards), loan decisions and schedules (Loans), message delivery (Notification)

Accountant and *Director* in this chapter are corporate user roles of a customer company (AccountMandate.role). They are not bank employees (Employee.role).

8.2 Feature table

ID	Feature	Actor	Priority
CBS-01	Account opening (individual and legal entity) with IBAN (International Bank Account Number) generation	Individual, legal entity user	Must
CBS-02	Corporate mandates and permissions, dual-signature setting	Legal entity admin	Must
CBS-03	Balance, holds and transaction history, statement export (CSV (comma-separated values), PDF (portable document format))	Customer, corporate user	Must
CBS-04	Funds reservation and posting interface for Cards, Loans and Accounting	System (Cards, Loans, Accounting)	Must
CBS-05	Account administration: freeze, unfreeze, close, change daily limit	Employee (SUPPORT, ADMIN)	Should
CBS-06	Login with one-time code (SCA (strong customer authentication)), session timeout, new-device alert	Customer, corporate user, employee	Must
PAY-01	Internal transfer between accounts in the bank	Customer, corporate user	Must
PAY-02	SEPA (Single Euro Payments Area) transfer, outgoing and incoming	Customer, corporate user	Must
PAY-03	SWIFT (Society for Worldwide Interbank Financial Telecommunication) transfer, outgoing and incoming	Customer, corporate user	Should
PAY-04	Dual-signature payment workflow	Corporate user (Accountant, Director)	Must
PAY-05	Saved recipients (beneficiaries)	Customer, corporate user	Could
PAY-06	Fee and FX (foreign exchange) quote shown before confirmation	Customer, corporate user	Must

Won't, deliberately excluded from this release: standing orders and scheduled payments; SEPA Instant transfers; bulk payment files; interest on savings accounts and overdrafts; accounts and payments in currencies other than EUR (euro), USD (US dollar) and GBP (British pound); cash deposits and withdrawals; SCA by biometrics or a banking app (only SMS (short message service) codes); real connections to the SEPA and SWIFT networks and to a rate provider (mocks are used).

8.3 Business rules

ID	Rule
BR-01	IBAN (International Bank Account Number): CBS (core banking system) generates the bank's own IBANs as LT (country code of Lithuania) + 18 digits (20 characters) with valid mod-97 (modulo 97, the standard bank-account check calculation) check digits, unique. A beneficiary IBAN can belong to another country: it must start with a known country code, have the length of that country (for example LT 20, DE 22 characters, never more than 34) and have valid mod-97 check digits.
BR-02	Available balance = balance - sum(ACTIVE holds). A payment needs availableBalance >= amount + fee.
BR-03	Daily limit: default EUR (euro) 10,000 outgoing per account per calendar day (Europe/Vilnius), kept in AccountSettings.dailyLimit. Completed payments and ACTIVE holds count.
BR-04	SCA (strong customer authentication): a 6-digit code is required for login, for every payment above EUR 500 (EUR equivalent), for every payment approval, for every mandate change, and for card activation and credit-card repayment (chapter 9). CBS generates the code and publishes OTP_REQUESTED; Notification delivers it by SMS (short message service). A code is valid 5 minutes and allows 3 attempts. SCA is done once per payment, by the person who authorizes it (the customer, or the Director in a dual-signature flow).
BR-05	Fees (in EUR, debited in the account currency at the quoted rate): internal EUR 0.00, SEPA (Single Euro Payments Area) EUR 0.20, SWIFT (Society for Worldwide Interbank Financial Telecommunication) EUR 5.00.
BR-06	Currency: SEPA payments are in EUR only. If the payment currency differs from the account currency, CBS converts at the quoted bank rate plus a 1.0% markup. The quote is fixed when the user confirms. Exchange rates are kept in an FxRate table with the currency pair, the rate and validFrom. In this project they are loaded from the default data and updated once a day by a mock rate provider.
BR-07	Dual signature: if AccountSettings.dualSignatureRequired is true, every outgoing payment needs approval by a user with APPROVE_PAYMENTS who is not the creator. A pending payment expires after 7 days.

BR-08	Status effects: account FROZEN blocks outgoing payments, incoming funds are still accepted; CLOSED rejects everything. Customer status RESTRICTED blocks outgoing payments and new accounts, incoming funds are accepted; BLOCKED blocks login and all operations.
BR-09	Closing an account needs balance 0.00 and no ACTIVE hold.
BR-10	Validation: amount above 0 with at most 2 decimals; beneficiary name 1-70 and purpose 1-140 characters; BIC (Bank Identifier Code) 8 or 11 characters, required for SWIFT.
BR-11	Transaction records: Transaction.amount is negative for a debit and positive for a credit. An internal transfer is one transaction on the source account with toAccountId set; the recipient's history shows it as a credit. Counterparty country is taken from the IBAN prefix (BIC characters 5-6 if there is no IBAN).
BR-12	Idempotency: every POST that creates a payment or posting carries an Idempotency-Key header; a repeated key returns the first result. The key is stored in Payment.idempotencyKey (payments) and Transaction.idempotencyKey (postings from Cards and Loans).
BR-13	Holds: ACTIVE while a payment is processed, CONSUMED when the money is deducted, RELEASED on failure, rejection, cancellation or expiry. The balance is deducted only when a hold is consumed.
BR-14	AML (anti-money laundering) contract: outgoing payments are screened before execution (synchronous call, 2 s timeout). If AML does not answer within 2 s or returns a 5xx error and the amount is above EUR 1,000, the transaction stays ON_HOLD and the call is repeated every minute, at most 5 times; after the fifth failure CBS publishes AML_SCREENING_FAILED and the transaction stays ON_HOLD until the AML officer decides. Up to EUR 1,000 the payment is executed and AML checks it afterwards. The screening call in section 11.8.1 also sends referenceId, initiatedByUserId and amountEur. referenceId is the Transaction id. initiatedByUserId is the user who created the payment. amountEur is the amount converted to EUR with the CBS exchange rate at that moment, without the FX markup. The 1,000 EUR limit compares amountEur. A repeated call uses the same referenceId, so AML returns the first decision.
BR-15	Sessions: after login CBS returns an access token valid 5 minutes and a refresh token. The app renews the access token with the refresh token; CBS refuses this when the last activity of the Session was more than 15 minutes ago, so a session ends after 15 minutes of inactivity (NFR-1). Logout ends the session at once.

8.4 Interfaces with other modules

Module	What is exchanged
CRM (and KYC (know your customer))	CBS (core banking system) consumes CUSTOMER_ACTIVATED and CUSTOMER_STATUS_CHANGED and keeps a status replica per customer (CustomerReplica). Mandate users must be ACTIVE with KYC APPROVED. CRM reads accounts and IBANs through read-only lookups. At sign-in CBS calls POST /internal/crm/users/authenticate to check the password (F6), because CRM owns the password hash; the password is never sent in an event.
AML (anti-money laundering)	CBS calls POST /internal/aml/screenings before each outgoing payment, consumes TRANSACTION_RELEASED and TRANSACTION_REJECTED (only for a transaction that is ON_HOLD), publishes AML_SCREENING_FAILED and TRANSACTION_COMPLETED.
Cards	Cards checks the account, reserves debit funds (hold plus pending transaction), then captures or cancels; credit-card repayments are debit postings of type CARD_REPAYMENT. CBS also validates one-time code tokens for Cards (POST /internal/cbs/sca/validate).
Loans	Loans credits loan payouts and debits installments through postings. An installment that cannot be debited returns INSUFFICIENT_FUNDS, and Loans marks it late.
Accounting	The nightly batch reads the day's completed transactions (GET /internal/cbs/transactions) and maps them with the table below.
Notification	CBS publishes the events in section 8.10.3; Notification turns them into messages and delivers one-time codes.
External	SEPA (Single Euro Payments Area) and SWIFT (Society for Worldwide Interbank Financial Telecommunication) networks through the Payment Gateway Interface.

CBS transaction (type, sign of amount)	Accounting transaction type
INTERNAL (debit, toAccountId set)	TransferInternal
SEPA debit / credit	SepaOut / SepaIn
SWIFT debit / credit	SwiftOut / SwiftIn
CARD debit	CardPayment
CARD_REPAYMENT debit	CardRepayment
LOAN credit / debit	LoanPayout / LoanInstallment

Note: a credit-card purchase has no CBS transaction. Accounting reads completed credit-card purchases from Cards (section 9.9.3).

8.5 Business flows

8.5.1 F1: Account opening and mandate assignment

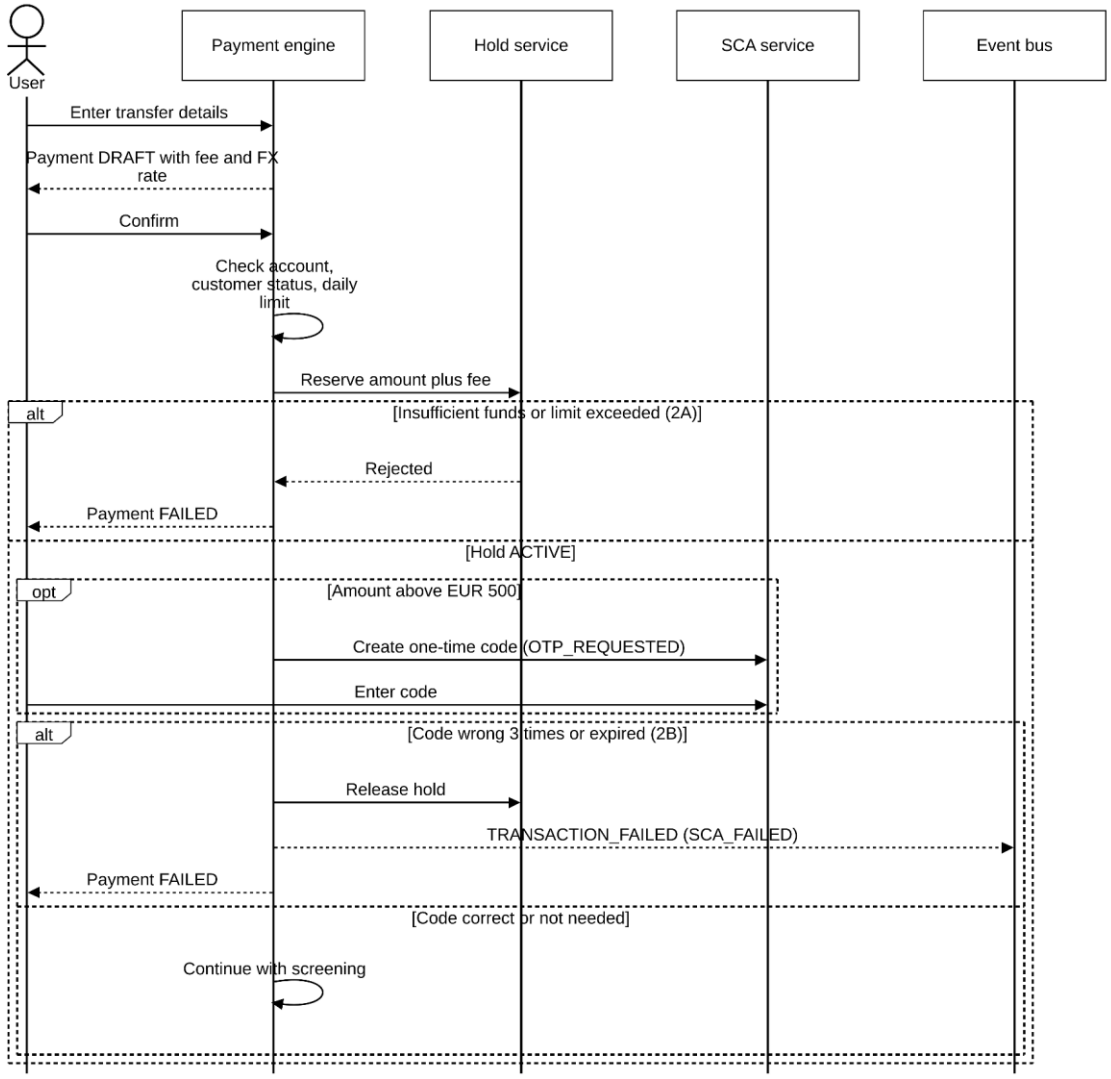
- Actors: Customer, Legal entity admin, CBS (Account Core Service), SCA (strong customer authentication) service.
- Preconditions: customer status ACTIVE (not RESTRICTED) and kycStatus = APPROVED.
- Main scenario:
  1. The customer requests an account with type (CURRENT, SAVINGS) and currency (EUR, USD, GBP).
  2. CBS creates the Account (OPEN) with a unique IBAN (International Bank Account Number) and AccountSettings with the default limit, and publishes ACCOUNT_OPENED. For individuals the flow ends here.
  3. *Legal entities:* the admin opens Manage mandates, selects a user, role and permissions (VIEW, CREATE_PAYMENTS, APPROVE_PAYMENTS) and may switch on dual signature.
  4. The admin enters a one-time code (BR-04). CBS saves the AccountMandate records and publishes MANDATE_CHANGED.
- Alternative 1A, invalid user: the user is not ACTIVE or not KYC-verified. CBS answers *"Assigned user must have an active and verified KYC (know your customer) status."* and the admin returns to step 3.
- Alternative 1B, code wrong or expired: nothing is saved.
- Alternative 1C, last admin or approver: a change is rejected if it leaves the account without an ADMIN, or with dual signature on and nobody holding APPROVE_PAYMENTS.
- Alternative 1D, customer restricted: the request in step 1 is rejected with *"Your profile is restricted. Contact support."*

8.5.2 F2: Transfer execution (internal, SEPA, SWIFT)

- Actors: User (customer, or Director in a dual-signature flow), Payment engine, Hold service, SCA service, AML (anti-money laundering), Payment gateway, Event bus.
- Preconditions: source account OPEN, customer not RESTRICTED, user has CREATE_PAYMENTS (or owns the account). A payment coming from F3 is already APPROVED, its hold exists and SCA is done.
- Main scenario:
  1. The user enters source account, recipient (IBAN, BIC (Bank Identifier Code) for SWIFT (Society for Worldwide Interbank Financial Telecommunication)), amount, currency and purpose. CBS creates a Payment (DRAFT) and shows fee and FX (foreign exchange) rate (PAY-06), calculated from the latest FxRate (BR-06).
  2. The user confirms. CBS checks account status, customer status, daily limit and available balance, then creates an AccountHold (ACTIVE) for amount plus fee.
  3. If the amount is above EUR (euro) 500 the user enters a one-time code.
  4. CBS creates the Transaction (PENDING) and calls AML with the request of rule BR-14. AML answers ALLOW.
  5. CBS routes the payment. *Internal:* moves the money to the target account. *SEPA/SWIFT:* sends the message to the payment gateway and waits for confirmation.
  6. On settlement CBS deducts the balance, sets the hold CONSUMED, Transaction = COMPLETED, Payment = EXECUTED, and publishes TRANSACTION_COMPLETED (one event per notified party for internal transfers).
- Alternative 2A, insufficient funds or limit exceeded (step 2): no hold is created, Payment = FAILED, message *"Insufficient available balance or daily limit exceeded."*
- Alternative 2B, code wrong 3 times or expired (step 3): hold RELEASED, Payment = FAILED, TRANSACTION_FAILED (reason SCA_FAILED).
- Alternative 2C, AML answers HOLD (step 4): Transaction = ON_HOLD, the hold stays ACTIVE. When an AML officer decides, AML publishes TRANSACTION_RELEASED (processing continues at step 5) or TRANSACTION_REJECTED (Transaction = REJECTED, Payment = FAILED, hold RELEASED). CBS acts on these events only while the transaction is ON_HOLD and ignores them otherwise. The customer is informed by AML's neutral message only; CBS publishes no other event, so the reason is never shown. The 60 s limit of 2E does not apply while the transaction is ON_HOLD.
- Alternative 2D, AML does not answer within 2 s or returns a 5xx error (step 4): rule BR-14: if amountEur is above EUR 1,000 the transaction is held and the call is repeated with the same referenceId; up to EUR 1,000 processing continues at step 5.

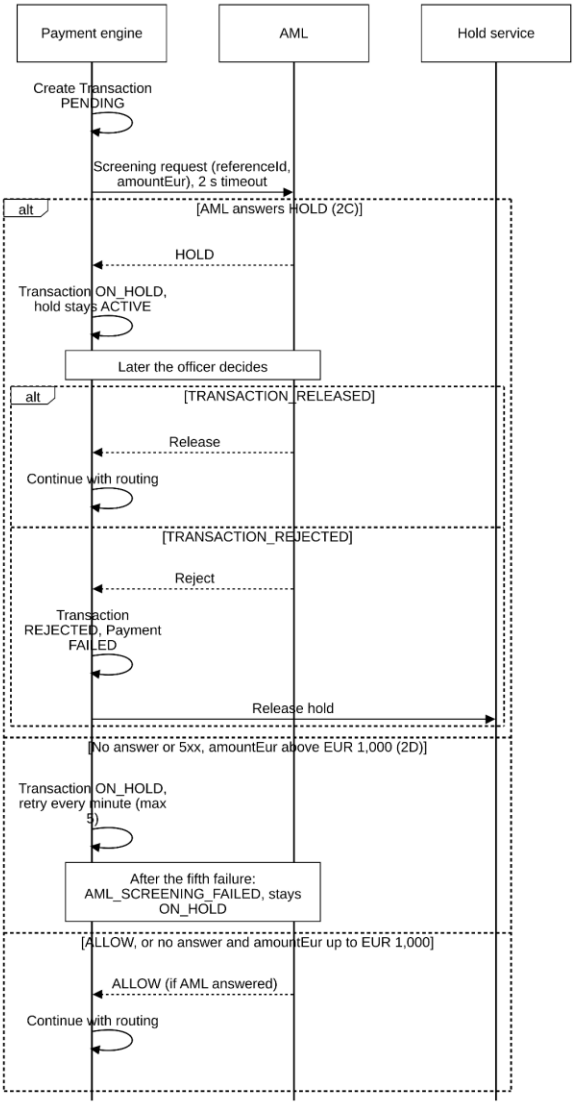
- Alternative 2E, gateway error or no confirmation within 60 s after the message is sent to the gateway (step 5): Transaction = FAILED, hold RELEASED (the balance was never deducted), Payment = FAILED, TRANSACTION_FAILED (reason GATEWAY_ERROR).
- Incoming payments: a SEPA (Single Euro Payments Area) or SWIFT message from the gateway is credited to the target IBAN if the account is OPEN or FROZEN; CBS creates a COMPLETED credit transaction with the sender's name, IBAN and country (counterpartyName, counterpartyIban, counterpartyCountry) and publishes TRANSACTION_COMPLETED (direction INCOMING). An unknown or CLOSED account is returned to the sender.

Part 1, steps 1 to 3: create the payment, reserve the money and check the one-time code.



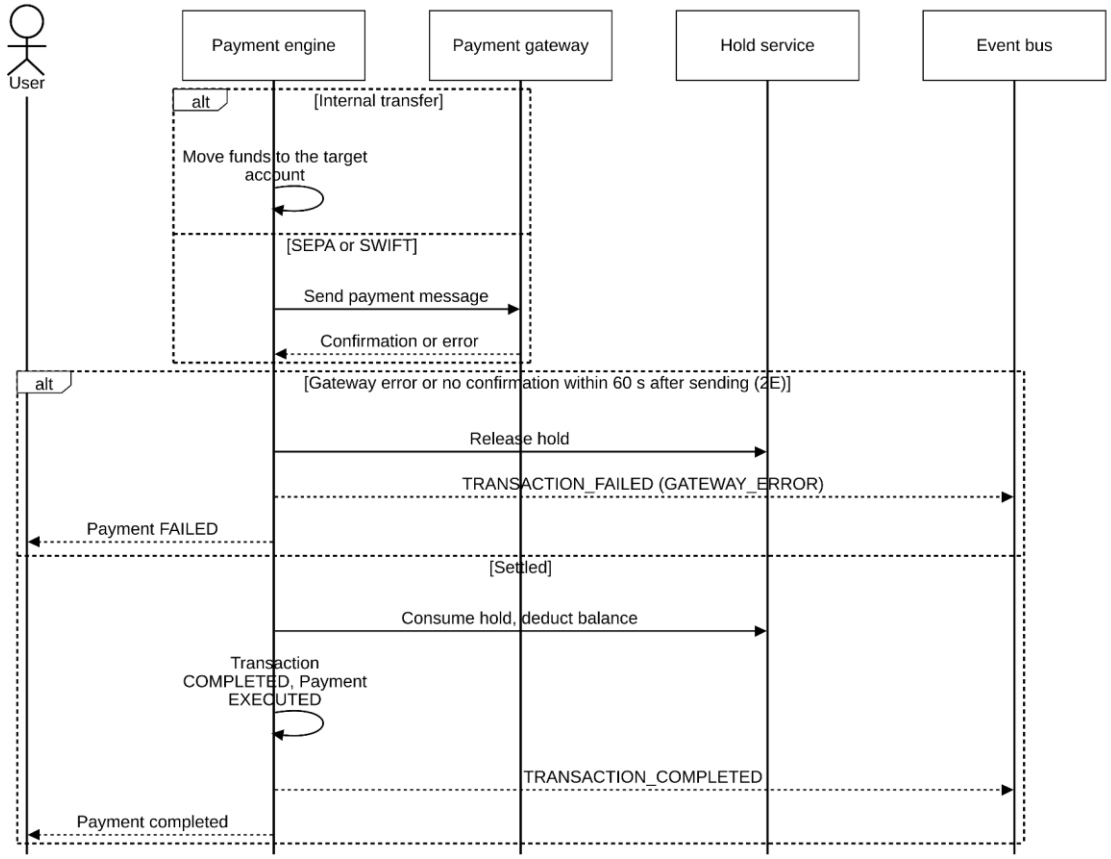
Sequence diagram F2, part 1, steps 1 to 3

Part 2, step 4: AML screening.



Sequence diagram F2, part 2, step 4

Part 3, steps 5 and 6: routing and settlement.

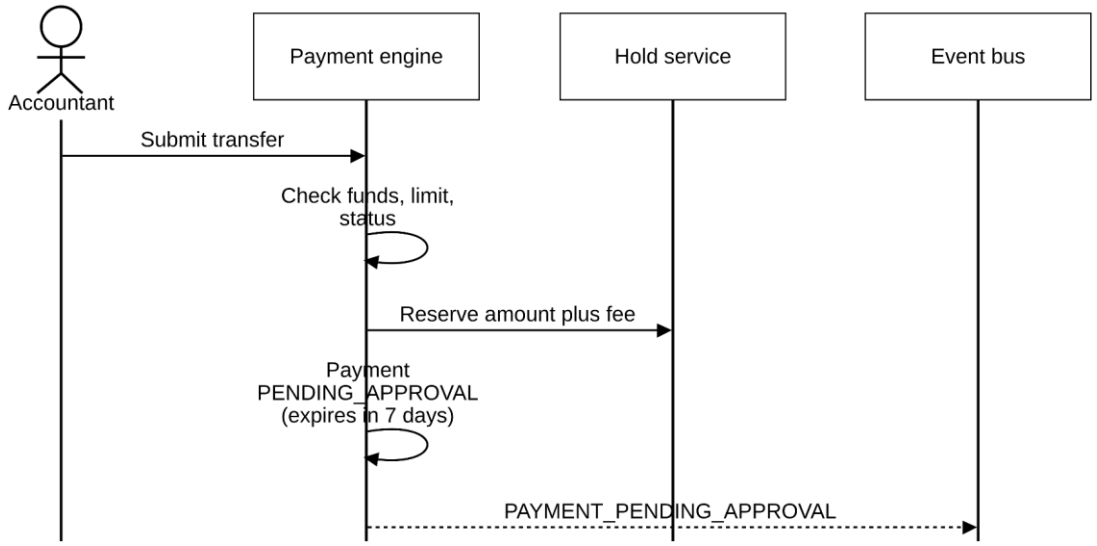


Sequence diagram F2, part 3, steps 5 and 6

8.5.3 F3: Legal entity payment with two signatures

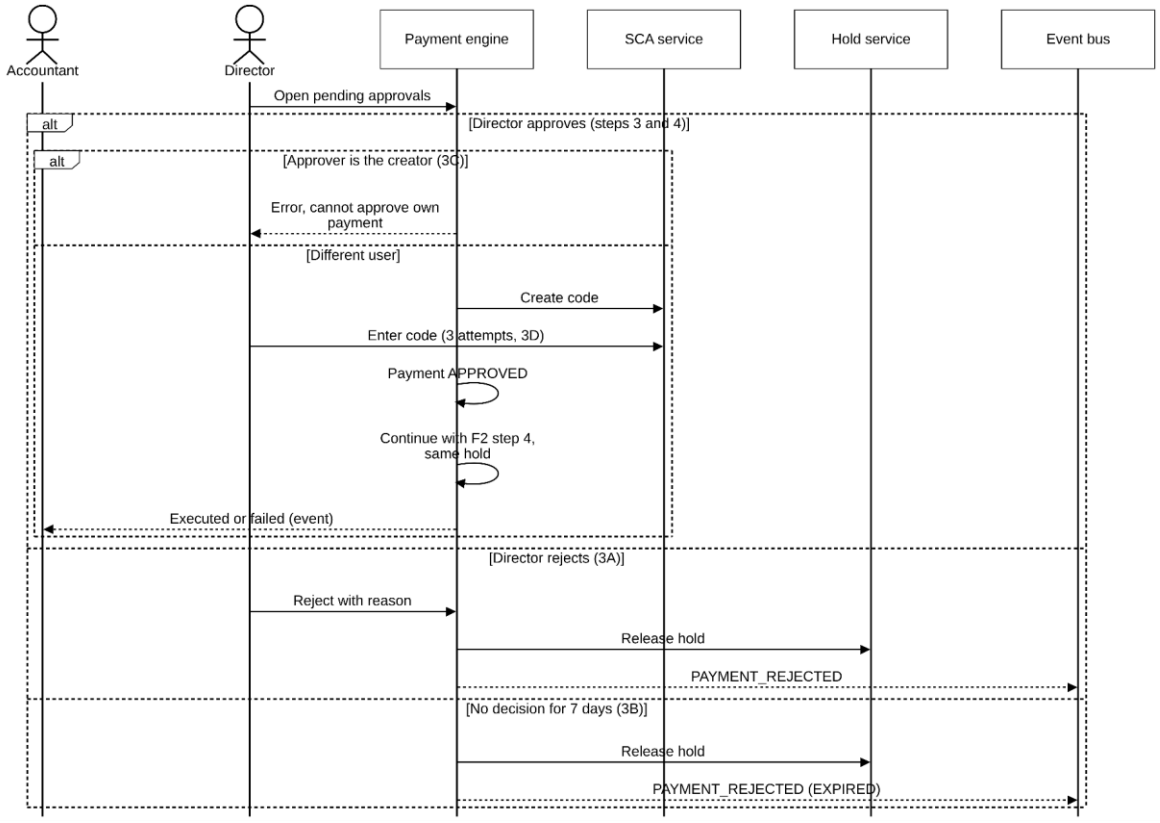
- Actors: Accountant (creator), Director (approver), Payment engine, Hold service, SCA service, Event bus.
- Preconditions: dualSignatureRequired = true; the Accountant has CREATE_PAYMENTS; a Director has APPROVE_PAYMENTS.
- Main scenario:
    1. The Accountant submits the transfer. CBS runs the checks of F2 step 2, creates the Payment as PENDING_APPROVAL with expiresAt = now + 7 days and the AccountHold, and publishes PAYMENT_PENDING_APPROVAL once for each user with APPROVE_PAYMENTS.
    2. The Director opens Pending approvals and reviews the payment.
    3. The Director approves and enters a one-time code. The approver must not be the creator; this is checked before the code is created.
    4. CBS sets Payment = APPROVED and continues with F2 step 4, reusing the existing hold.
- Alternative 3A, rejection: the Director enters a reason; Payment = REJECTED, hold RELEASED, PAYMENT_REJECTED to the Accountant.
- Alternative 3B, expiry: after 7 days without a decision Payment = EXPIRED, hold RELEASED, PAYMENT_REJECTED (reason EXPIRED).
- Alternative 3C, creator approves: CBS refuses with *“A payment cannot be approved by the user who created it.”*
- Alternative 3D, code wrong 3 times or expired: the challenge is FAILED, nothing else changes and the payment stays PENDING_APPROVAL. The Director can approve again with a new code until the payment expires.

Part 1, step 1: the Accountant submits the payment.



Sequence diagram F3, part 1

Part 2, steps 2 to 4 and alternatives 3A to 3D: the Director decides.



Sequence diagram F3, part 2

8.5.4 F4: Funds reservation and posting for Cards and Loans

- Actors: Cards, Loans, CBS (Account Core Service, Hold service).
- Preconditions: the target account is OPEN (for debits also not FROZEN) and its customer is not RESTRICTED.
- Main scenario, card purchase:
    1. Cards checks the account status, then reserves funds for the authorization (referenceId = card authorization id). CBS creates an ACTIVE hold and a PENDING transaction (type CARD) and returns holdId and transactionId.
    2. When Cards approves the purchase, it publishes CARD_TRANSACTION_AUTHORIZED (section 9.3.3). When the purchase completes, Cards calls capture. CBS consumes the hold, deducts the balance and sets Transaction = COMPLETED. CBS publishes no TRANSACTION_COMPLETED for card transactions.

- Main scenario, Loans: Loans asks CBS to credit a loan payout (type LOAN, COMPLETED at once, TRANSACTION_COMPLETED published) and later to debit each installment. A credit-card repayment is a debit posting of type CARD_REPAYMENT requested by Cards.
- Alternative 4A, insufficient funds: CBS answers INSUFFICIENT_FUNDS; no hold or transaction is created. Cards declines the purchase, Loans marks the installment late.
- Alternative 4B, cancel or expiry: Cards calls *cancel* or the hold passes expiresAt; the hold is RELEASED and the transaction FAILED.
- Alternative 4C, account not open: CBS answers ACCOUNT_NOT_OPEN; Loans keeps the payout pending.

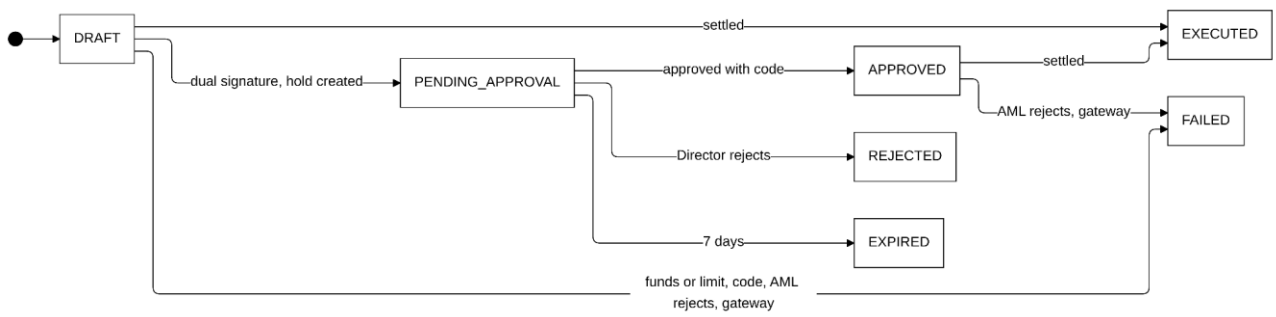
8.5.5 F5: Account administration (employee)

A short action-response flow, described without a diagram. Actors: employee with role SUPPORT or ADMIN. Steps: (1) the employee finds the customer in CRM (customer relationship management) and opens the account on screen W-CBS-10; (2) chooses *Freeze*, *Unfreeze*, *Close* or a new daily limit and enters a mandatory reason; (3) CBS validates the change, updates the account, writes an AuditLogEntry to the shared audit log (section 8.8) and publishes ACCOUNT_STATUS_CHANGED for status changes. Alternatives: 5A closing is rejected while the balance is not 0.00 or a hold is ACTIVE (*"Account cannot be closed: balance must be zero and no active holds."*); 5B an invalid status change (for example freezing a CLOSED account) is rejected; 5C a daily limit above EUR 100,000 needs the role ADMIN.

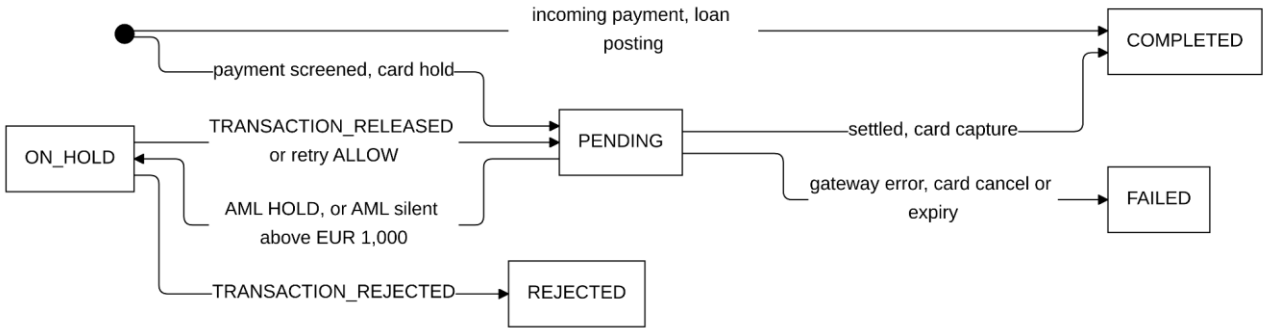
8.5.6 F6: Login and strong customer authentication

- Actors: User (customer or bank employee), CBS (SCA service), Notification (delivers the SMS (short message service)).
- Preconditions: the user is registered in CRM (customer relationship management), which owns the password hash. Employee users are loaded from the default data of CRM.
- Main scenario:
  1. The user enters e-mail and password. CBS calls POST /internal/crm/users/authenticate (section 7.5.3); CRM checks the password and returns the user and customer status. CBS continues only if the user is ACTIVE and the customer is not BLOCKED or CLOSED. A customer who is still onboarding (no CustomerReplica yet) can log in, so that the KYC (know your customer) documents can be uploaded; until CUSTOMER_ACTIVATED, CBS account and payment endpoints answer CUSTOMER_NOT_ACTIVE.
  2. CBS creates an OtpChallenge (purpose LOGIN) and publishes OTP_REQUESTED (userId, code, purpose); Notification sends the code by SMS.
  3. The user enters the code. CBS checks it, creates a Session and returns an access token and a refresh token (rule BR-15: the session ends after 15 minutes of inactivity). CBS updates UserDevice; a device not seen before also publishes NEW_DEVICE_LOGIN (userId, deviceName, city).
- Alternative 6A, wrong code: the attempt is counted; after 3 attempts the challenge is FAILED and the user starts again.
- Alternative 6B, expired code: after 5 minutes the user requests a new code (at most 3 per 15 minutes).
- Alternative 6C, wrong password, locked user or blocked customer: the same neutral message *"Sign-in failed."* is shown.
- Operations: payments, approvals, mandate changes, card activation and credit-card repayment use the same challenge with another purpose and referenceId; a successful check returns a single-use scaToken bound to that purpose and reference, valid for 5 minutes. Cards checks the token with POST /internal/cbs/sca/validate.
- Employees log in to the back-office with the same flow. CBS keeps a read-only copy of the Employee list (EmployeeReplica), loaded from the same default data as AML (anti-money laundering), and puts the employee role into the token at login.

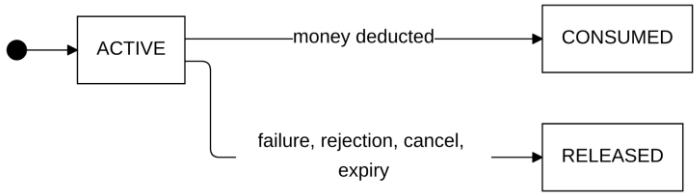
8.6 Status models



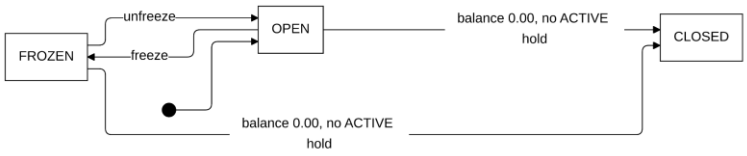
Status model of Payment



Status model of Transaction



Status model of AccountHold



Status model of Account

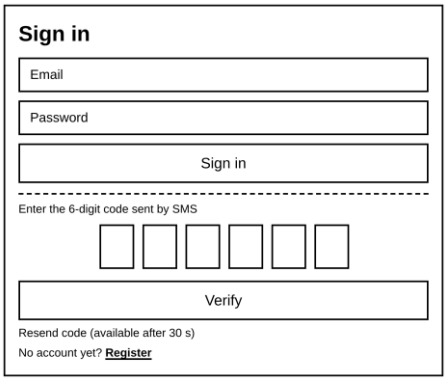
While a payment is processed it keeps its status (DRAFT or APPROVED). The API (application programming interface) returns the payment together with transactionStatus, and the app shows a payment whose transaction is ON_HOLD as *Under review*.

8.7 Wireframes

Each screen has its own picture, with a table of its elements next to it. Elements are named as they appear in the picture. Customer screens are mobile screens with the bottom navigation from the user interface principles (Home, Payments, Cards, Loans, More); Saved recipients, Pending approvals and Mandates are opened from More. The one-time code dialog is the shared component from the user interface principles. W-CBS-10 is a back-office screen for bank employees.

8.7.1 Sign in

Flow: Sign in -> enter the code -> Verify -> Accounts. Register -> registration screen of CRM (customer relationship management).



W-CBS-01 Sign in and one-time code

Element	What happens when it is used
E-mail, Password	Checked in F6 step 1.
Sign in	Starts F6 and shows the code boxes.
Code boxes	Six-digit SMS (short message service) code.
Verify	Completes F6 and opens Accounts (W-CBS-02).
Resend code	Available after 30 s; limit in F6, 6B.
Register	Opens the registration screen of CRM (chapter 7); CBS only stores the login credentials (F6).

8.7.2 Accounts

Flow: Accounts -> + Open -> Open new account -> Create -> Accounts. Accounts -> tap an account -> Account details.



W-CBS-02 Accounts

Element	What happens when it is used
+ Open	Opens W-CBS-03. Disabled with a hint when the customer is RESTRICTED.
Account card	Type, currency, IBAN (International Bank Account Number), balance and available balance (BR-02). A tap opens W-CBS-04.
Frozen	Shown for FROZEN accounts; Transfer is disabled on W-CBS-04.



W-CBS-03 Open account

Element	What happens when it is used
Account type, Currency	CURRENT or SAVINGS; EUR (euro), USD (US dollar) or GBP (British pound).
Error banner	Shows the rejection of F1 step 1 (alternative 1D).
Create	Creates the account (F1) and returns to Accounts with the new card.
Cancel	Closes the sheet without changes.

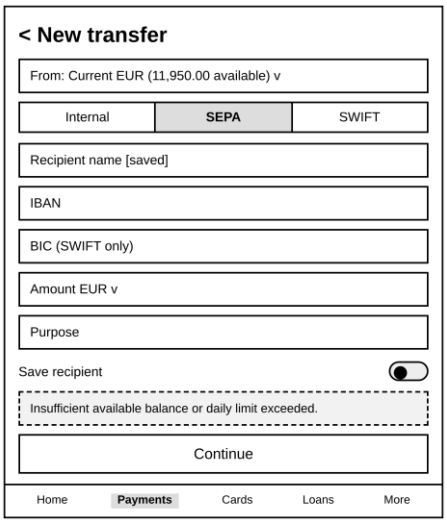


W-CBS-04 Account details

Element	What happens when it is used
Balance, Available	Refreshed on every open.
Transfer	Opens W-CBS-05 with this account preselected; hidden without CREATE_PAYMENTS.
Export	CSV (comma-separated values) or PDF (portable document format) of the filtered rows (up to 10,000).
Date range, Type and amount, Search	Filter the list on createdAt, Transaction.type and absolute amount, and search counterparty and purpose; they apply on input.
Transaction row	Date, type, signed amount, status; a tap shows counterparty, fee and reference.
Load more	Loads the next 50 rows.

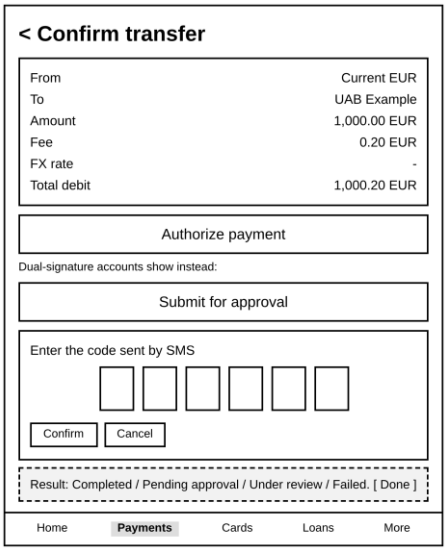
8.7.3 Transfer

Flow: Account details -> Transfer -> New transfer -> Continue -> Confirm transfer -> Authorize payment -> code -> Confirm -> Done -> Account details. On a dual-signature account: Submit for approval instead of Authorize payment. From New transfer: recipient icon -> Saved recipients -> choose a recipient.



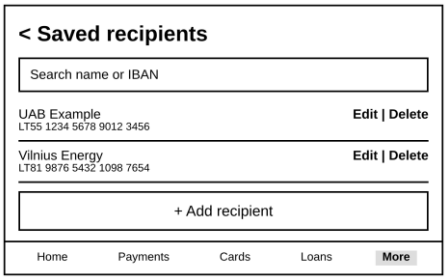
W-CBS-05 New transfer

Element	What happens when it is used
From	Accounts where the user has CREATE_PAYMENTS, with available balance.
Internal, SEPA, SWIFT	Payment type (SEPA = Single Euro Payments Area, SWIFT = Society for Worldwide Interbank Financial Telecommunication); BIC (Bank Identifier Code) is only for SWIFT and EUR only for SEPA.
Recipient name	The saved-recipient icon opens W-CBS-07 in selection mode; the choice fills name, IBAN and BIC.
IBAN, BIC	IBAN is validated on blur (BR-01) with an inline error; BIC is visible and required only for SWIFT.
Amount, Purpose	Amount above 0 with 2 decimals (another currency adds an FX (foreign exchange) line on W-CBS-06); purpose is required, up to 140 characters.
Save recipient	Creates a Beneficiary when the payment is submitted.
Error banner	Server errors such as INSUFFICIENT_FUNDS or DAILY_LIMIT_EXCEEDED.
Continue	Validates, creates the DRAFT payment with quote and opens W-CBS-06.



W-CBS-06 Confirm transfer

Element	What happens when it is used
Summary	Source, recipient, amount, fee, FX rate and total debit (PAY-06).
Authorize payment	Starts F2; above EUR 500 the code dialog opens. Not shown on dual-signature accounts.
Submit for approval	Shown instead of Authorize payment on dual-signature accounts; starts F3 without a code.
Code boxes, Confirm	Six-digit code, 5 minute timer, 3 attempts; Confirm checks it and continues F2 or F3.
Cancel	Closes the dialog; the payment stays unauthorized and its hold is released.
Result	Completed, Pending approval, Under review (ON_HOLD, neutral text) or Failed with the reason; Done opens W-CBS-04.

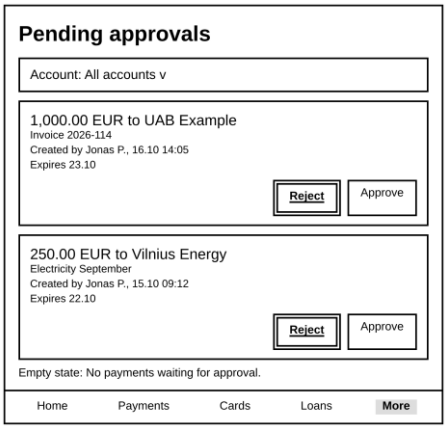


W-CBS-07 Saved recipients

Element	What happens when it is used
Search	Filters by name or IBAN.
Recipient row	In selection mode a tap returns the recipient to W-CBS-05; otherwise *Edit* and *Delete* (with confirmation).
+ Add recipient	Form with name, IBAN and BIC; saves a Beneficiary.

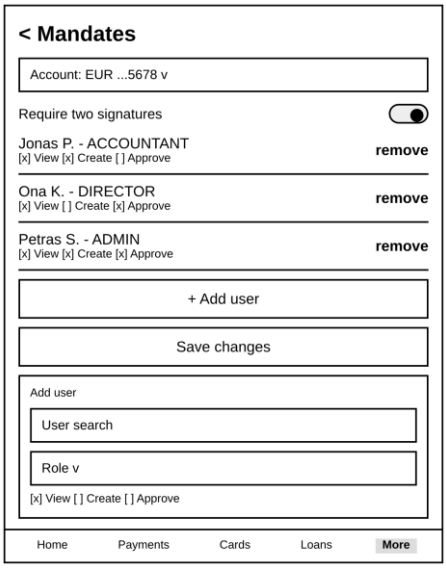
8.7.4 Company accounts

Flow: More -> Pending approvals -> Approve or Reject. More -> Mandates -> change users -> Save changes -> code.



W-CBS-08 Pending approvals

Element	What happens when it is used
Account	Filters pending payments by account.
Payment card	Amount, recipient, purpose, creator, creation time and expiry.
Reject	Opens a dialog with a mandatory reason; confirming rejects the payment (F3, 3A).
Approve	Disabled for own payments; opens the code dialog; success continues F3 step 4.
Empty state	“No payments waiting for approval.”

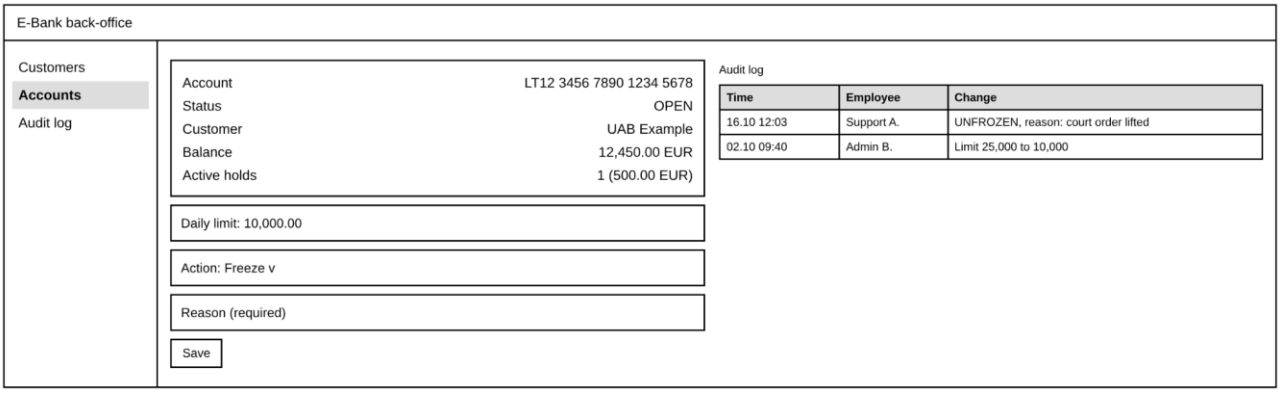


W-CBS-09 Manage mandates

Element	What happens when it is used
Account	Chooses the account whose mandates are edited.
Require two signatures	Sets dualSignatureRequired; needs at least one user with Approve.
Mandate row	User, role and permission boxes; *remove* marks the mandate for removal. At least one user keeps the role ADMIN (1C).
+ Add user	Dialog with user search, role and permissions; invalid users are rejected (1A).
Save changes	Asks for a code, then saves (F1 step 4).

8.7.5 Back-office

Flow: Back-office menu -> Accounts -> choose the account -> Action, limit and Reason -> Save.

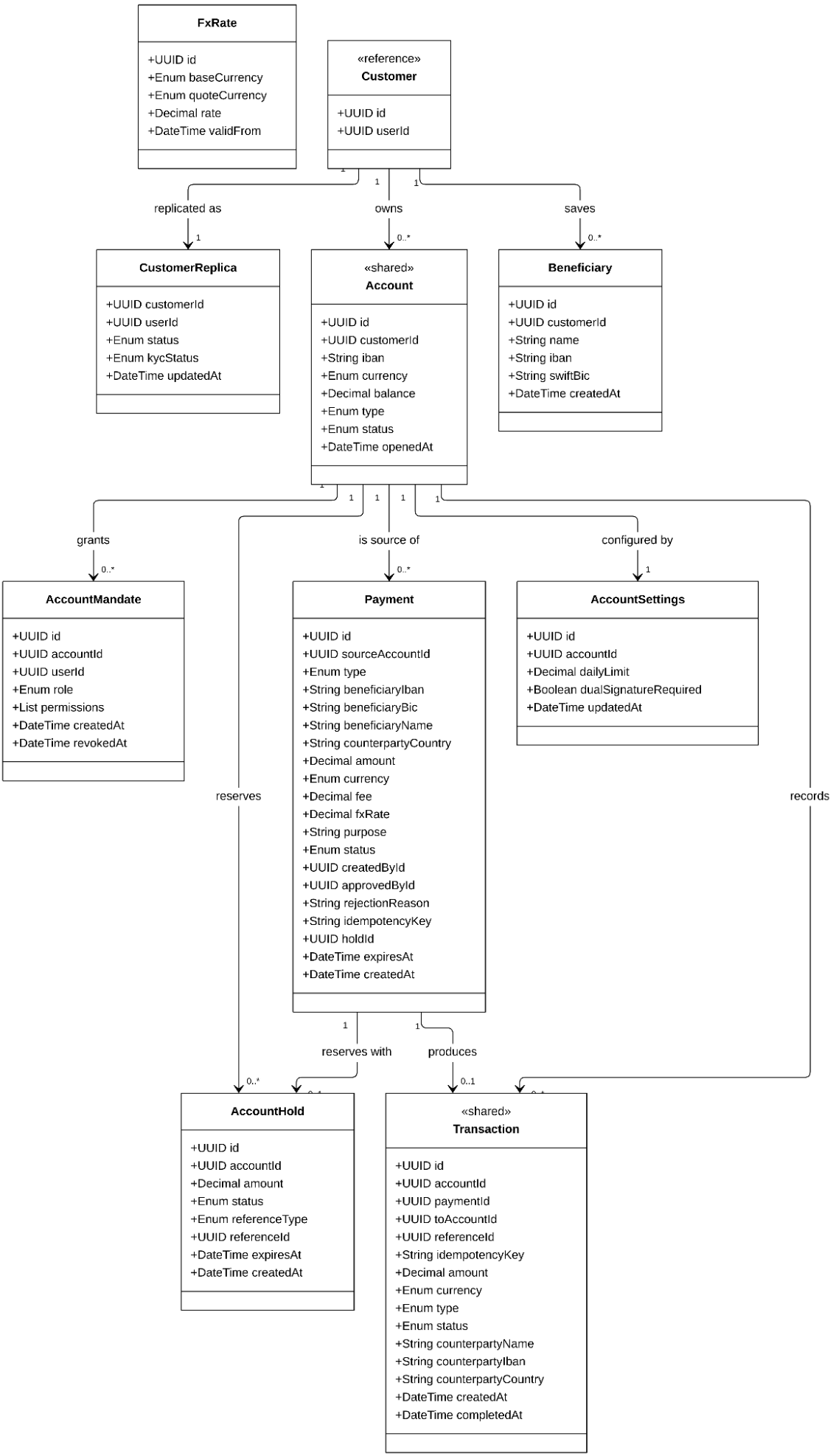


W-CBS-10 Account administration (employee)

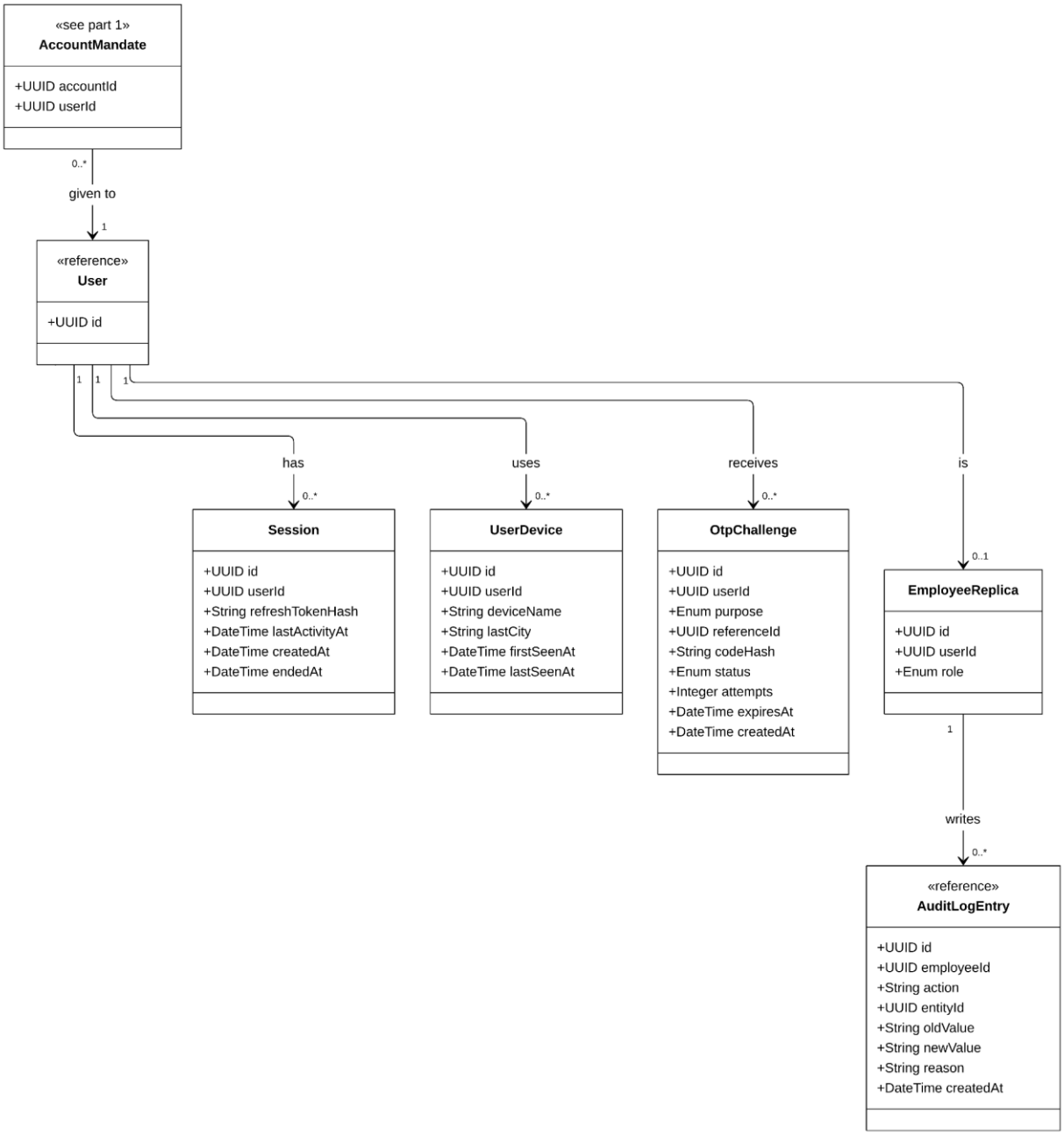
Element	What happens when it is used
Account box	IBAN, status, customer, balance, and number and total of ACTIVE holds.
Daily limit	Above EUR 100,000 only ADMIN may save.
Action	Freeze, Unfreeze or Close; invalid options are disabled.
Reason	Mandatory; written to the audit log.
Save	Runs F5; shows the rejection text on error.
Audit log	Last changes: time, employee, change.

8.8 Data model

Account and Transaction are shared entities; other modules reference them by id. CBS (core banking system) added the nullable fields toAccountId, paymentId and referenceId, the counterparty fields, completedAt, the type value CARD_REPAYMENT and the status ON_HOLD to Transaction, which AML (anti-money laundering), Accounting, Cards and Notification need. paymentId is empty for postings made by Cards and Loans, which have no Payment; their referenceId holds the card authorization or loan id. idempotencyKey (rule BR-12) is stored on Payment and on Transaction, so a repeated request can be recognised. Customer and User belong to other modules, and AuditLogEntry to the shared audit log (AML chapter, section 11.5). CBS owns the login data (Session, UserDevice, OtpChallenge; the password hash belongs to CRM), keeps EmployeeReplica, a read-only copy of the Employee list, and FxRate, the exchange rates (BR-06). The diagram is split into two parts.


FxRate
+UUID id
+Enum baseCurrency
+Enum quoteCurrency
+Decimal rate
+DateTime validFrom
«reference»
Customer
+UUID id
+UUID userId
replicated as
owns
saves
CustomerReplica
+UUID customerId
+UUID userId
+Enum status
+Enum kycStatus
+DateTime updatedAt
«shared»
Account
+UUID id
+UUID customerId
+String iban
+Enum currency
+Decimal balance
+Enum type
+Enum status
+DateTime openedAt
Beneficiary
+UUID id
+UUID customerId
+String name
+String iban
+String swiftBic
+DateTime createdAt
grants
is source of
configured by
reserves
records
AccountMandate
+UUID id
+UUID accountId
+UUID userId
+Enum role
+List permissions
+DateTime createdAt
+DateTime revokedAt
Payment
+UUID id
+UUID sourceAccountId
+Enum type
+String beneficiaryIban
+String beneficiaryBic
+String beneficiaryName
+String counterpartyCountry
+Decimal amount
+Enum currency
+Decimal fee
+Decimal fxRate
+String purpose
+Enum status
+UUID createdById
+UUID approvedById
+String rejectionReason
+String idempotencyKey
+UUID holdId
+DateTime expiresAt
+DateTime createdAt
AccountSettings
+UUID id
+UUID accountId
+Decimal dailyLimit
+Boolean dualSignatureRequired
+DateTime updatedAt
reserves with
produces
AccountHold
+UUID id
+UUID accountId
+Decimal amount
+Enum status
+Enum referenceType
+UUID referenceId
+DateTime expiresAt
+DateTime createdAt
«shared»
Transaction
+UUID id
+UUID accountId
+UUID paymentId
+UUID toAccountId
+UUID referenceId
+String idempotencyKey
+Decimal amount
+Enum currency
+Enum type
+Enum status
+String counterpartyName
+String counterpartyIban
+String counterpartyCountry
+DateTime createdAt
+DateTime completedAt

CBS and Payment class diagram, part 1: accounts and payments



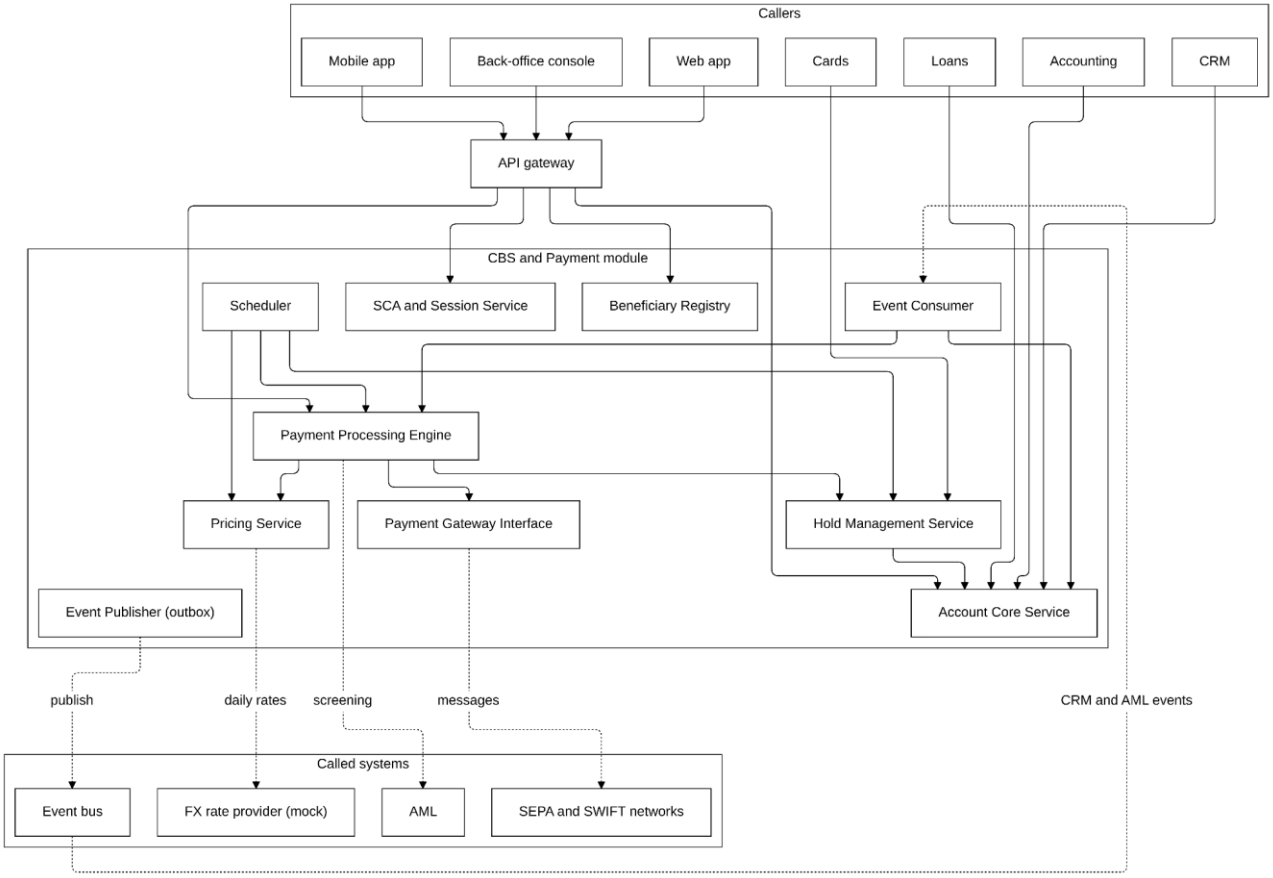
CBS and Payment class diagram, part 2: login, employees and audit

Enumeration	Values
Account.currency, Payment.currency	EUR, USD, GBP
Account.type / status	CURRENT, SAVINGS / OPEN, FROZEN, CLOSED
AccountHold.status / referenceType	ACTIVE, RELEASED, CONSUMED / PAYMENT, CARD_AUTHORIZATION
AccountMandate.role / permissions	ACCOUNTANT, DIRECTOR, ADMIN / VIEW, CREATE_PAYMENTS, APPROVE_PAYMENTS
Payment.type / status	INTERNAL, SEPA, SWIFT / DRAFT, PENDING_APPROVAL, APPROVED, REJECTED, EXPIRED, EXECUTED, FAILED
Transaction.type / status	INTERNAL, SEPA, SWIFT, CARD, CARD_REPAYMENT, LOAN / PENDING, COMPLETED, FAILED, REJECTED, ON_HOLD
OtpChallenge.purpose / status	LOGIN, PAYMENT, PAYMENT_APPROVAL, MANDATE_CHANGE, CARD_ACTIVATION, CARD_REPAYMENT / PENDING, VERIFIED, EXPIRED, FAILED
FxRate.baseCurrency, quoteCurrency	EUR, USD, GBP
EmployeeReplica.role	SUPPORT, COMPLIANCE, LOAN_OFFICER, ACCOUNTANT, ADMIN

Constraints: Account.iban unique and balance >= 0; Payment.idempotencyKey unique per (createdById, idempotencyKey) and Transaction.idempotencyKey unique when set; Transaction.paymentId refers to the Payment

that produced it; Transaction.completedAt is set when the status becomes COMPLETED; AccountSettings.accountId unique; one active mandate per (accountId, userId); Beneficiary unique per (customerId, iban); Payment.approvedById differs from createdById; Payment.beneficiaryBic required for SWIFT; AccountHold.expiresAt is set only for card holds; FxRate unique per (baseCurrency, quoteCurrency, validFrom); CustomerReplica.status uses the CRM (customer relationship management) CustomerStatus values and kycStatus the KYC (know your customer) status values.

8.9 Components



CBS and Payment components

For readability the diagram leaves out two kinds of arrows: every service writes its events through the Event Publisher (outbox), and the Payment Processing Engine also calls the SCA and Session Service and the Account Core Service. The SCA and Session Service also calls CRM (customer relationship management) for the password check.

Component	Responsibility
Account Core Service	Accounts, balances, postings, mandates, settings, daily limits, customer status replica and the transaction query used by Accounting.
Hold Management Service	Creates, consumes, releases and expires balance holds atomically so concurrent payments cannot overspend an account.
Payment Processing Engine	Payment lifecycle (draft, approval, screening, routing, settlement), dual-signature orchestration, AML (anti-money laundering) calls and retries.
Pricing Service	Calculates fee and FX (foreign exchange) rate from the latest FxRate with markup and returns the quote shown to the user.
Beneficiary Registry	Stores, validates and returns saved recipients.
Payment Gateway Interface	Formats outgoing SEPA (Single Euro Payments Area) and SWIFT (Society for Worldwide Interbank Financial Telecommunication) messages, receives incoming ones and maps confirmations and errors.
SCA (strong customer authentication) and Session Service	Login (the password check is done by CRM), one-time codes, scaToken validation, sessions and refresh tokens with the 15 minute timeout (BR-15), device tracking, employee role in the token.
Event Publisher (outbox)	Saves each event in the same database transaction as the change and publishes it at least once.

Event Consumer	Receives CRM (customer relationship management) events for the customer replica and the AML events TRANSACTION_RELEASED and TRANSACTION_REJECTED (acted on only for ON_HOLD transactions); ignores an eventId it has already processed.
Scheduler	Repeats AML calls every minute (BR-14), expires dual-signature payments after 7 days and card holds at expiresAt, and loads the exchange rates once a day from the mock rate provider.

8.10 API contract

Conventions are in the System architecture chapter (JSON (JavaScript Object Notation), /api/v1, error body, event envelope). Endpoints marked SCA (strong customer authentication) need the header X-SCA-Token. Money is a string with two decimals.

8.10.1 Customer endpoints

Method and path	Purpose	Request (main fields)	Success
POST /api/v1/auth/login	Start login (F6)	email, password, deviceName	200 challengeId
POST /api/v1/auth/verify	Finish login	challengeId, code	200 accessToken, expiresIn (300 s), refreshToken
POST /api/v1/auth/refresh	Renew access token (BR-15)	refreshToken	200 accessToken, expiresIn
POST /api/v1/auth/logout	End session	refreshToken	204
POST /api/v1/sca/challenges	Create code for an operation	purpose, referenceId	201 challengeId
POST /api/v1/sca/challenges/{id}/verify	Check code	code	200 scaToken
GET /api/v1/accounts	Own accounts	none	200 list with availableBalance
POST /api/v1/accounts	Open account (F1)	type, currency	201 Account
GET /api/v1/accounts/{id}	Account details	none	200 Account, settings
GET /api/v1/accounts/{id}/transactions	History	query: from, to, type, minAmount, maxAmount, q, page, size (max 50)	200 page of Transaction
GET /api/v1/accounts/{id}/statement	Export	query: from, to, format (csv, pdf)	200 file (max 10,000 rows)
GET /api/v1/accounts/{id}/mandates	List mandates	none	200 list
PUT /api/v1/accounts/{id}/mandates	Replace mandates (SCA)	dualSignatureRequired, list of userId, role, permissions	200 list
POST /api/v1/payments	Create draft with quote	sourceAccountId, type, beneficiaryName, beneficiaryIban, beneficiaryBic, amount, currency, purpose, saveBeneficiary	201 Payment with fee, fxRate, totalDebit
POST /api/v1/payments/{id}/submit	Confirm (SCA above EUR (euro) 500, not for dual signature)	none	200 Payment (PENDING_APPROVAL, EXECUTED or FAILED) with transactionStatus; ON_HOLD is shown as under review
GET /api/v1/payments	List payments	query: status, accountId, page, size	200 page
POST /api/v1/payments/{id}/approve	Approve (SCA)	none	200 Payment
POST /api/v1/payments/{id}/reject	Reject	reason	200 Payment (REJECTED)
GET/POST/PUT/DELETE /api/v1/beneficiaries	Saved recipients	name, iban, swiftBic	200, 201, 204

8.10.2 Employee and internal endpoints

Method and path	Purpose	Request / response
PATCH /api/v1/admin/accounts/{id}/status	Freeze, unfreeze, close (SUPPORT, ADMIN)	status, reason / 200 Account

PATCH /api/v1/admin/accounts/{id}/settings	Change daily limit	dailyLimit, reason / 200 settings
GET /internal/cbs/accounts/{id}	Status check for Cards, Loans	200 id, customerId, currency, status, balance, availableBalance
GET /internal/cbs/accounts/by-iban/{iban}	IBAN (International Bank Account Number) lookup for CRM (customer relationship management)	200 accountId, customerId, status
GET /internal/cbs/customers/{customerId}/accounts	Products for Customer 360	200 list
POST /internal/cbs/sca/validate	Check a one-time code token for Cards	scaToken, purpose, referenceId / 200 valid, or 401 SCA_CODE_INVALID
POST /internal/cbs/holds	Reserve card funds	accountId, amount, currency, referenceType, referenceId, expiresAt / 201 holdId, transactionId
POST /internal/cbs/holds/{id}/capture	Consume hold	finalAmount (optional, not above reserved) / 200 Transaction
POST /internal/cbs/holds/{id}/cancel	Release hold	none / 200
POST /internal/cbs/postings/credit	Loan payout	accountId, amount, currency, type LOAN, referenceId / 201 Transaction
POST /internal/cbs/postings/debit	Installment, card repayment	accountId, amount, currency, type LOAN or CARD_REPAYMENT, referenceId / 201 Transaction
GET /internal/cbs/transactions	Accounting batch	query: from, to (UTC (Coordinated Universal Time) instants, compared with completedAt), status / 200 list with fee and fxRate from the payment. Accounting sends the Europe/Vilnius day boundaries converted to UTC, so each transaction belongs to exactly one day.

8.10.3 Events

Events use the common envelope. userId is the user to notify (the customer's user unless stated).

Event	Published when	Payload fields
ACCOUNT_OPENED	Account created	userId, customerId, accountId, iban, currency, type
ACCOUNT_STATUS_CHANGED	Frozen, unfrozen, closed	userId, customerId, accountId, iban, oldStatus, newStatus
MANDATE_CHANGED	Mandate saved or removed	userId (affected user), accountId, iban, role, permissions
PAYMENT_PENDING_APPROVAL	Dual-signature payment created	userId (one event per approver), paymentId, accountId, amount, currency, beneficiaryName
PAYMENT_REJECTED	Rejected or expired	userId (creator), paymentId, amount, currency, beneficiaryName, reason (REJECTED or EXPIRED)
TRANSACTION_COMPLETED	INTERNAL, SEPA (Single Euro Payments Area), SWIFT (Society for Worldwide Interbank Financial Telecommunication), LOAN settled	userId, customerId, transactionId, accountId, type, amount, currency, amountEur, direction, counterpartyName, counterpartyCountry
TRANSACTION_FAILED	Code failure or gateway error	userId, paymentId, transactionId, amount, currency, beneficiaryName, reason (SCA_FAILED or GATEWAY_ERROR)
AML_SCREENING_FAILED	Fifth failed AML (anti-money laundering) call	transactionId, customerId
OTP_REQUESTED	Code created	userId, code, purpose
NEW_DEVICE_LOGIN	Login from a new device	userId, deviceName, city

CBS (core banking system) consumes CUSTOMER_ACTIVATED, CUSTOMER_STATUS_CHANGED (CRM), TRANSACTION_RELEASED and TRANSACTION_REJECTED (AML). The screening call to AML is defined in the AML chapter.

CBS acts on TRANSACTION_RELEASED and TRANSACTION_REJECTED only when the transaction is ON_HOLD. For a transaction in any other status, it ignores the event.

8.10.4 Error codes

Errors use the common body {"error": {"code": "...", "message": "..."}}.

Code	HTTP (Hypertext Transfer Protocol) status	Meaning
VALIDATION_ERROR, INVALID_IBAN	400	Field rules (BR-01, BR-10) failed.
SCA_REQUIRED, SCA_CODE_INVALID, SESSION_EXPIRED	401	Token missing, expired or used; wrong or expired code; refresh refused after 15 minutes of inactivity.
PERMISSION_DENIED, SAME_USER_APPROVAL, CUSTOMER_RESTRICTED, CUSTOMER_NOT_ACTIVE	403	Missing permission or role; approver is the creator; customer restricted; customer still onboarding.
ACCOUNT_NOT_FOUND	404	No such account or no access.
ACCOUNT_NOT_OPEN, INVALID_STATE, LAST_ADMIN, ACCOUNT_NOT_EMPTY	409	Account closed or frozen for debits; wrong status for the action; mandate rule; close rejected.
INSUFFICIENT_FUNDS, DAILY_LIMIT_EXCEEDED, MANDATE_USER_INVALID	422	Business rule failed.
GATEWAY_UNAVAILABLE	502	Payment network did not answer.
INTERNAL_ERROR, SERVICE_UNAVAILABLE	500, 503	Server error; callers treat it as a temporary failure (section 3.2).

9 Card management

9.1 Scope and ownership

Cards manages card orders, status, limits, and approval or decline of card purchases. The card network and physical card production are outside the system: the network is a mock interface (section 1.4).

Owned by this module	Only referenced (owned by other modules)
Cards, card numbers, limits, orders, credit balances of credit cards, purchase authorizations and repayment records	Customer status (CRM, customer relationship management), accounts, available funds, holds, transactions and one-time codes (CBS, core banking system), screening (AML, anti-money laundering), credit-card limit decisions (Loans), message delivery (Notification)

9.2 Feature table and business rules

ID	Feature	Actor	Priority
CARD-01	Order, activate and view a debit card linked to the customer's open EUR (euro) account	Customer	Must
CARD-02	Change the per-purchase and daily limits within the bank maximums	Customer	Should
CARD-03	Temporarily block or unblock a card, or permanently block a lost or stolen card and order a replacement	Customer	Must
CARD-04	Approve or decline a purchase using card and customer status, limits, the AML result, and either CBS available funds or available card credit	Card network, System	Must
CARD-05	Use a credit card within its approved limit and repay the used credit from a bank account	Customer	Should
CARD-06	Apply for a credit card, with the limit decided by Loans	Customer	Should

Won't, deliberately excluded from this release:

- Prepaid and virtual cards, and payments through mobile wallets.
- PIN (personal identification number) management and security codes.
- Physical card production and delivery.

- Cash withdrawals and purchases in currencies other than EUR.
- Interest and monthly bills for credit cards.

ID	Rule
CD-BR-01	Eligibility: only ACTIVE cards of an ACTIVE customer can be used. A debit card spends from the customer's own OPEN EUR account; a credit card spends from its approved EUR credit limit.
CD-BR-02	Card number: this version uses a mock number. Cards generates a unique 16-digit PAN (primary account number): the fixed test prefix 400000, nine random digits and a check digit calculated with the Luhn algorithm (the standard card-number check). The PAN is stored only encrypted with AES-256 (Advanced Encryption Standard; panEncrypted). A keyed hash (panHash) is used to find the card when the network sends the number. last4 is copied from the PAN. No API (application programming interface) or event returns the full number.
CD-BR-03	Expiry: expiresAt is the last day of the month, 36 months after the card is issued. The Scheduler sets the status EXPIRED the next day. A replacement gets a new number, last4 and expiry.
CD-BR-04	Spending limits: a new card has a per-purchase limit of 500 EUR and a daily limit of 1,500 EUR. The customer can change them up to the bank maximums of 5,000 EUR per purchase and 10,000 EUR per day. The per-purchase limit cannot be above the daily limit. The daily limit counts APPROVED and COMPLETED purchases of the card in the calendar day (Europe/Vilnius). The CBS daily limit of the account (BR-03) also applies to debit cards.
CD-BR-05	Credit: the requested limit is 100 to 10,000 EUR. Only Loans decides the limit. Available credit = creditLimit - outstandingAmount - pendingAmount.
CD-BR-06	Reservations: a debit hold (expiresAt on POST /internal/cbs/holds) and a credit reservation are valid for 7 days. If the card network reports neither completion nor cancellation by then, the reservation is released.
CD-BR-07	One-time codes: activation and repayment use the CBS code (BR-04: six digits, valid 5 minutes, 3 attempts) with the purposes CARD_ACTIVATION and CARD_REPAYMENT.
CD-BR-08	Repeated requests: a repeated networkRequestId returns the saved answer. Customer orders and repayments carry an Idempotency-Key.

9.3 Business flows

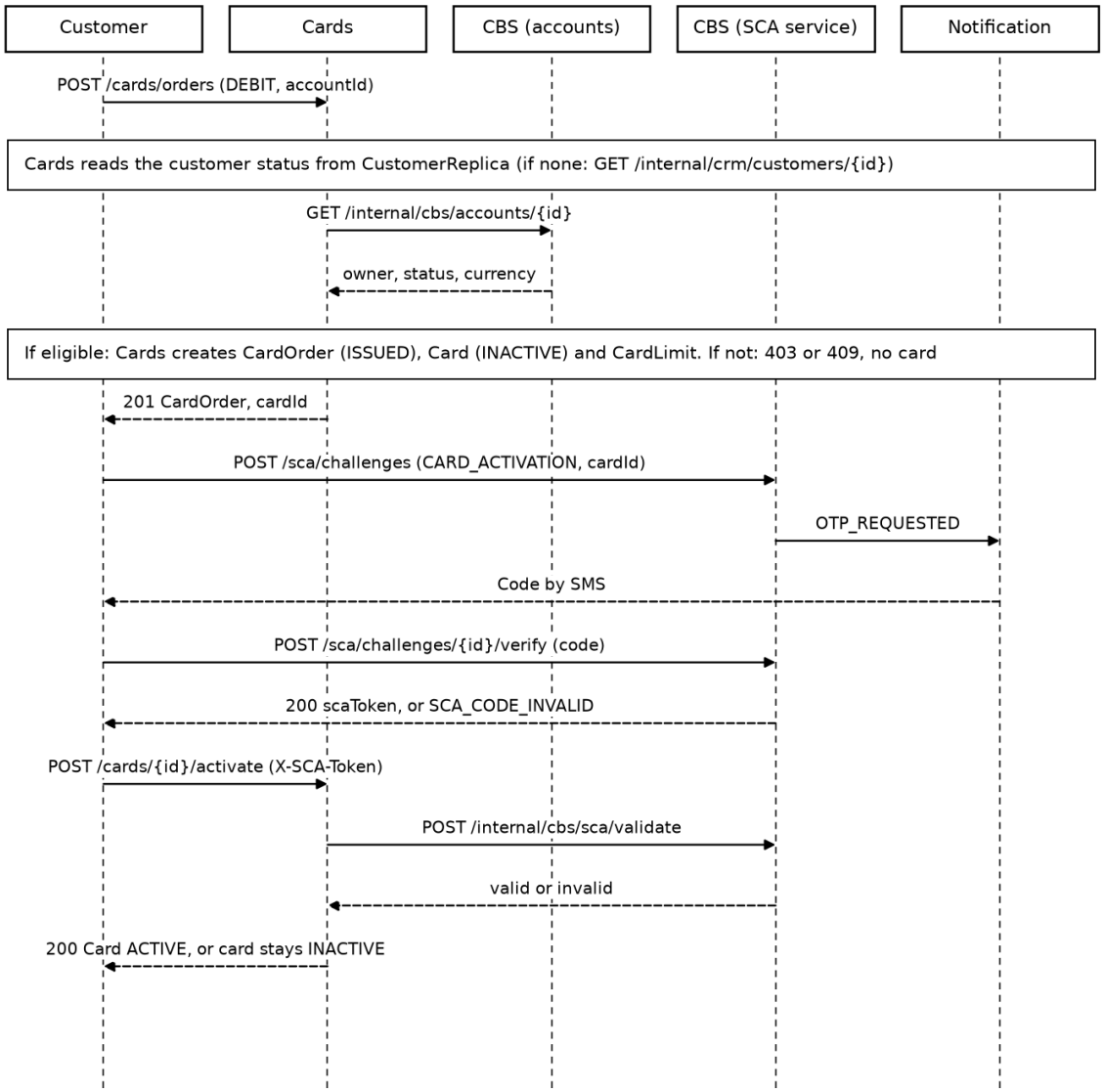
9.3.1 Order and activate

- Actors: Customer, Cards, CBS (accounts and one-time codes), Notification (delivers the code).
- Preconditions: the customer is signed in and ACTIVE. For a debit card, the customer has an open EUR account. For a credit card, the order follows section 9.3.5.

Main scenario:

1. The customer opens CD3, chooses Debit and an own open EUR account, and orders the card.
2. Cards reads the customer status from its replica and checks the account with GET /internal/cbs/accounts/{id}: owner, OPEN and EUR.
3. Cards creates the CardOrder (ISSUED), the Card (INACTIVE) with its number and expiry (CD-BR-02, CD-BR-03) and the CardLimit with the default limits (CD-BR-04).
4. The app asks CBS for a code with purpose CARD_ACTIVATION and the card id as the reference. CBS publishes OTP_REQUESTED, and Notification sends the code by SMS (short message service).
5. The customer enters the code. CBS checks it and returns a scaToken (SCA: strong customer authentication).
6. The app sends the activation with the token. Cards validates it with CBS and sets the card to ACTIVE.

- Alternative 1A, customer or account not eligible: Cards answers CUSTOMER_NOT_ELIGIBLE or ACCOUNT_NOT_OPEN and creates no card.
- Alternative 1B, wrong or expired code: the card stays INACTIVE. After 3 wrong codes the customer requests a new one.
- Alternative 1C, no customer replica yet: Cards reads the status with GET /internal/crm/customers/{id} (section 7.10.3).



Order and activate a debit card

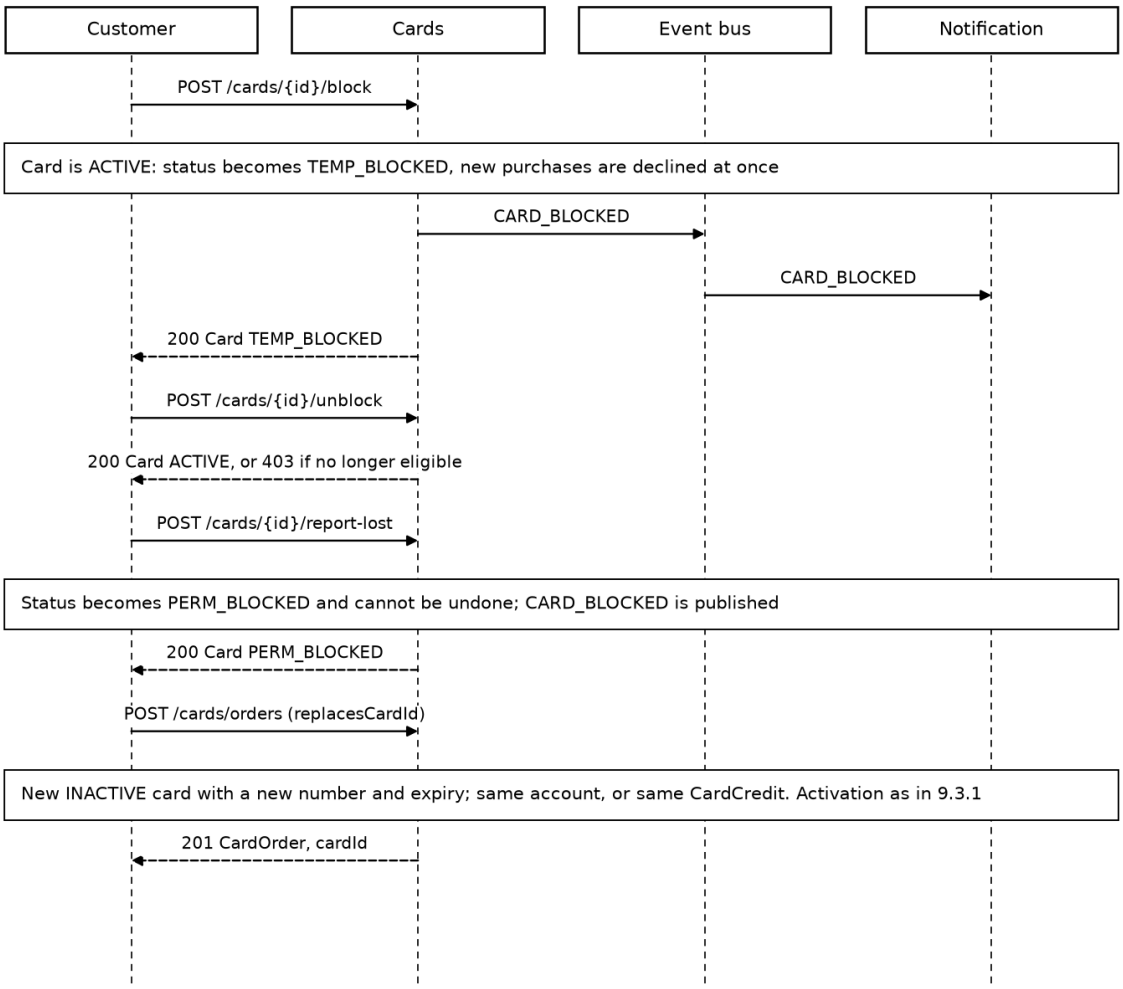
9.3.2 Block or replace

- Actors: Customer, Cards, Notification.
- Preconditions: the card belongs to the customer. A temporary block needs an ACTIVE card; unblock needs a TEMP_BLOCKED card.

Main scenario:

1. The customer chooses Block card on CD2. Cards sets the status to TEMP_BLOCKED. New authorizations are declined at once, and CARD_BLOCKED is published.
2. The customer chooses Unblock. If the customer is still eligible (and, for debit, the linked account is open), the card becomes ACTIVE.
3. The customer chooses Report lost or stolen and confirms. The status becomes PERM_BLOCKED, which cannot be undone, and CARD_BLOCKED is published.
4. The customer orders a replacement. Cards creates a new INACTIVE card with replacesCardId set, linked to the same account or the same CardCredit. A credit-card replacement keeps the same credit limit and outstanding balance. Activation continues as in section 9.3.1 from step 4.

- Alternative 2A, repeated block: the status does not change and no new event is published.
- Alternative 2B, not eligible at unblock: Cards answers CUSTOMER_NOT_ELIGIBLE or ACCOUNT_NOT_OPEN, and the card stays TEMP_BLOCKED.



Block, unblock, report lost and replace a card

9.3.3 Authorize a purchase

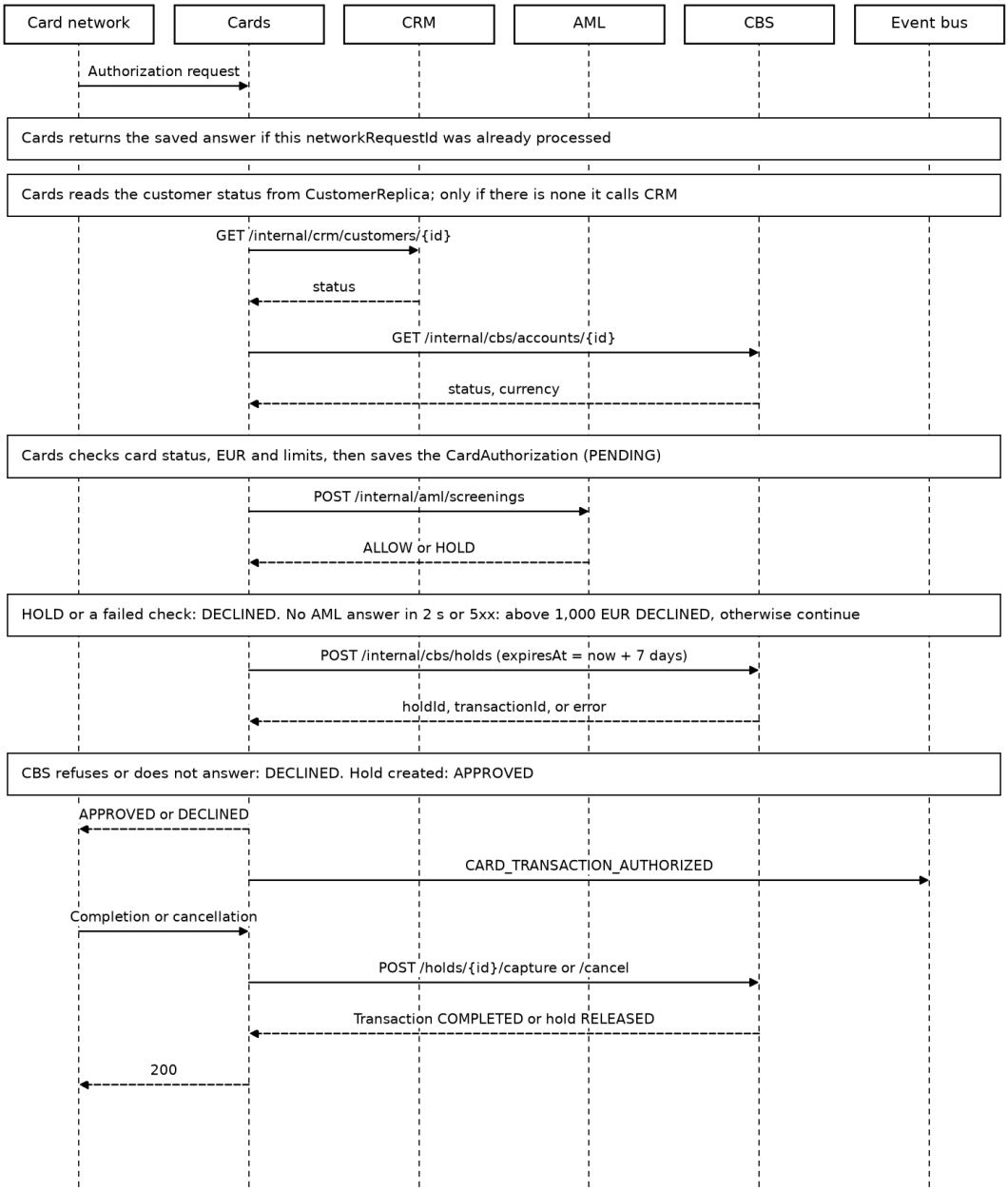
- Actors: Card network, Cards, AML, CBS (debit cards).
- Preconditions: the network sends a unique networkRequestId with the card number, amount, currency and merchant.

Main scenario:

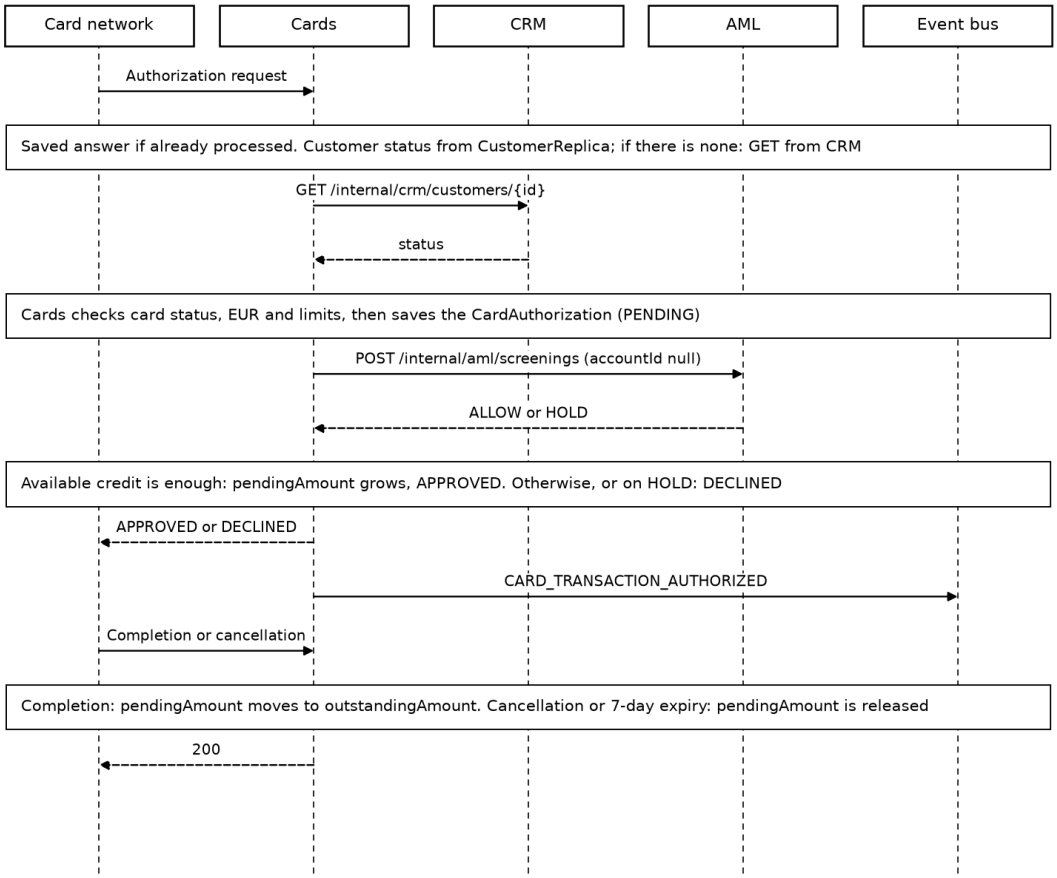
1. Cards returns the saved answer if this networkRequestId was already processed.
2. Cards finds the card and checks card and customer status, the EUR currency and the spending limits. For a debit card it also checks the account with GET /internal/cbs/accounts/{id}.
3. Cards first saves the CardAuthorization (PENDING), then calls AML as described in section 11.8.1, with source CARD, referenceId set to the CardAuthorization id, initiatedByUserId set to the card holder, and amountEur (purchases are in EUR, so it equals the amount). accountId is null for a credit card.
4. Cards reserves the money. For a debit card it calls POST /internal/cbs/holds with referenceType CARD_AUTHORIZATION and expiresAt 7 days from now, and stores holdId and transactionId. For a credit card it adds the amount to pendingAmount if the available credit is enough.
5. Cards saves APPROVED, answers the network and publishes CARD_TRANSACTION_AUTHORIZED.
6. The card network reports completion. For a debit card, Cards calls capture on the hold, and CBS sets the transaction to COMPLETED. For a credit card, the amount moves from pendingAmount to outstandingAmount. The status becomes COMPLETED.

- Alternative 3A, a check in step 2 fails: the attempt is DECLINED with a reason code, and AML is not called.
- Alternative 3B, AML returns HOLD: Cards declines this attempt; the card is not blocked. A later AML decision does not charge the customer.
- Alternative 3C, AML does not answer: if AML does not answer within 2 seconds or returns a 5xx error, Cards does not retry: above 1,000 EUR it declines the attempt, and at 1,000 EUR or less it approves it. AML then checks it later from CARD_TRANSACTION_AUTHORIZED.

- Alternative 3D, funds cannot be reserved: CBS answers INSUFFICIENT_FUNDS, DAILY_LIMIT_EXCEEDED or ACCOUNT_NOT_OPEN, or does not answer, or the available credit is too low. The attempt is DECLINED.
- Alternative 3E, cancellation: the network reports a cancellation. Cards calls cancel on the hold, or releases pendingAmount for a credit card. The status becomes CANCELLED.
- Alternative 3F, no report within 7 days: CBS releases the hold itself (section 8.5.4, 4B). The Scheduler sets the status EXPIRED and releases pendingAmount for a credit card. A later report gets INVALID_STATE.



Debit-card purchase: authorization and completion



Credit-card purchase: authorization and completion

9.3.4 Repay used credit

- Actors: Customer, Cards, CBS.
- Preconditions: the credit card has an outstanding amount, and the customer has an own open EUR account.

Main scenario:

1. The customer opens CD4, enters the amount and chooses the account. Cards checks that the amount is not above outstandingAmount and creates the CardRepayment (PENDING).
2. The customer confirms with a one-time code (purpose CARD_REPAYMENT, reference the repayment id) and the app sends the scaToken to Cards, which validates it with CBS.
3. Cards calls POST /internal/cbs/postings/debit with type CARD_REPAYMENT, and the repayment id as referenceId and Idempotency-Key.
4. Only after CBS confirms success, Cards lowers outstandingAmount by the same amount, so the available credit grows, and sets the CardRepayment to COMPLETED.

- Alternative 4A, amount above the outstanding balance: Cards answers AMOUNT_ABOVE_OUTSTANDING.
- Alternative 4B, CBS refuses: on INSUFFICIENT_FUNDS or ACCOUNT_NOT_OPEN the repayment is FAILED and the outstanding amount does not change.
- Alternative 4C, CBS does not answer: Cards retries with the same Idempotency-Key and backoff (section 3.2). If it still fails, the repayment is FAILED.

This system does not calculate interest or monthly bills.



Repay used credit

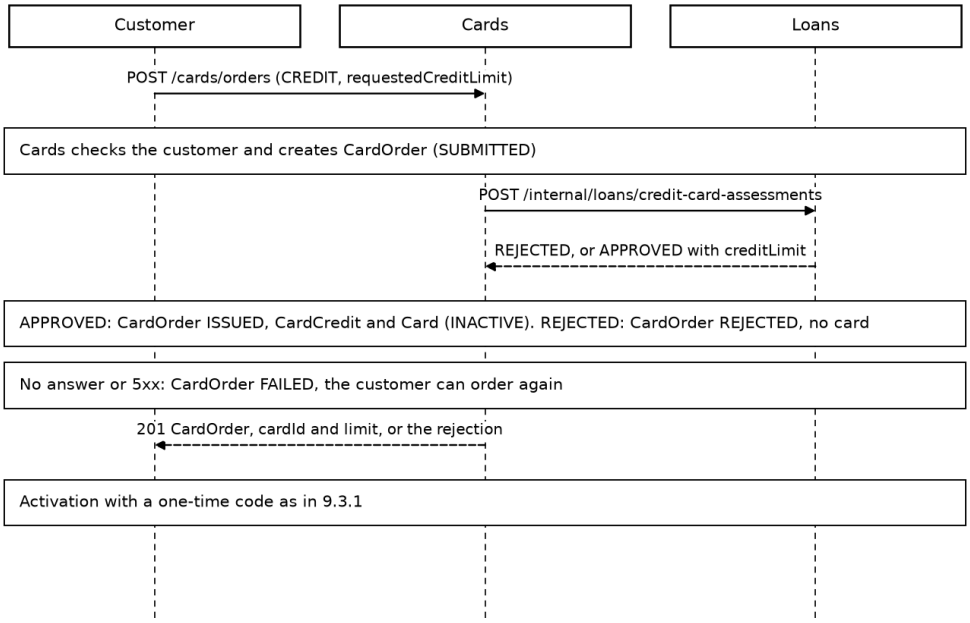
9.3.5 Credit-card approval

- Actors: Customer, Cards, Loans.
- Preconditions: the customer is signed in and ACTIVE.

Main scenario:

1. The customer opens CD3, chooses Credit, enters the requested limit and orders the card. Cards creates the CardOrder (SUBMITTED).
2. Cards calls POST /internal/loans/credit-card-assessments with customerId and requestedLimit (section 10.8.2).
3. Loans returns APPROVED with a creditLimit in EUR. Cards stores it in approvedCreditLimit, creates the CardCredit and the INACTIVE Card, and sets the order to ISSUED. Cards does not calculate or change the approved limit.
4. Activation continues as in section 9.3.1 from step 4.

- Alternative 5A, rejection: Loans returns REJECTED. The order becomes REJECTED and no card is created.
- Alternative 5B, Loans does not answer: on a timeout or a 5xx error the order becomes FAILED, and the customer can order again.

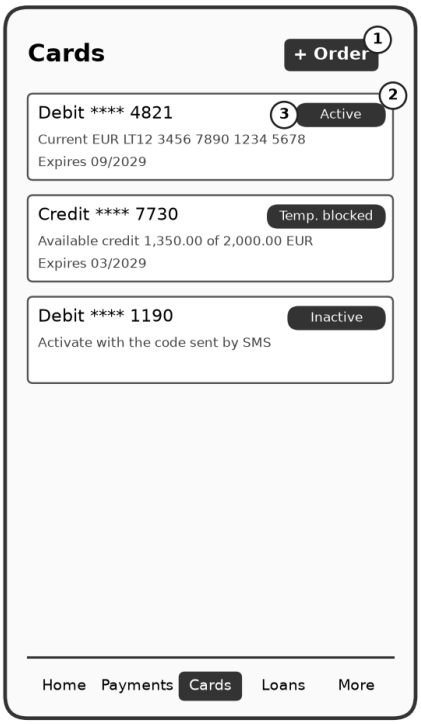


Credit-card approval

9.4 Wireframes

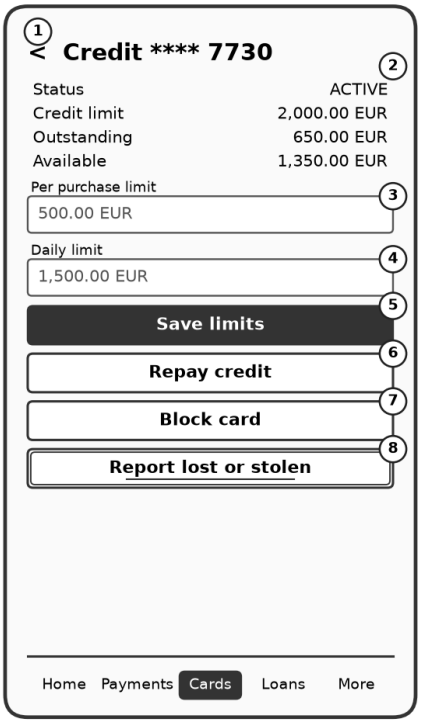
The screens are part of the customer mobile app (bottom navigation item Cards). Transitions: Cards opens CD1; a card opens CD2, and an inactive card opens the activation part of CD3; + Order opens CD3; Activate and Order card return to CD1; Repay credit opens CD4, and Confirm returns to CD2.

CD1 My cards



Ref	Element	What happens when it is used
CD1.1	+ Order	Opens CD3.
CD1.2	Card row	Shows type, last 4 digits, account or available credit, expiry and a status label. Opens CD2.
CD1.3	Status label	Active, Inactive, Temp. blocked, Perm. blocked, Expired or Cancelled. An inactive card shows how to activate it.

CD2 Card details



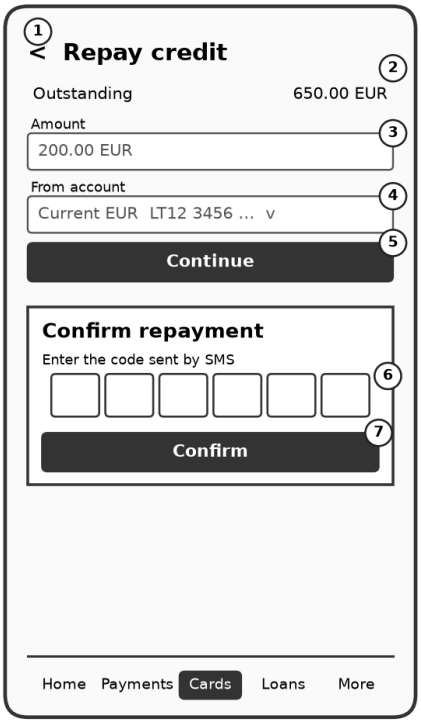
Ref	Element	What happens when it is used
CD2.1	Back	Returns to CD1.
CD2.2	Card summary	Read-only status. Credit limit, outstanding and available credit are shown for credit cards only.
CD2.3	Per purchase limit	Up to 5,000 EUR (CD-BR-04). A higher value shows an error under the field.
CD2.4	Daily limit	Up to 10,000 EUR, and not below the per-purchase limit. An invalid value shows an error under the field.
CD2.5	Save limits	Saves both limits and stays on CD2.
CD2.6	Repay credit	Credit cards only. Opens CD4.
CD2.7	Block card / Unblock	A temporary block takes effect immediately and publishes CARD_BLOCKED; the button changes to Unblock.
CD2.8	Report lost or stolen	Confirmation dialog, then a permanent block; a replacement can be ordered.

CD3 Order card and activate



Ref	Element	What happens when it is used
CD3.1	Back	Returns to CD1.
CD3.2	Debit / Credit tabs	Choose the card type and show the fields of that type.
CD3.3	Linked account	Debit only: the customer’s own open EUR accounts.
CD3.4	Requested limit	Credit only: 100 to 10,000 EUR (CD-BR-05). An invalid value shows an error under the field.
CD3.5	Credit notice	Credit only: explains that the bank assesses the application.
CD3.6	Order card	Sends the order. An issued card shows the activation part; a rejection shows a message.
CD3.7	Code boxes	The six-digit code sent by SMS (one-time code dialog, section 5.2).
CD3.8	Activate	A correct code activates the card and returns to CD1; a wrong or expired code leaves it inactive.

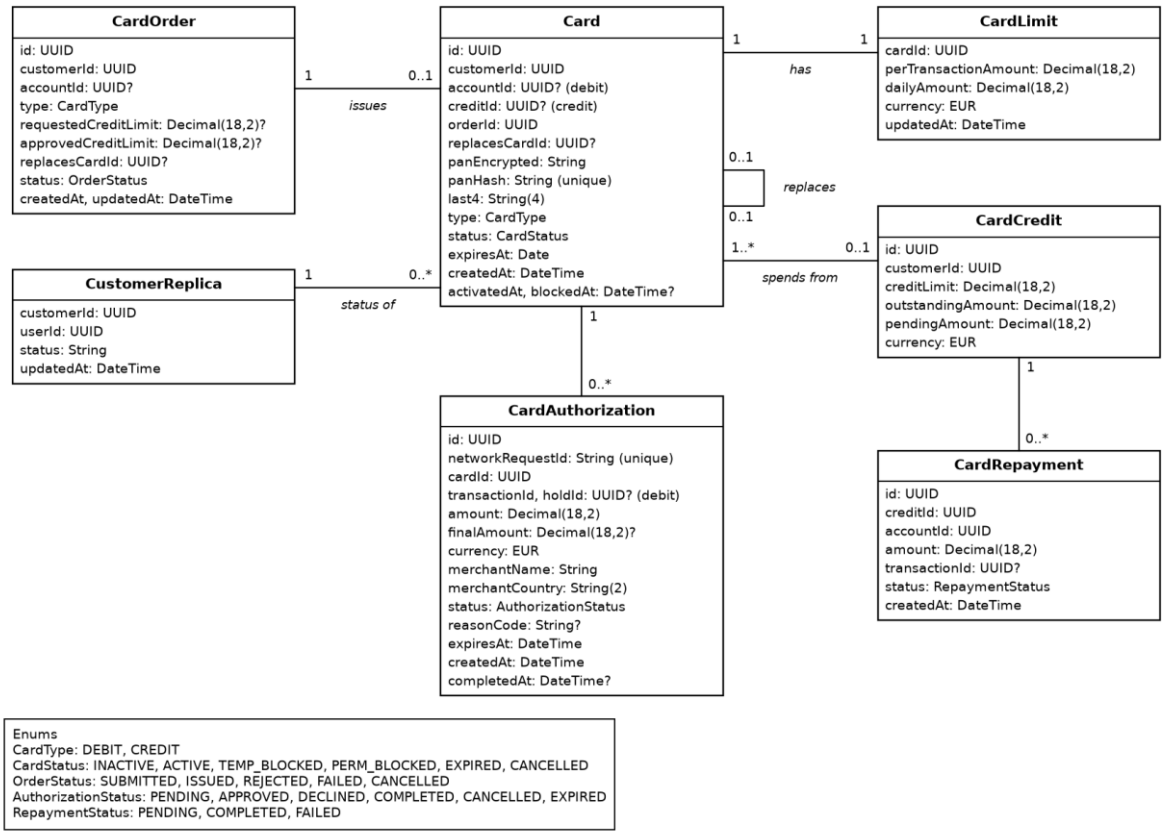
CD4 Repay credit



Ref	Element	What happens when it is used
CD4.1	Back	Returns to CD2 without a repayment.
CD4.2	Outstanding	Read-only amount that can be repaid.
CD4.3	Amount	Not above the outstanding amount. A higher value shows an error under the field.
CD4.4	From account	The customer’s own open EUR accounts.
CD4.5	Continue	Creates the repayment and sends the code by SMS.
CD4.6	Code boxes	The six-digit code (one-time code dialog, section 5.2).
CD4.7	Confirm	Repays the amount and returns to CD2 with the updated amounts. CBS errors appear in a banner above the button.

9.5 Data model

These are Cards records. IDs for customers, accounts, and CBS transactions are references, not shared database tables. UUID means universally unique identifier. Decimal(18,2) means up to 18 digits in total, with two after the decimal point.



Card data model

Entity	Fields
Card	id: UUID, customerId: UUID, accountId: UUID? (debit), creditId: UUID? (credit), orderId: UUID, replacesCardId: UUID?, panEncrypted: String, panHash: String (unique), last4: String(4), type: CardType, status: CardStatus, expiresAt: Date, createdAt: DateTime, activatedAt: DateTime?, blockedAt: DateTime?
CardLimit	cardId: UUID, perTransactionAmount: Decimal(18,2), dailyAmount: Decimal(18,2), currency: EUR, updatedAt: DateTime
CardOrder	id: UUID, customerId: UUID, accountId: UUID?, type: CardType, requestedCreditLimit: Decimal(18,2)?, approvedCreditLimit: Decimal(18,2)?, replacesCardId: UUID?, status: OrderStatus, createdAt: DateTime, updatedAt: DateTime
CardCredit (credit only)	id: UUID, customerId: UUID, creditLimit: Decimal(18,2), outstandingAmount: Decimal(18,2), pendingAmount: Decimal(18,2), currency: EUR
CardAuthorization	id: UUID, networkRequestId: String (unique), cardId: UUID, transactionId: UUID? (debit), holdId: UUID? (debit), amount: Decimal(18,2), finalAmount: Decimal(18,2)?, currency: EUR, merchantName: String, merchantCountry: String(2), status: AuthorizationStatus, reasonCode: String?, expiresAt: DateTime, createdAt: DateTime, completedAt: DateTime?
CardRepayment	id: UUID, creditId: UUID, accountId: UUID, amount: Decimal(18,2), transactionId: UUID?, status: RepaymentStatus, createdAt: DateTime
CustomerReplica	customerId: UUID, userId: UUID, status: String, updatedAt: DateTime. A copy fed by CRM events.

Relationships: one customer may have many cards; one account may be linked to many debit cards. Each card has one spending-limit record and many authorization records. A credit card refers to one CardCredit record; a replacement refers to that same record. An order issues at most one card. Lists and events expose only last4, never a full card number.

9.6 Interfaces with other modules

Interaction	Information exchanged
Customer ↔ Cards	Card order, activation, limits, block/unblock, replacement and repayment; Cards returns the updated card or order status.

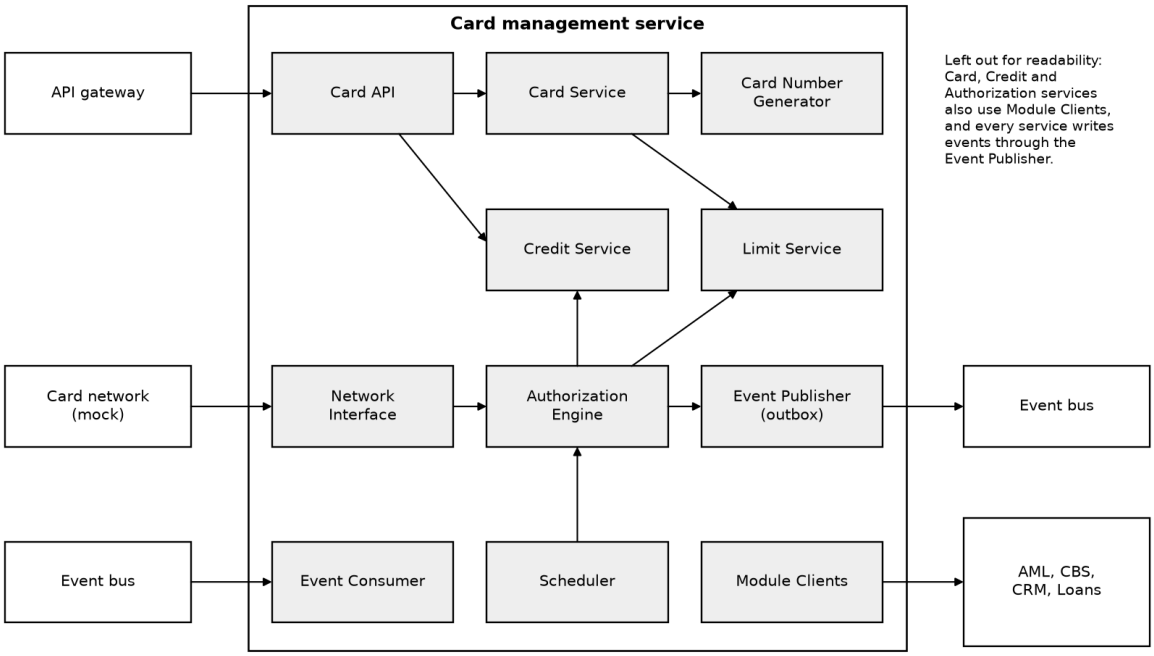
Interaction	Information exchanged
Card network ↔ Cards	Purchase request with a unique request ID, card, merchant and amount; Cards returns APPROVED or DECLINED. Repeating the same request ID returns the same result. The network also reports completion and cancellation.
Cards ↔ CRM	CRM events feed the customer replica. Cards calls GET /internal/crm/customers/{id} only when it has no replica. CRM reads the customer's cards for Customer 360.
Cards → Loans	Cards sends customerId and requestedLimit; Loans returns REJECTED, or APPROVED with the limit in EUR.
Cards → AML	Cards calls POST /internal/aml/screenings before reserving funds; AML returns ALLOW or HOLD.
Cards → CBS	CBS checks debit accounts, reserves debit funds (hold), captures or cancels it, records repayments and validates one-time code tokens. Cards stores the credit limit, outstanding credit and pending card purchases.
Cards → Notification and AML	CARD_TRANSACTION_AUTHORIZED contains userId, customerId, cardAuthorizationId, cardId, transactionId for debit cards, amount, currency, amountEur, merchantName, merchantCountry and cardLast4. CARD_BLOCKED contains userId, customerId, cardId and cardLast4, and goes to Notification only.
Accounting → Cards	Accounting reads the completed credit-card purchases of the day in the nightly batch (section 9.9.3).

9.7 Non-functional requirements

The baseline is in chapter 4. These targets are specific to Cards.

ID	Area	Requirement
CD-NFR-1	Performance	Cards answers an authorization in under 1 s (p95 (the value that 95% of measurements do not exceed), the value 95% of measurements do not exceed) and never later than 3 s.
CD-NFR-2	Reliability	A repeated request never reserves money twice. An AML or CBS failure never approves a purchase above 1,000 EUR.
CD-NFR-3	Security	The card number is encrypted with AES-256 (Advanced Encryption Standard), with keys in the key management service. Screens, logs and events show only the last 4 digits.
CD-NFR-4	Status changes	A saved block stops the next authorization. A customer status change reaches Cards within 5 s (p95, goal G3).
CD-NFR-5	Audit	Activation, block, unblock, limit changes and repayments are written to the audit log (NFR-1).

9.8 Components



Card management components

Component	Responsibility
Card API	Customer endpoints: checks the token, the owner of the card and the request fields.
Network Interface	Receives authorization, completion and cancellation messages from the card network and returns saved answers for repeated requests.
Card Service	Orders, activation, block, unblock, replacement and expiry of cards.
Card Number Generator	Creates the card number, last4 and expiry, and encrypts the number (CD-BR-02, CD-BR-03).
Authorization Engine	Runs the checks, the AML call and the reservation of a purchase, and handles completion, cancellation and expiry.
Limit Service	Stores limits, checks them against the bank maximums and counts daily spending.
Credit Service	Credit-card orders, available credit, pending and outstanding amounts, and repayments.
Module Clients	Calls to AML, CBS, CRM and Loans with timeouts and retries (section 3.2).
Event Publisher (outbox)	Saves each event in the same database transaction as the change and publishes it at least once.
Event Consumer	Receives CRM events for the customer replica; ignores an event it has already processed.
Scheduler	Expires reservations after 7 days and sets expired cards to EXPIRED.

9.9 API contract

Conventions are in section 3.2. Endpoints marked SCA need the header X-SCA-Token. Endpoints marked Idempotent need the header Idempotency-Key.

9.9.1 Customer endpoints

Method and path	Purpose	Request (main fields)	Success
GET /api/v1/cards	Own cards	none	200 list with id, type, last4, status, expiresAt, account or available credit
GET /api/v1/cards/{id}	Card details	none	200 Card, limits, credit amounts
POST /api/v1/cards/orders	Order a card or replacement (Idempotent)	type, accountId (debit), requestedCreditLimit (credit), replacesCardId (optional)	201 CardOrder, cardId

Method and path	Purpose	Request (main fields)	Success
POST /api/v1/cards/{id}/activate	Activate (SCA)	none	200 Card
PUT /api/v1/cards/{id}/limits	Change limits	perTransactionAmount, dailyAmount	200 CardLimit
POST /api/v1/cards/{id}/block	Temporary block	none	200 Card
POST /api/v1/cards/{id}/unblock	Remove the temporary block	none	200 Card
POST /api/v1/cards/{id}/report-lost	Permanent block	none	200 Card
POST /api/v1/cards/{id}/repayments	Start a repayment (Idempotent)	accountId, amount	201 CardRepayment
POST /api/v1/cards/{id}/repayments/{repaymentId}/confirm	Confirm a repayment (SCA)	none	200 CardRepayment, updated credit amounts

9.9.2 Card network interface (mock)

Method and path	Purpose	Request (main fields)	Success
POST /network/v1/authorizations	Purchase request	networkRequestId, pan, amount, currency, merchantName, merchantCountry	200 authorizationId, result (APPROVED or DECLINED), reasonCode
POST /network/v1/authorizations/{id}/completion	Purchase completed	finalAmount (optional, not above the reserved amount)	200
POST /network/v1/authorizations/{id}/cancellation	Purchase cancelled	none	200

Reason codes of a declined purchase: CARD_NOT_ACTIVE, CUSTOMER_NOT_ACTIVE, ACCOUNT_NOT_OPEN, CURRENCY_NOT_SUPPORTED, LIMIT_EXCEEDED, AML_HOLD, AML_UNAVAILABLE, INSUFFICIENT_FUNDS, DAILY_LIMIT_EXCEEDED, INSUFFICIENT_CREDIT, CBS_UNAVAILABLE.

9.9.3 Internal endpoints

Method and path	Used by	Request / response
GET /internal/cards/customers/{customerId}/cards	CRM, Customer 360	200 list with id, type, last4, status, expiresAt
GET /internal/cards/credit-purchases	Accounting, nightly batch	query: from, to (UTC, Coordinated Universal Time, compared with completedAt) / 200 list with authorizationId, cardId, customerId, finalAmount, currency, completedAt

Calls Cards makes to other modules: POST /internal/aml/screenings (11.8.1); GET /internal/cbs/accounts/{id}, POST /internal/cbs/holds, POST /internal/cbs/holds/{id}/capture, POST /internal/cbs/holds/{id}/cancel, POST /internal/cbs/postings/debit and POST /internal/cbs/sca/validate (8.10.2); GET /internal/crm/customers/{customerId} (7.10.3); POST /internal/loans/credit-card-assessments (10.8.2).

9.9.4 Events

Event published	Published when	Payload fields
CARD_TRANSACTION_AUTHORIZED	A purchase is APPROVED	userId, customerId, cardAuthorizationId, cardId, transactionId (debit cards), amount, currency, amountEur, merchantName, merchantCountry, cardLast4
CARD_BLOCKED	A temporary or permanent block is saved	userId, customerId, cardId, cardLast4

Event consumed	From	Used for
CUSTOMER_ACTIVATED	CRM	Creates the CustomerReplica.
CUSTOMER_STATUS_CHANGED	CRM	Updates the CustomerReplica. At CLOSED, all cards of the customer become CANCELLED.

9.9.5 Error codes

Errors use the common body {"error": {"code": "...", "message": "..."}}.

Code	HTTP (Hypertext Transfer Protocol) status	Meaning
VALIDATION_ERROR	400	A required field is missing or invalid.
SCA_REQUIRED, SCA_CODE_INVALID	401	Token missing, expired or used.
PERMISSION_DENIED, CUSTOMER_NOT_ELIGIBLE	403	Not the owner of the card; customer not ACTIVE.
CARD_NOT_FOUND	404	No such card or no access.
ACCOUNT_NOT_OPEN, INVALID_STATE	409	Account not open or not EUR; wrong card or authorization status for the action.
LIMIT_ABOVE_MAXIMUM, CREDIT_LIMIT_OUT_OF_RANGE, AMOUNT_ABOVE_OUTSTANDING, INSUFFICIENT_FUNDS	422	Business rule failed (CD-BR-04, CD-BR-05, section 9.3.4).
INTERNAL_ERROR, SERVICE_UNAVAILABLE	500, 503	Server error or a module is unavailable; the caller treats it as a temporary failure (section 3.2).

10 Loans and Accounting

10.1 Scope and ownership

Loans handles loan applications, credit scoring and decisions, offers and e-signing, payout, installment schedules, collection and late-payment handling. It also assesses credit-card applications for Cards. Accounting owns the chart of accounts, posting rules, the general ledger, the nightly posting and reconciliation, discrepancy handling, financial reports and customer statements.

Owned by this module	Only referenced (owned by other modules)
Loan applications, credit scores and decisions, offers, loans, installment schedules; ledger accounts, posting rules, LedgerEntry, discrepancies, financial reports, customer statements	Customer identity and status (CRM), identity verification (KYC), accounts, balances and transactions (CBS), screening (AML (anti-money laundering)), credit cards (Cards), message delivery (Notification)

Loan officer and Accountant in this chapter are bank employee roles (Employee.role).

10.2 Feature tables

Loans

ID	Feature	Actor	Priority
LOAN-01	Loan application	Customer	Must
LOAN-02	Credit scoring and decision	System, Loan officer	Must
LOAN-03	Offer, e-signing and payout	Customer, System	Must
LOAN-04	Installment schedule and collection	Customer, System	Should
LOAN-05	Late payment handling	System	Should
LOAN-06	Loan calculator	Customer	Could

Accounting

ID	Feature	Actor	Priority
ACC-01	End-of-day ledger posting	System	Must
ACC-02	General ledger view	Accountant	Must
ACC-03	Reconciliation and discrepancy handling	System, Accountant	Must
ACC-04	Financial reports	Accountant	Should

ACC-05	Chart of accounts management	Accountant	Should
ACC-06	Customer monthly statement	Customer	Could

10.3 Interfaces with other modules

Module	What is exchanged
CRM / KYC	Loans consumes CUSTOMER_STATUS_CHANGED and keeps the customer status. An application needs an ACTIVE customer with KYC (know your customer) APPROVED. CRM (customer relationship management) reads a customer's loans with GET /internal/loans/customers/{customerId}/loans for Customer 360.
CBS	Loans calls POST /internal/cbs/postings/credit for the payout and POST /internal/cbs/postings/debit for each installment, and GET /internal/cbs/accounts/{id} to check the account. Accounting calls GET /internal/cbs/transactions in the nightly batch.
Cards	Cards sends a credit-card application; Loans returns REJECTED or APPROVED with a creditLimit in EUR. Cards does not change the limit. Accounting reads the completed credit-card purchases with GET /internal/cards/credit-purchases in the nightly batch.
Notification	Loans publishes LOAN_INSTALLMENT_DUE and LOAN_INSTALLMENT_MISSED. Accounting publishes RECONCILIATION_DISCREPANCY, which is used in the Accounting back-office and the audit log, so no message is sent.
Users	Customer (applications, offers, schedule, statements), Loan officer (review), Accountant (ledger, reconciliation, reports, chart of accounts).

10.4 Business flows

10.4.1 Loan application

A customer with approved KYC submits a loan application. The system validates it and calculates a credit score. [1] A low score (below 400) is rejected automatically, and a high score (above 700) is approved automatically. A score in between goes to a loan officer, who approves it or rejects it with a reason. Once approved, the customer reviews the offer and either signs or declines it. After signing, the system asks CBS (core banking system) to pay out the money, and the installment schedule is created.

Credit score calculation

The score is a number from 300 to 850 calculated from the application data and the customer's existing loans:

1. New payment estimate = amount / termMonths (interest is ignored in the score).

2. Existing payments = the sum of the monthly installments of the customer's other loans with status DISBURSED (0 if the customer has none). Loans reads them from its own data.

3. Payment-to-income ratio = (new payment estimate + existing payments) / monthly income.

4. Score = 800 − round(1000 × ratio), limited to the range 300–850.

Complete formula, in one line:

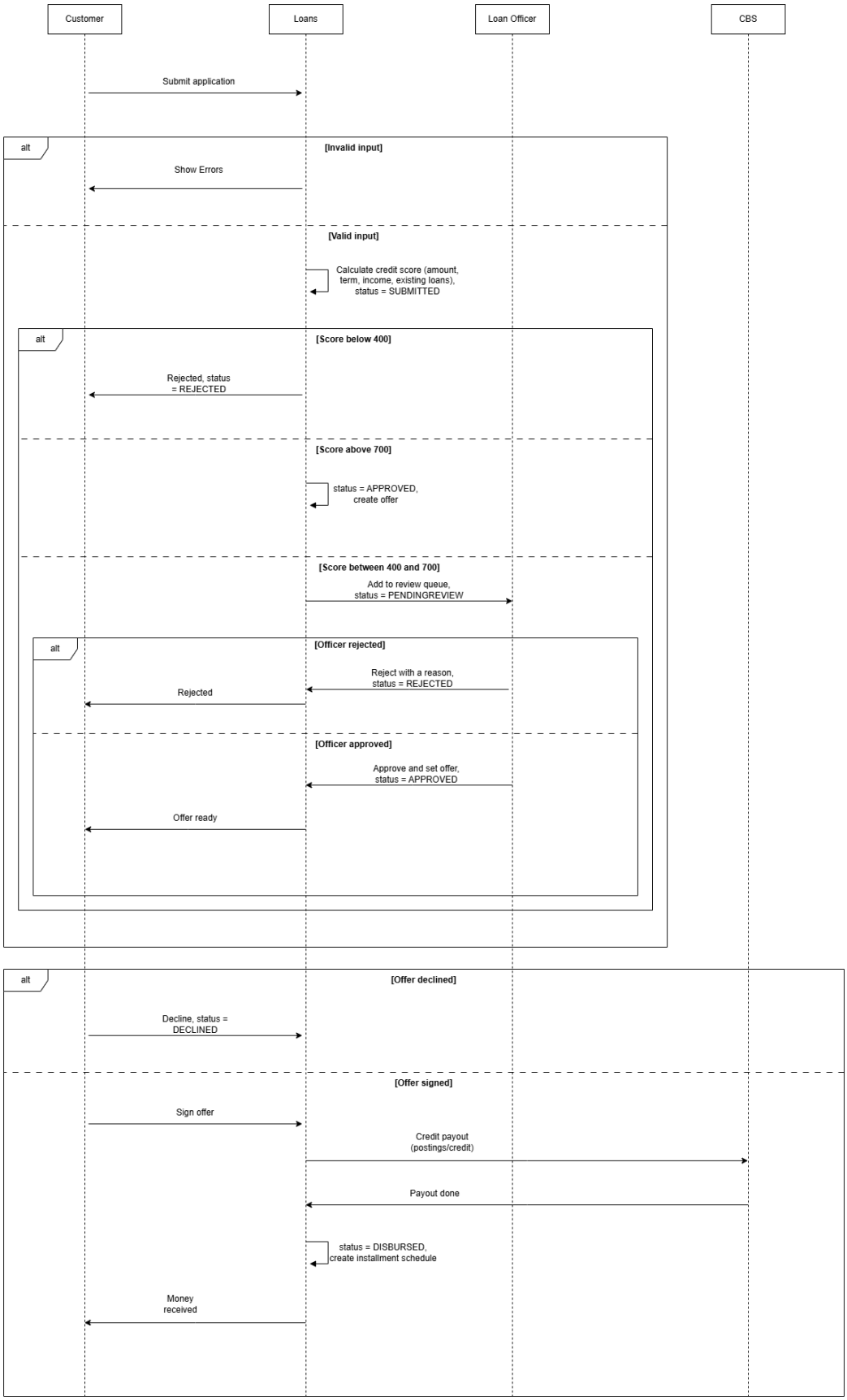
score = 800 − round(1000 × ((amount / termMonths + existingPayments) / monthlyIncome)), and the result is never below 300 or above 850.

Where: amount and termMonths come from the application, monthlyIncome is the income stated in the application, and existingPayments is the sum of the monthly installments of the customer's other DISBURSED loans (0 if none).

Example: monthlyIncome 3,000, amount 6,000, termMonths 24, existingPayments 300. New payment = 6,000 / 24 = 250. Ratio = (250 + 300) / 3,000 = 0.183. Score = 800 − round(1000 × 0.183) = 800 − 183 = 617. This is between 400 and 700, so the application goes to a loan officer.

The lower the share of income that goes to loan payments, the higher the score. An application with income 0 is rejected at once. The decision thresholds are the ones above: below 400 rejected, above 700 approved, 400–700 goes to a loan officer. The calculated score and its inputs (income, new payment, existing payments) are stored with the decision (see LA-NFR-3).

Monthly income	Amount / term	New payment	Existing payments	Ratio	Score	Decision
3,000	6,000 / 24 months	250.00	0.00	0.083	717	Approved automatically
3,000	6,000 / 24 months	250.00	300.00	0.183	617	Loan officer
1,000	6,000 / 24 months	250.00	0.00	0.250	550	Loan officer
600	6,000 / 24 months	250.00	0.00	0.417	383	Rejected automatically

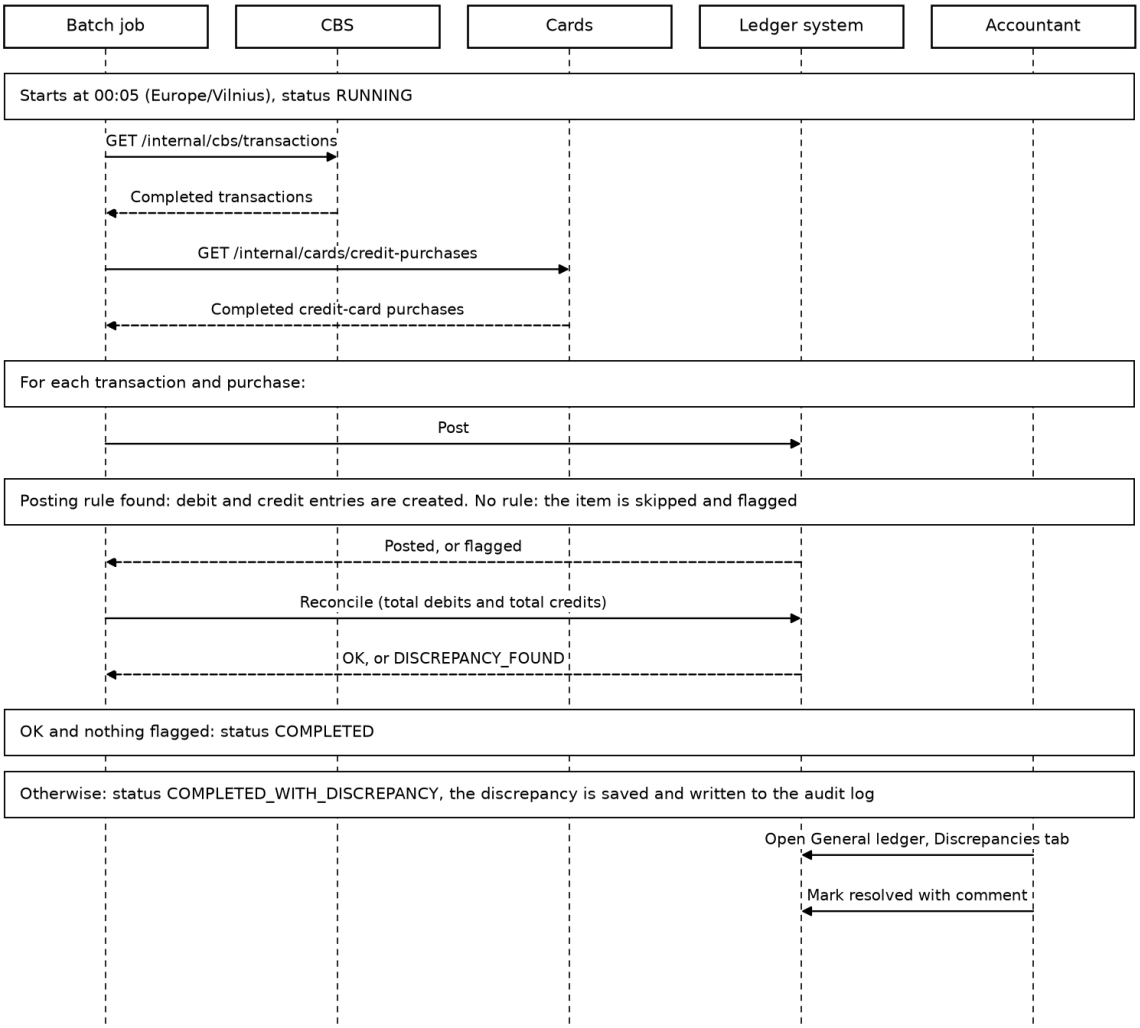


Loan application flow

10.4.2 End-of-day posting and reconciliation

Every night at 00:05 (Europe/Vilnius) the batch job collects the completed transactions of the previous calendar day from CBS and the completed credit-card purchases of the same day from Cards, and posts each one to the ledger as a debit entry and a credit entry, using the posting rules. Transactions without a rule are skipped and flagged. The system

then compares total debits with total credits. If they match, the run is complete. If not, the discrepancy is saved with the run date, shown to employees with the role ACCOUNTANT in General ledger and reports, and written to the audit log. The accountant investigates and resolves it there. No customer notification is sent. The day is chosen by the time a transaction was completed (from 00:00:00 up to, but not including, 00:00:00 of the next day), not by the time the batch starts, so every transaction belongs to exactly one day, including those completed between 23:59 and 00:00. The batch calls GET /internal/cbs/transactions with from and to set to the Europe/Vilnius midnights converted to UTC (Coordinated Universal Time); CBS compares them with Transaction.completedAt.



End-of-day posting and reconciliation

Posting rules map the CBS transaction types (see the CBS chapter) to ledger entries. Transactions without a rule are skipped and flagged. The CBS type CARD_REPAYMENT (credit-card repayment from a bank account) is posted with the rule CardRepayment: debit customer account liability, credit card receivable. A completed credit-card purchase has no CBS transaction; the batch reads it from Cards (section 9.9.3) and posts it with the rule CardCreditPurchase: debit card receivable, credit card network settlement. Its LedgerEntry.transactionId holds the card authorization id. A loan installment (LoanInstallment) is posted as one amount: CBS records only the total amount of the debit, so Accounting does not split it into principal and interest. Every posting rule therefore has one debit ledger account and one credit ledger account for the full amount.

10.5 Non-functional requirements

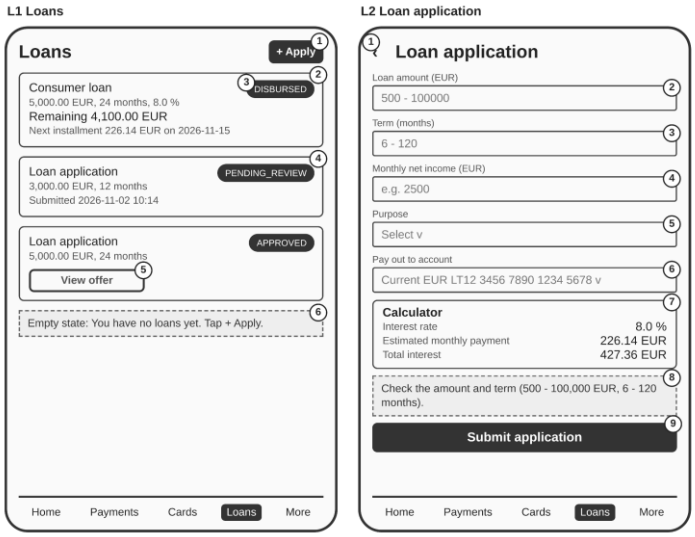
ID	Area	Requirement	Target
LA-NFR-1	Performance	The EOD (end of day) batch posts and reconciles all the day's transactions.	Within 1 hour for up to 100,000 transactions
LA-NFR-2	Accuracy	Debits equal credits for every posted transaction. Money amounts are stored as Decimal with 2 decimal places, never as floating point.	0 unbalanced transactions
LA-NFR-3	Audit	Every loan decision (auto or officer) and every discrepancy resolution is logged with user, time, and old and new status.	100% of decisions and resolutions logged

LA-NFR-4	Security	Posted ledger entries cannot be edited or deleted. Corrections are only possible through a new entry.	No update or delete operation exists

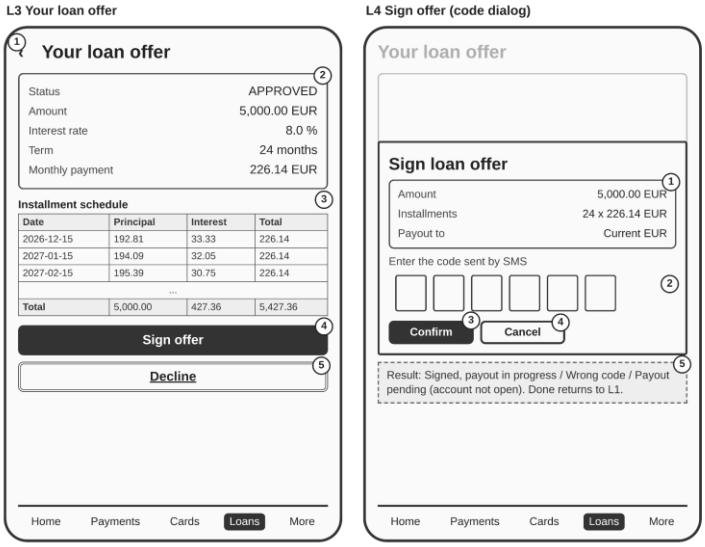
10.6 Wireframes

10.6.1 Customer

Customer screens are mobile screens with the bottom navigation defined in the UI (user interface) principles (Loans tab); the numbers in the pictures match the table. Transitions: L1 opens L2 (+ Apply) and L3 (View offer); L2 returns to L1 after Submit application or Back; L3 opens L4 (Sign offer) or returns to L1 (Decline, Back); L4 returns to L1 when done, or to L3 on Cancel. The one-time code dialog is the shared component described in the UI principles.



Wireframes L1 and L2



Wireframes L3 and L4

Ref	Element	What happens when it is used
L1.1	+ Apply	Opens L2. Disabled with a hint when the customer is not ACTIVE or KYC is not APPROVED (CUSTOMER_NOT_ELIGIBLE).
L1.2	Loan card	Shows amount, term, interest rate, remaining principal and the next installment (date and amount).
L1.3	Status badge	Status of the loan or application: PENDING_REVIEW, APPROVED, REJECTED, DISBURSED or DECLINED.
L1.4	Application card	Shows amount and term of an application that is not yet disbursed. A REJECTED application also shows the reason.
L1.5	View offer	Shown only for an APPROVED application. Opens L3.
L1.6	Empty state	Shown when the customer has no loans and no applications.
L2.1	Back	Returns to L1 and discards the entered data.

L2.2	Loan amount	500 to 100,000 EUR.
L2.3	Term	6 to 120 months.
L2.4	Monthly net income	Used in the credit score (see Loan application).
L2.5	Purpose	Selected from a list.
L2.6	Pay out to account	One of the customer's own OPEN accounts; the loan is paid out to it.
L2.7	Calculator	Updates when the amount or term changes; shows the interest rate, the estimated monthly payment and the total interest. It is an estimate only (GET /api/v1/loans/calculator).
L2.8	Banner	Shown when a field is empty or out of range; Submit application stays disabled.
L2.9	Submit application	Sends the application (POST /api/v1/loans/applications) and returns to L1, where the new application appears with its status.
L3.1	Back	Returns to L1.
L3.2	Offer	Status, amount, interest rate, term and monthly payment of the approved offer.
L3.3	Installment schedule	Date, principal, interest and total of each installment, with the totals in the last row (GET /api/v1/loans/{id}/schedule).
L3.4	Sign offer	Opens L4.
L3.5	Decline	Declines the offer (POST /api/v1/loans/offers/{id}/decline); the status becomes DECLINED and L1 opens.
L4.1	Summary	Amount, number and size of the installments, and the account that receives the payout.
L4.2	Code	Six-digit code sent by SMS (shared one-time code dialog).
L4.3	Confirm	Signs the offer (POST /api/v1/loans/offers/{id}/sign). Loans then asks CBS to pay out the money.
L4.4	Cancel	Closes the dialog and returns to L3.
L4.5	Result	Signed, payout in progress; wrong code; or payout pending when the account is not open (ACCOUNT_NOT_OPEN). Done returns to L1.

10.6.2 Loan officer

Loan officer screens are back-office screens with the left menu defined in the UI principles; the numbers in the pictures match the table. Transitions: LR1 opens LR2 (click a row); Approve or Reject in LR2 returns to LR1 and the list refreshes.



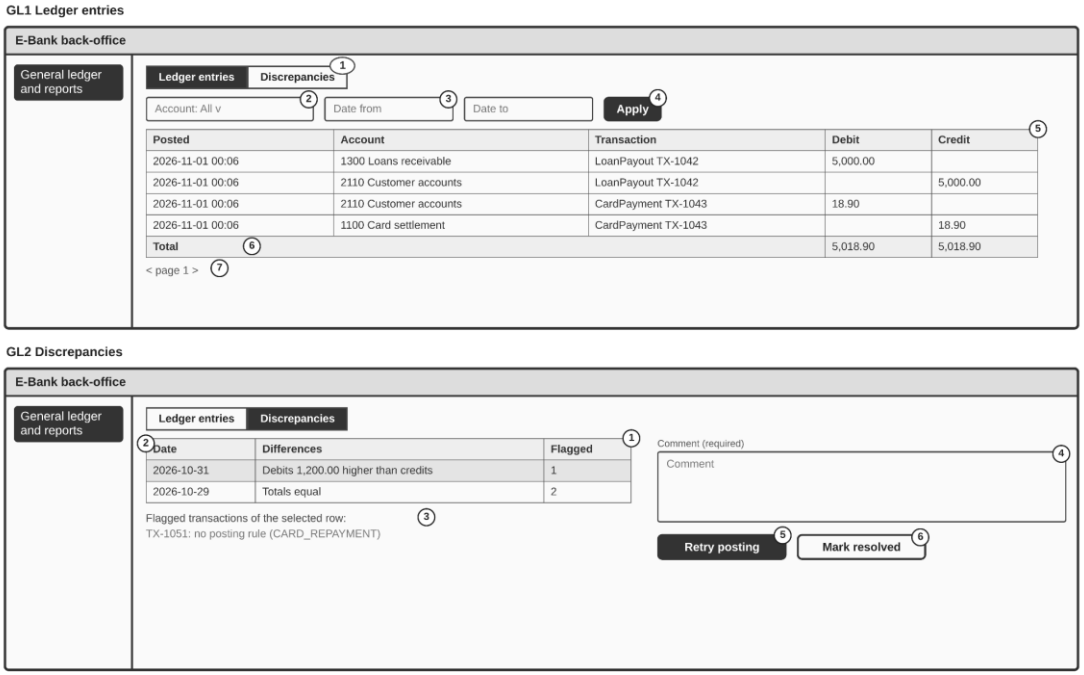
Wireframes LR1 and LR2

Ref	Element	What happens when it is used
LR1.1	Applications table	Applications waiting for review, status PENDING_REVIEW (GET /api/v1/admin/loans/applications?status=PENDING_REVIEW).
LR1.2	Score	Credit score calculated by the system (see Loan application).

LR1.3	Row	Click opens LR2.
LR1.4	Empty state	Shown when no application is waiting for review.
LR2.1	Back	Returns to LR1.
LR2.2	Application	Customer, amount, term, monthly net income and purpose from the application.
LR2.3	Credit score	The score and its inputs: new payment, existing payments and the payment-to-income ratio.
LR2.4	Offered amount	Prefilled with the requested amount; the officer may change it. Required for Approve.
LR2.5	Interest rate	Required for Approve.
LR2.6	Term	Prefilled with the requested term; required for Approve.
LR2.7	Reason	Required for Reject; an empty reason returns COMMENT_REQUIRED. Stored with the decision (LA-NFR-3).
LR2.8	Approve	Approves with the offered amount, interest rate and term (POST /api/v1/admin/loans/applications/{id}/approve). The customer can see the offer; LR1 opens.
LR2.9	Reject	Rejects with the reason (POST /api/v1/admin/loans/applications/{id}/reject); LR1 opens.

10.6.3 Accountant

Accountant screens are back-office screens; GL1 and GL2 are two tabs of General ledger and reports. The accountant opens GL2 to see new discrepancies; Retry posting or Mark resolved refreshes the table.



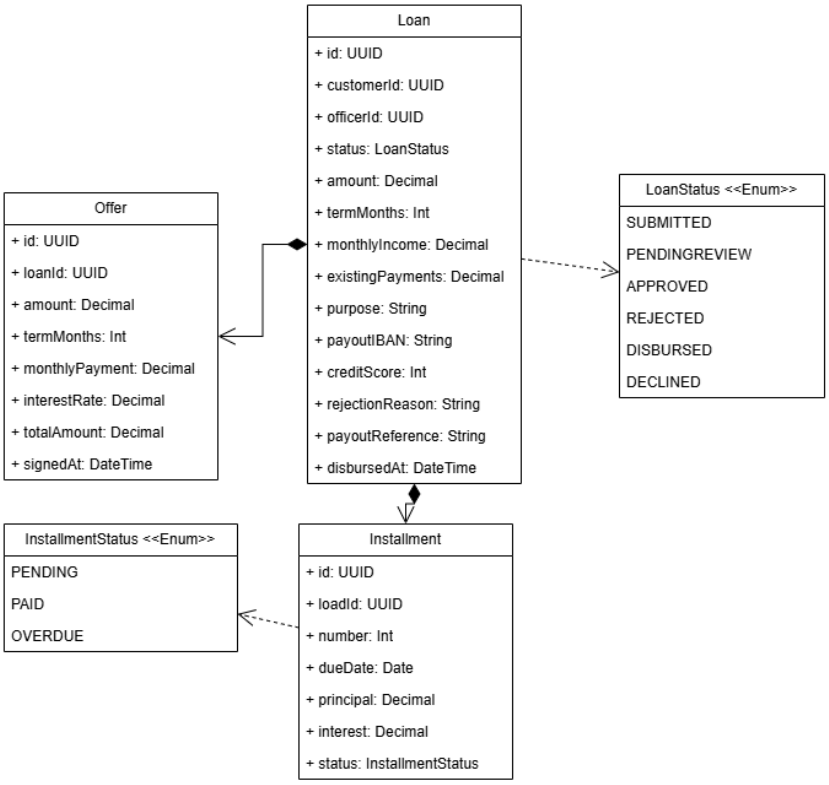
Wireframes GL1 and GL2

Ref	Element	What happens when it is used
GL1.1	Tabs	Switch between Ledger entries (GL1) and Discrepancies (GL2).
GL1.2	Account	Filters by ledger account; default is all accounts.
GL1.3	Date from / Date to	Filter by posting date.
GL1.4	Apply	Reloads the table (GET /api/v1/admin/ledger/entries).
GL1.5	Entries table	Posted entries with account, transaction, debit and credit. Posted entries cannot be edited or deleted (LA-NFR-4).
GL1.6	Total	Sums of debit and credit of the filtered entries; they are equal when the posting is balanced.
GL1.7	Pagination	Moves between pages of entries.
GL2.1	Discrepancies table	Reconciliation runs with differences or flagged transactions.
GL2.2	Selected row	Selecting a row shows its flagged transactions.
GL2.3	Flagged transactions	Transactions skipped because no posting rule exists, with the reason.
GL2.4	Comment	Required for Mark resolved (COMMENT_REQUIRED).

GL2.5	Retry posting	Posts the flagged transactions again (POST /api/v1/admin/ledger/postings/{id}/retry), for example after a posting rule was added.
GL2.6	Mark resolved	Closes the discrepancy with the comment (POST /api/v1/admin/ledger/discrepancies/{id}/resolve); the table refreshes.

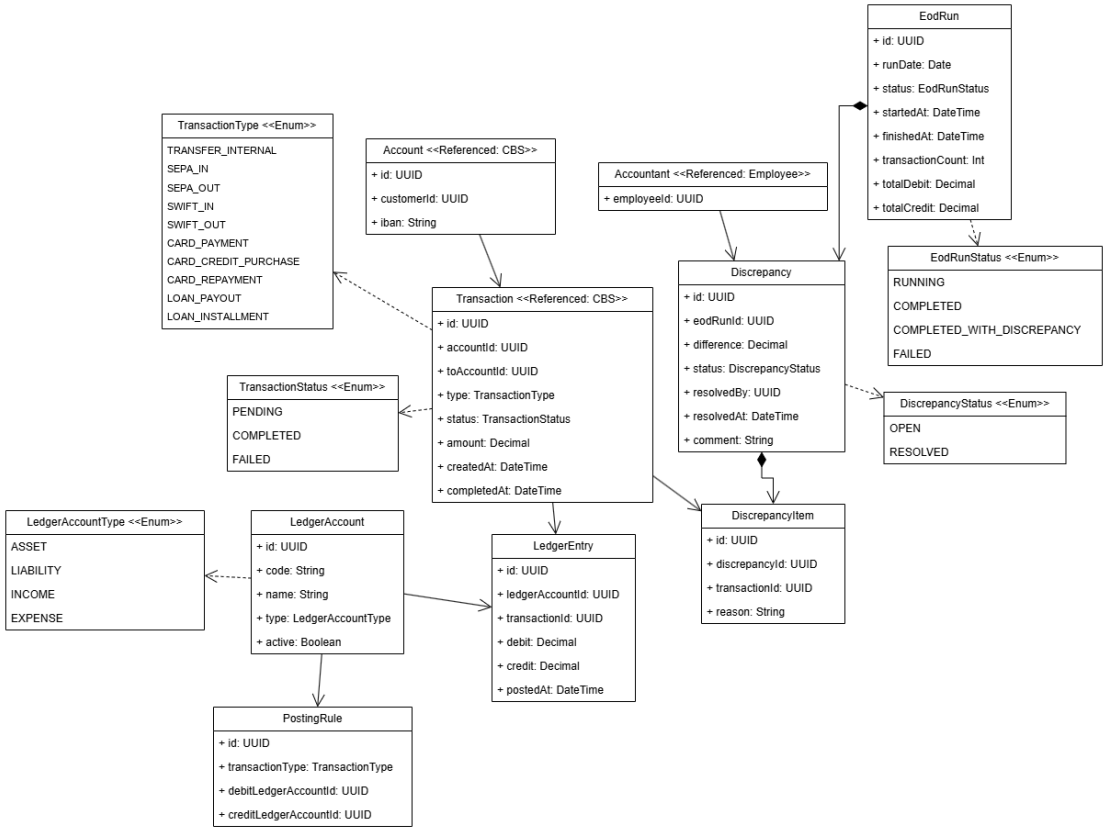
10.7 Data model

10.7.1 Loan



Loan data model

10.7.2 Accounting



Accounting data model

10.8 API contract

Conventions are in the System architecture chapter (JSON (JavaScript Object Notation), /api/v1, error body, event envelope). Employee endpoints require the role shown in the table.

10.8.1 Customer endpoints

Method and path	Purpose	Request / response
POST /api/v1/loans/applications	Submit a loan application	amount, termMonths, income, purpose, accountId / 201 application
GET /api/v1/loans/applications/{id}	Application and offer status	200 application
POST /api/v1/loans/offers/{id}/sign	Sign the offer (one-time code)	code / 200 loan
POST /api/v1/loans/offers/{id}/decline	Decline the offer	none / 200
GET /api/v1/loans	List the customer’s loans	200 list
GET /api/v1/loans/{id}/schedule	Installment schedule	200 list of installments
GET /api/v1/loans/calculator	Loan calculator	query: amount, termMonths / 200 monthlyPayment, totalCost
GET /api/v1/statements/{month}	Monthly statement	200 statement

10.8.2 Employee and internal endpoints

Method and path	Purpose	Request / response
GET /api/v1/admin/loans/applications?status=PENDING_REVIEW	Applications waiting for review (LOAN_OFFICER)	200 list
POST /api/v1/admin/loans/applications/{id}/approve	Approve	offeredAmount, interestRate, termMonths / 200 application
POST /api/v1/admin/loans/applications/{id}/reject	Reject	reason (required) / 200 application
GET /api/v1/admin/ledger/entries	General ledger view (ACCOUNTANT)	query: from, to, ledgerAccountId / 200 list
POST /api/v1/admin/ledger/postings/{id}/retry	Retry a failed posting	none / 200

POST /api/v1/admin/ledger/discrepancies/{id}/resolve	Mark a discrepancy resolved	comment (required) / 200
GET /api/v1/admin/reports/{type}	Financial reports	query: from, to / 200 report
GET, POST, PATCH /api/v1/admin/ledger/accounts	Chart of accounts management	ledger account fields / 200 or 201
POST /internal/loans/credit-card-assessments	Credit-card limit decision for Cards	customerId, requestedLimit / 200 decision (REJECTED or APPROVED), creditLimit
GET /internal/loans/customers/{customerId}/loans	Loan list for Customer 360 (CRM)	200 list with id, amount, status, nextDueDate

10.8.3 Events

Events use the common envelope.

Event	Published by	Consumed by
LOAN_INSTALLMENT_DUE, LOAN_INSTALLMENT_MISSED	Loans	Notification
RECONCILIATION_DISCREPANCY	Accounting	Accounting back-office and audit log

Payload fields of the Loans events:

Event	Payload fields
LOAN_INSTALLMENT_DUE	userId, customerId, loanId, amount, dueDate
LOAN_INSTALLMENT_MISSED	userId, customerId, loanId, amount, dueDate

10.8.4 Error codes

Errors use the common body {"error": {"code": "...", "message": "..."}}.

Code	HTTP	Meaning
CUSTOMER_NOT_ELIGIBLE	403	Customer is not ACTIVE or KYC is not APPROVED
COMMENT_REQUIRED	422	Reject reason or resolution comment is missing
INSUFFICIENT_FUNDS	502	CBS could not debit the installment; Loans marks it late
ACCOUNT_NOT_OPEN	502	CBS could not credit the payout; Loans keeps it pending

11 AML

11.1 Scope and ownership

The AML (Anti-Money Laundering) system checks customer transactions for signs of money laundering and terrorist financing. It helps compliance officers investigate suspicious cases and report them to the Financial Crime Investigation Service (FNTT), Lithuania's financial intelligence unit (FIU). Banks are required by law to do this.

AML connects to the rest of the bank in two ways. Before a transaction is executed, the Payment system and Card management ask AML for a decision and wait for the answer. After a transaction is completed, AML receives the transaction event and looks for patterns across several transactions.

Module boundaries:

- Sanctions and PEP (politically exposed person) screening during onboarding is requested by KYC (SCREENING_REQUESTED) and answered by AML (SCREENING_COMPLETED, CLEAR or HIT). AML matches the person or company against its watchlist (section 11.7); KYC keeps the decision and the customer risk level. AML uses the risk level in its transaction score.
- AML does not read other modules' databases. It keeps its own copies of the transaction and customer data it needs (AmlTransaction and AmlCustomer, section 11.7), filled only from screening requests and events.
- AML does not send messages to customers itself. It publishes events, and the Notification system sends the messages.
- FNTT is an external system. It is described as an interface in section 11.8.4, so it can be replaced with a mock during implementation.

11.2 Feature table

ID	Feature	Actor	Priority
AML-01	Before an outgoing payment or card transaction is executed, the system checks it against the active rules and returns ALLOW or HOLD.	Payment system, Card management	Must
AML-02	The system returns HOLD when the risk score is 70 or more (AmlConfig holdScore), or when a matched rule has the action HOLD.	System	Must
AML-03	The system returns HOLD for every new outgoing transaction of a Customer who has an open AmlCase, or an AmlAlert with status NEW and held = true.	System	Must
AML-04	Within 5 minutes after a transaction that was not screened before execution is completed, the system checks it against all active rules, including the multi-transaction rule R2 (24-hour window).	System	Should
AML-05	The system creates an alert with a risk score for every transaction that matches at least one rule.	System	Must
AML-06	The AML officer sees the alerts in the AML task queue, sorted by risk score.	AML officer	Must
AML-07	The AML officer takes an alert. After that, no other officer can take it.	AML officer	Must
AML-08	The AML officer sees the case details: triggered rules, transaction, customer risk level, 90-day history and earlier cases.	AML officer	Must
AML-09	The AML officer releases a held transaction, and the transaction is executed. If the transaction was already executed, the officer closes the case as a false positive. A comment is required.	AML officer	Must
AML-10	The AML officer reports a case to FNTT. A held transaction is rejected, and the FNTT reference is saved. A comment is required.	AML officer	Must
AML-11	The customer is never shown that a report to FNTT was made.	System	Must
AML-12	The system records every action on cases (employee, time, action, values before and after). Records cannot be changed or deleted.	System	Must
AML-13	The AML officer filters the queue by status, rule, score range and date range, and searches it by customer name, IBAN (International Bank Account Number) or alert ID.	AML officer	Should
AML-14	The system matches a person or company from SCREENING_REQUESTED against the sanctions and PEP watchlist and answers with SCREENING_COMPLETED.	KYC, System	Must

Won't (deliberately excluded from this release):

- A screen for editing the rules, the configuration, the high-risk country list or the watchlist. They are loaded from the default data in section 11.7.
- Automatic updates of the sanctions and PEP lists from external sources.
- Machine-learning risk scoring.
- Reports to FNTT without an officer's decision.

11.3 Business flows

11.3.1 F1: Screen a transaction

Actors: Payment system, Card management, AML system.

Preconditions: at least one active AmlRule exists, and the Customer has a KYC risk level (LOW, MEDIUM or HIGH) in AmlCustomer.

Basic course:

1. The Payment system creates a Transaction with status PENDING and sends AML a screening request (section 11.8.1).
2. The AML system saves the transaction in AmlTransaction and loads the Customer's risk level and all active AmlRules.
3. The AML system checks the transaction against each active AmlRule. The default rules are:
    - R1, large amount: amountEur is more than 15,000 EUR (euro).
    - R2, structuring: the current transfer and the Customer's outgoing transfers completed in the 24 hours before it (a rolling window over all accounts of the Customer) are counted together. R2 matches when there are 5 or more such transfers, each below 15,000 EUR, and their total is above 15,000 EUR.

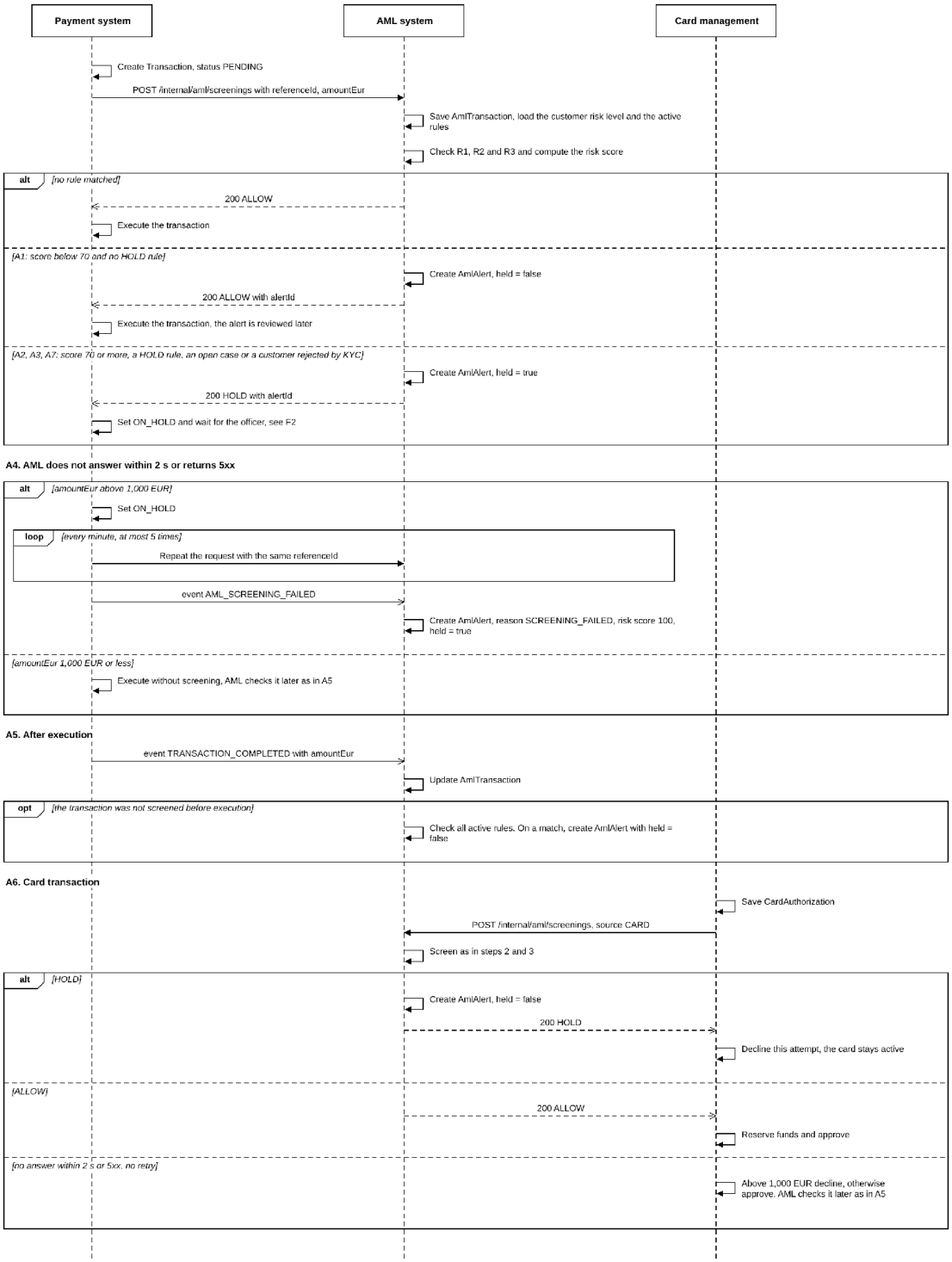
    - R3, high-risk country: the counterparty's country (for cards, the merchant's country) is in the HighRiskCountry table. The starting list contains KP (North Korea) and IR (Iran), as ISO 3166-1 alpha-2 codes.
4. The AML system finds no matching rule and returns ALLOW.
5. The Payment system executes the transaction.

Amounts: all thresholds are in EUR. The caller sends amountEur: the amount converted with the CBS (core banking system) exchange rate at the time of the transaction, without the FX (foreign exchange) markup. AML compares only amountEur with the thresholds.

Risk score: the sum of the weights of the matched rules, multiplied by the customer risk factor (LOW 1.0, MEDIUM 1.5, HIGH 2.0), with a maximum of 100. Example: R3 (weight 70) for a MEDIUM customer gives 70 × 1.5 = 105, so the score is 100. The weights and factors are in section 11.7.

Alternate courses:

- A1. A rule matches, but the risk score is below 70 and no matched rule has the action HOLD. The AML system creates an AmlAlert with held = false and returns ALLOW. The Payment system executes the transaction. The alert appears in the AML task queue to be reviewed afterwards. Alerts with held = false do not stop new transactions.
- A2. The risk score is 70 or more, or a matched rule has the action HOLD. The AML system creates an AmlAlert with held = true and returns HOLD. The Payment system sets the transaction status to ON_HOLD. The transaction stays ON_HOLD, with its hold active, until the AML officer decides in F2; the 60-second limit of NFR-2 does not apply to it.
- A3. The Customer already has an open AmlCase, or an AmlAlert with status NEW and held = true. The AML system returns HOLD, whatever the risk score, and creates an AmlAlert with held = true. If a case is open, the new alert is linked to it; otherwise it is linked when an officer takes the Customer's first alert (F2, step 3).
- A4. The AML system does not answer within 2 seconds, or returns a 5xx error. If amountEur is above 1,000 EUR, the Payment system sets the transaction to ON_HOLD and repeats the request every minute, up to 5 times (BR-14). A repeat carries the same referenceId, so it never creates a second alert. After the fifth failure, the Payment system publishes AML_SCREENING_FAILED, and the AML system creates an AmlAlert for manual review (reason SCREENING_FAILED, no rule codes, risk score 100, held = true). If amountEur is 1,000 EUR or less, the Payment system executes the transaction, and AML checks it later as in A5.
- A5. The Payment system publishes TRANSACTION_COMPLETED. The AML system updates the transaction in AmlTransaction, or creates it if the transaction was executed without screening. A transaction that was not screened before execution (A4 with 1,000 EUR or less, and A6) is checked against all active rules, as in step 3. If a rule matches, the AML system creates an AmlAlert with held = false; nothing is held.
- A6. Card transaction. Card management calls AML before it reserves funds, with source = CARD and referenceId = the CardAuthorization id, and the AML system screens it as in steps 2 and 3. HOLD means that Cards declines this attempt; the card itself is not blocked (section 9.3.3). The AmlAlert is created with held = false, because no transaction waits for a decision, and AML never publishes TRANSACTION_RELEASED or TRANSACTION_REJECTED for card alerts. If AML does not answer within 2 seconds or returns a 5xx error, Cards does not retry: above 1,000 EUR it declines the attempt; at 1,000 EUR or less it approves it, and AML checks it afterwards from CARD_TRANSACTION_AUTHORIZED (A5).
- A7. The Customer was rejected by KYC (kycStatus = REJECTED in AmlCustomer). The AML system returns HOLD and creates an AmlAlert with held = true. Normally this does not happen, because such a customer never becomes ACTIVE and CBS does not accept their payments.



AML F1: screen a transaction

11.3.2 F2: Investigate a suspicious case

Actors: AML officer (Employee with the role COMPLIANCE), AML system, Payment system, Notification system, FNTT.

Preconditions: an AmlAlert with status NEW exists, and the AML officer is logged in to the back-office.

Basic course (the activity turns out to be legitimate):

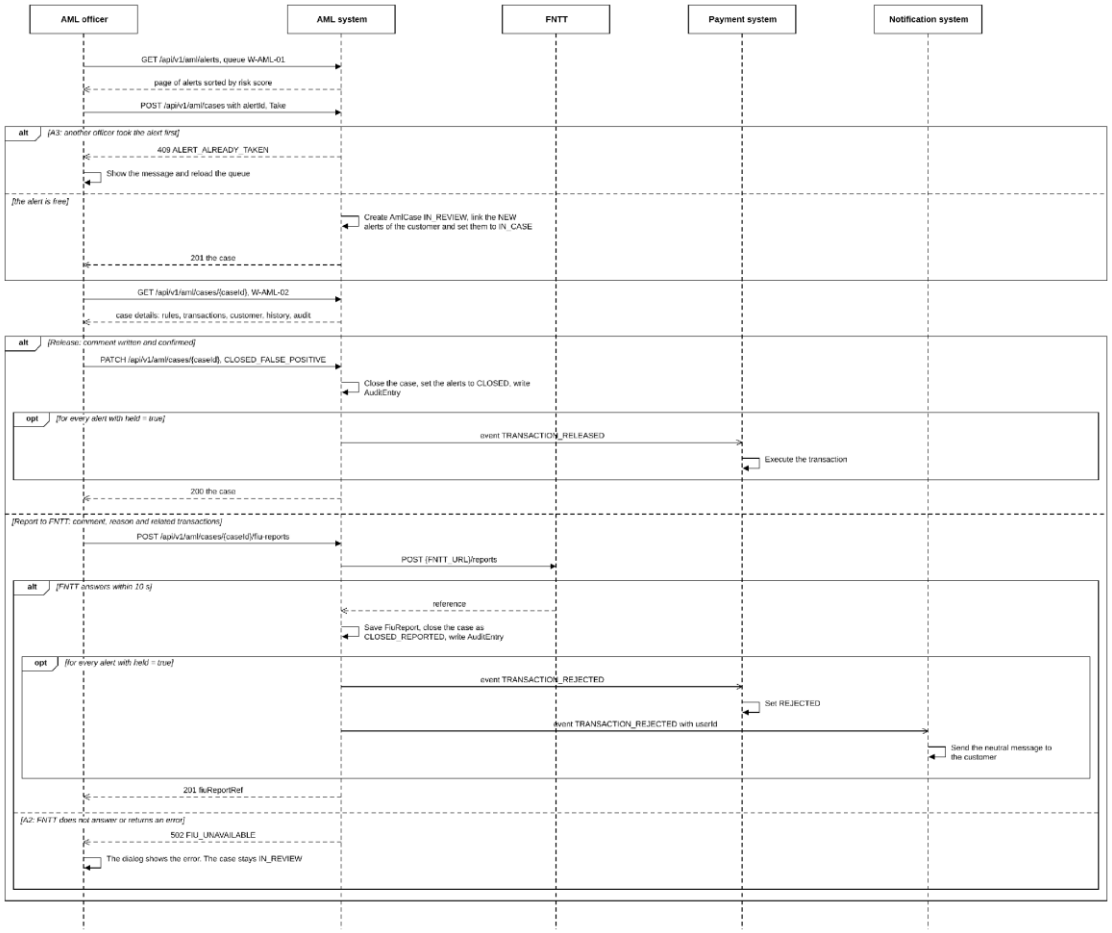
0. The AML officer clicks AML queue in the back-office menu. The system shows the AML task queue (W-AML-01), sorted by risk score.
1. The AML officer clicks Take on an AmlAlert.
2. The system creates an AmlCase with status IN_REVIEW and assigns it to the AML officer. It links this alert and all other NEW alerts of the same Customer to the case and sets their status to IN_CASE. It opens Case details (W-AML-02).

3. The AML officer reviews the triggered rules, the transaction, the Customer panel, the 90-day history and the earlier cases.
4. The AML officer types a comment, clicks Release and confirms.
5. The system closes the AmlCase as CLOSED_FALSE_POSITIVE and sets its alerts to CLOSED. For every alert of the case with held = true, it publishes TRANSACTION_RELEASED. It returns the AML officer to the AML task queue.
6. The Payment system executes the released transactions.

Every action in this flow is written to the audit log (AuditEntry, section 11.7), with the values before and after.

Alternate courses:

- A1. The suspicion is confirmed. At step 5, the AML officer types a comment and clicks Report to FNTT. In the dialog, the officer types the reason, selects the related transactions from the Customer's transactions of the last 90 days and clicks Submit. The system sends the report to FNTT (section 11.8.4), saves the returned reference in FiuReport, closes the AmlCase as CLOSED_REPORTED and sets its alerts to CLOSED. For every alert with held = true, it publishes TRANSACTION_REJECTED. The Payment system sets those transactions to REJECTED, and the Customer receives the message "Your payment could not be executed. Please contact the bank." Nothing the Customer can see mentions the report, because AML law prohibits this (tipping-off).
- A2. FNTT does not answer within 10 seconds or returns an error. At A1, the system shows "FNTT is unavailable. The report was not sent. Please try again." The AmlCase stays IN_REVIEW, and the held transactions stay ON_HOLD.
- A3. Another officer took the alert first. At step 2, the system shows "This alert has already been taken" and reloads the AML task queue.
- A4. No alert of the case has held = true: its transactions were already executed (F1, A1 and A5) or declined (F1, A6). At step 5 the button is labelled Close as false positive. The system closes the case as in step 6 but publishes no event. If the officer reports the case instead (A1), the report is sent and the case is closed as CLOSED_REPORTED, but no TRANSACTION_REJECTED is published. CBS ignores TRANSACTION_RELEASED and TRANSACTION_REJECTED for a transaction that is not ON_HOLD.
- A5. The alert was created from AML_SCREENING_FAILED (F1, A4). The triggered rules panel shows "Screening failed". The officer decides as in the basic course or in A1.



AML F2: investigate a suspicious case

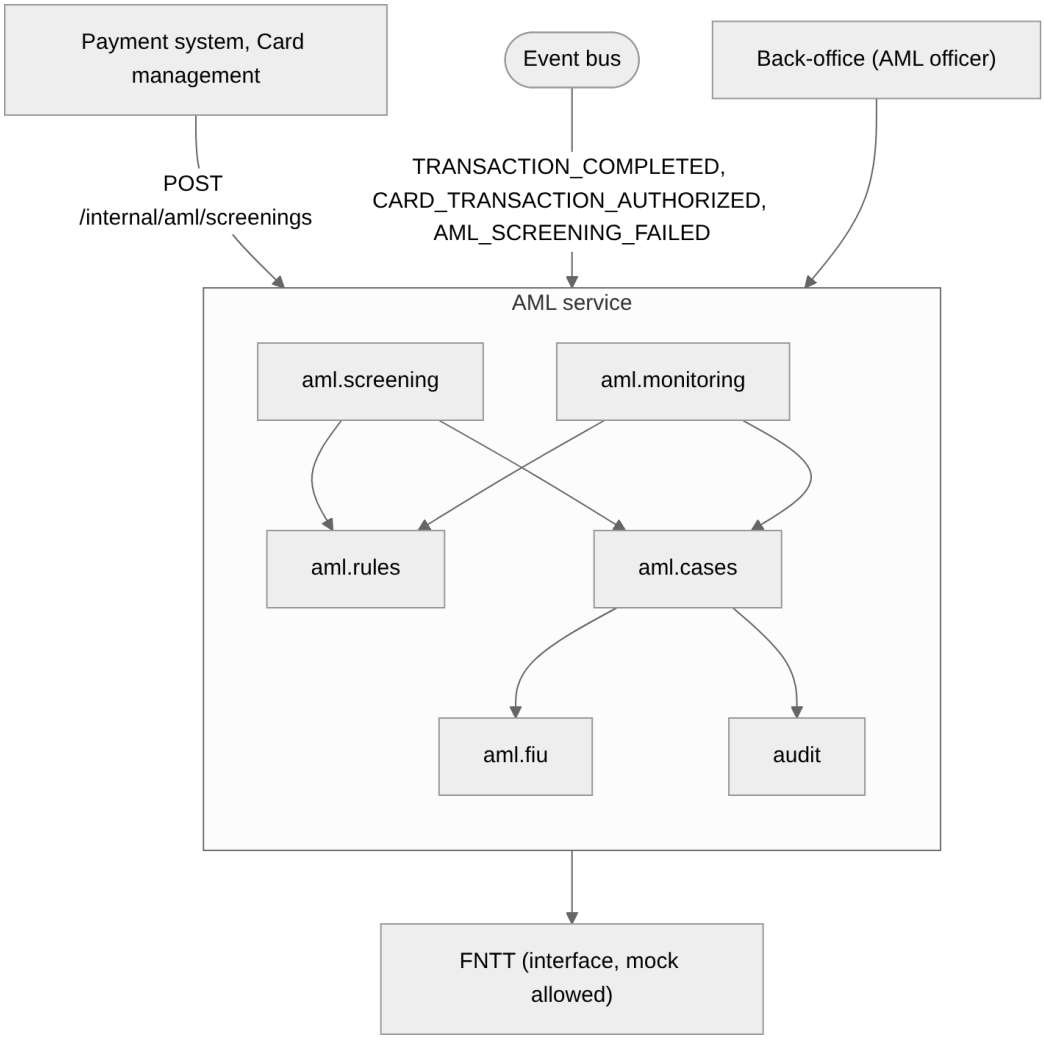
11.4 Non-functional requirements

These apply to this module. The requirements common to the whole system are in section 4.

ID	Requirement
AML-NFR-01	95% of screening decisions are returned within 500 ms, and every decision within 2 seconds.
AML-NFR-02	If AML does not answer, no outgoing transaction above 1,000 EUR is executed until it has been checked.
AML-NFR-03	A new AmlAlert appears in the AML task queue within 10 seconds after it is created. W-AML-01 reloads the alert list every 5 seconds while it is open.
AML-NFR-04	Cases, alerts, FNTT reports and audit records are kept for 8 years after the case is closed, as required by the Lithuanian AML law. An AmlTransaction row is kept as long as an alert or an FNTT report refers to it; other rows are deleted after 90 days.
AML-NFR-05	Only employees with the role COMPLIANCE can open AML screens and endpoints. Others receive 403 PERMISSION_DENIED.
AML-NFR-06	A repeated screening request with the same referenceId returns the first decision and creates no second alert.

11.5 High-level architecture

The module consists of the packages below; the layering of the whole system is in section 3.3.



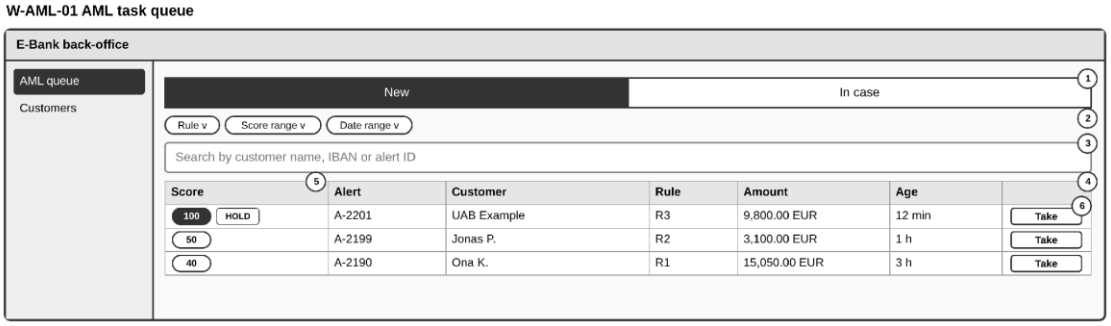
AML packages

Package	Responsibility
aml.screening	Handles the synchronous screening request and returns ALLOW or HOLD within 2 seconds.
aml.rules	Loads the rules and the configuration and checks a transaction and the customer's recent history against them. Matches persons and companies from SCREENING_REQUESTED against the watchlist.
aml.monitoring	Listens to TRANSACTION_COMPLETED, CARD_TRANSACTION_AUTHORIZED and the KYC and CRM (customer relationship management) customer events. Keeps AmlTransaction and AmlCustomer up to date and checks transactions that were not screened before execution (F1, A5).

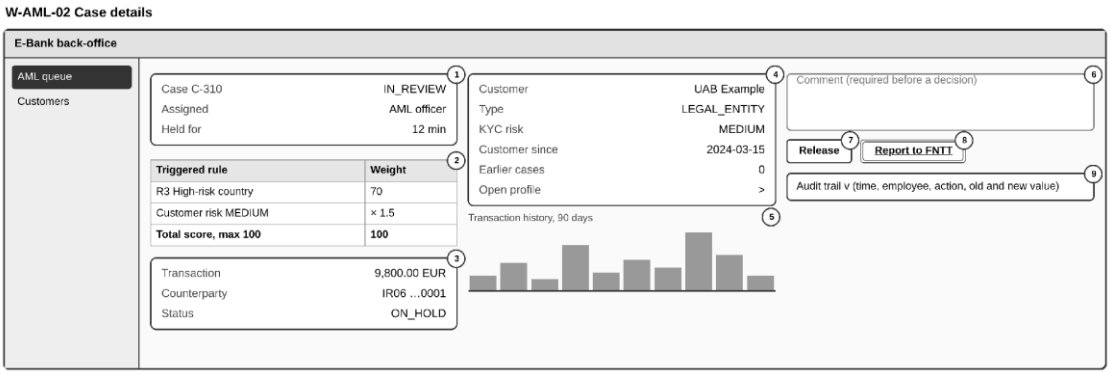
Package	Responsibility
aml.cases	Manages alerts and cases and the back-office actions on them.
aml.fiu	Builds the FNTT report and sends it through the FNTT interface.
audit	Writes and reads the audit log. Other modules can also use it for their back-office actions.

11.6 UI wireframes

Both screens are back-office screens for employees with the role COMPLIANCE. The left menu shows only the items allowed for the employee's role (section 5.1); Customers opens the CRM customer search.



W-AML-01 AML task queue



W-AML-02 Case details

11.6.1 W-AML-01 AML task queue

The officer's starting screen. It shows a table of alerts, filters, a search field and a button for taking an alert.

#	Element	What happens when it's used
1	Tabs: New, In case	Filters the table by alert status (NEW or IN_CASE)
2	Filters: rule, score range, date range	Reloads the table with the selected filters
3	Search	Searches by customer name, IBAN or alert ID and reloads the table
4	Alert row: score, alert ID, customer, rule, amount with currency, age, HOLD label	Opens W-AML-02 for the case, if the alert already belongs to one. The HOLD label is shown when held = true. An alert from AML_SCREENING_FAILED shows "Screening failed" in the Rule column
5	Score	Dark for 70 and above, outlined for 40 to 69, dashed for below 40. Clicking the column header sorts the table
6	Take	Assigns the alert to the officer, creates a case and opens W-AML-02. If another officer took it first, shows "This alert has already been taken" and reloads the table

An empty tab shows "No alerts." The table reloads every 5 seconds (AML-NFR-03).

11.6.2 W-AML-02 Case details

#	Element	What happens when it's used
1	Case header: case ID, status, assigned officer, how long the transaction has been held	Read only. "Held for" is shown only while a transaction of the case is ON_HOLD
2	Triggered rules with weights and the total score	Read only. For an alert from AML_SCREENING_FAILED it shows "Screening failed"

#	Element	What happens when it's used
3	Transaction: amount, counterparty, type, time, status	Read only
4	Customer: name, type, KYC risk level, customer since, earlier cases	The "Open profile" link opens the CRM Customer 360
5	Transaction history, 90 days	Bar chart of the daily outgoing totals in EUR. Read only
6	Comment	The decision buttons are disabled until a comment is written
7	Release	Opens a confirmation dialog. After confirmation, the case closes as a false positive, the held transactions are released, and the officer goes back to W-AML-01. When no transaction of the case is held, the button is labelled Close as false positive and releases nothing
8	Report to FNTT	Opens the Report to FNTT dialog with a "Reason for suspicion" text area, a checkbox list of the Customer's transactions of the last 90 days, and the buttons Cancel and Submit. After Submit, the case closes as reported and the officer goes back to W-AML-01. If FNTT does not answer, the dialog shows an error and stays open
9	Audit trail	Expands the list of actions on this case: time, employee, action, old and new value

A closed case opens read only: elements 6–8 are hidden, and any action returns "This case has already been closed".

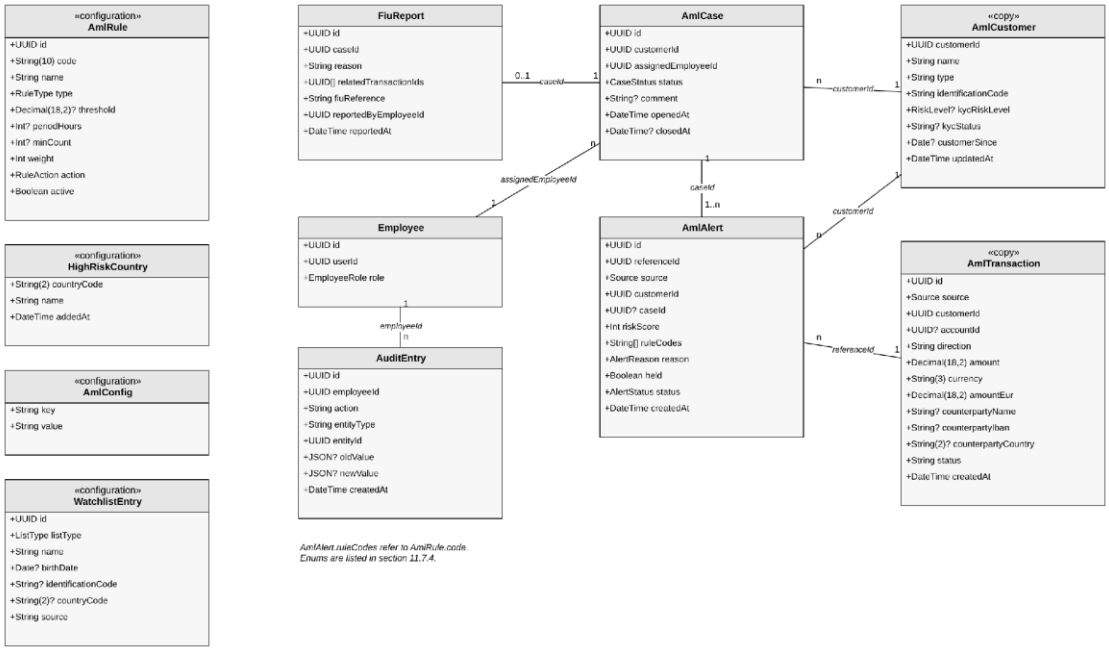
All screen texts are message keys with LT and EN values (NFR-5). The error code returned by the API (for example ALERT_ALREADY_TAKEN) is the key of the message shown.

11.6.3 Screen transitions

From	Action	To
Back-office menu	AML queue	W-AML-01
W-AML-01	Take, or click a row that has a case	W-AML-02
W-AML-02	Release, Close as false positive or Report to FNTT completed	W-AML-01
W-AML-02	Open profile	CRM Customer 360

11.7 Data model

AML owns every entity below. User, Customer and Transaction are shared entities owned by other modules (section 3.4); AML refers to them only by id and keeps its own copies, AmlTransaction and AmlCustomer. Transaction.status (CBS) includes ON_HOLD and REJECTED for this module. Employee is a shared entity owned by this module: employees are loaded from the default data, and other modules read the employee's role from the employee token. A "?" after a type marks a field that can be empty.



AML data model

11.7.1 Configuration data

Loaded from the default data at deployment. There is no screen for editing it (section 11.2, Won't).

Entity	Fields
AmlRule	id: UUID, code: String(10) unique, name: String, type: RuleType, threshold: Decimal(18,2)? (EUR), periodHours: Int?, minCount: Int?, weight: Int (1–100), action: RuleAction, active: Boolean
HighRiskCountry	countryCode: String(2) unique (ISO 3166-1 alpha-2), name: String, addedAt: DateTime
AmlConfig	key: String unique, value: String
WatchlistEntry	id: UUID, listType: ListType, name: String, birthDate: Date?, identificationCode: String? (personal code or company registration number), countryCode: String(2)?, source: String

AmlRule.periodHours and minCount are used only by VELOCITY rules; threshold is not used by HIGH_RISK_COUNTRY rules.

Default rules:

Code	Name	Type	Threshold (EUR)	Period (h)	Min count	Weight	Action
R1	Large amount	AMOUNT_THRESHOLD	15000	–	–	40	ALERT
R2	Structuring	VELOCITY	15000	24	5	50	ALERT
R3	High-risk country	HIGH_RISK_COUNTRY	–	–	–	70	HOLD

Default configuration (AmlConfig):

Key	Value	Used in
holdScore	70	AML-02; F1, A1 and A2
maxScore	100	Risk score (F1)
riskFactor.LOW, riskFactor.MEDIUM, riskFactor.HIGH	1.0, 1.5, 2.0	Risk score (F1)
historyDays	90	W-AML-02 history and the Report to FNTT dialog
fiuTimeoutSec	10	F2, A2
unreferencedTransactionDays	90	AML-NFR-04

The 2-second timeout, the 1,000 EUR limit and the retries belong to the callers (BR-14 in CBS, section 9.3.3 in Cards).

Default high-risk countries: KP (North Korea) and IR (Iran).

Watchlist matching (AML-14): names are normalised (upper case, no diacritics, no punctuation, single spaces). An entry matches when the normalised names are equal and, when both sides have a birth date or an identification code, those are equal too. A match on the name alone is still a HIT, and the KYC officer decides (section 6). The default data contains test entries only:

List	Name	Birth date	Identification code	Country
SANCTIONS	SANCTIONS TEST PERSON	1970-01-01	–	KP
SANCTIONS	SANCTIONS TEST COMPANY LTD	–	TEST-0001	IR
PEP	PEP TEST PERSON	1965-05-05	–	LT

11.7.2 Business data

Entity	Fields
AmlAlert	id: UUID, referenceId: UUID (the Transaction id or the CardAuthorization id), source: Source, customerId: UUID, caseId: UUID?, riskScore: Int (0–100), ruleCodes: String[] (empty for SCREENING_FAILED), reason: AlertReason, held: Boolean (true when the transaction waits ON_HOLD for an officer's decision), status: AlertStatus, createdAt: DateTime
AmlCase	id: UUID, customerId: UUID, assignedEmployeeId: UUID, status: CaseStatus, comment: String? (required before the case is closed), openedAt: DateTime, closedAt: DateTime?
FiuReport	id: UUID, caseId: UUID unique, reason: String, relatedTransactionIds: UUID[], fiuReference: String, reportedByEmployeeId: UUID, reportedAt: DateTime
AuditEntry	id: UUID, employeeId: UUID, action: String (for example CASE_TAKEN, CASE_RELEASED, CASE_REPORTED), entityType: String, entityId: UUID, oldValue: JSON (JavaScript Object Notation)?, newValue: JSON?, createdAt: DateTime. Rows are never updated or deleted
Employee	id: UUID, userId: UUID unique, role: EmployeeRole

One AmlCase has one or more alerts; an alert belongs to at most one case. A case has at most one FiuReport. An alert refers to one AmlTransaction through referenceId.

11.7.3 Copies of other modules' data

Entity	Fields	Filled from
AmlTransaction	id: UUID (= referenceId), source: Source, customerId: UUID, accountId: UUID?, direction: INCOMING/OUTGOING, amount: Decimal(18,2) (never negative), currency: String(3), amountEur: Decimal(18,2), counterpartyName: String?, counterpartyIban: String?, counterpartyCountry: String(2)?, status: PENDING/ON_HOLD/COMPLETED/REJECTED/DECLINED, createdAt: DateTime	Screening requests, TRANSACTION_COMPLETED, CARD_TRANSACTION_AUTHORIZED and AML's own decisions
AmlCustomer	customerId: UUID, name: String, type: INDIVIDUAL/LEGAL_ENTITY, identificationCode: String (personal code or company registration number), kycRiskLevel: RiskLevel?, kycStatus: APPROVED/REJECTED?, customerSince: Date?, updatedAt: DateTime	SCREENING_REQUESTED with partyRole = SUBJECT (name, type, code); KYC_APPROVED and KYC_REJECTED (risk level, status); CUSTOMER_ACTIVATED (customerSince)

11.7.4 Enums

Enum	Values
EmployeeRole	SUPPORT, OPERATIONS, COMPLIANCE, AUDITOR, LOAN_OFFICER, ACCOUNTANT, ADMIN
RuleType	AMOUNT_THRESHOLD, VELOCITY, HIGH_RISK_COUNTRY
RuleAction	ALERT, HOLD
AlertStatus	NEW, IN_CASE, CLOSED
AlertReason	RULE_MATCH, SCREENING_FAILED
CaseStatus	IN_REVIEW, CLOSED_FALSE_POSITIVE, CLOSED_REPORTED
ScreeningDecision	ALLOW, HOLD
Source	PAYMENT, CARD
RiskLevel	LOW, MEDIUM, HIGH
ListType	SANCTIONS, PEP

11.8 API contract

All endpoints use JSON. Amounts are decimal strings with two decimal places, for example "15000.00". Times are in ISO 8601 (the international date and time format) UTC (Coordinated Universal Time), for example "2026-10-11T10:42:00Z". Request fields are required unless marked optional.

Every error has the common body (section 3.2):

{ "error": { "code": "COMMENT_REQUIRED", "message": "A comment is required before a decision." } }

Back-office endpoints need an employee token with the role COMPLIANCE. Without a valid token they return 401 UNAUTHORIZED; with another role, 403 PERMISSION_DENIED (AML-NFR-05). Internal endpoints are only reachable between services.

11.8.1 Screening (internal)

POST /internal/aml/screenings

Request:

```
{
  "referenceId": "uuid",
  "source": "PAYMENT",
  "customerId": "uuid",
  "accountId": "uuid",
  "initiatedByUserId": "uuid",
  "direction": "OUTGOING",
  "amount": "15200.00",
  "currency": "EUR",
  "amountEur": "15200.00",
  "counterpartyName": "Example GmbH",
  "counterpartyIban": "DE89370400440532013000",
  "counterpartyCountry": "DE",
  "createdAt": "2026-10-11T10:42:00Z"
}
```

Field	Meaning
referenceId	For source = PAYMENT, the Transaction id; for source = CARD, the CardAuthorization id. A repeated request with the same referenceId returns the first decision
source	PAYMENT or CARD
customerId	The Customer who sends the money
accountId (optional)	The debited account; null for credit cards
initiatedByUserId	The user who started the payment, or the card holder. TRANSACTION_REJECTED is sent to this user
direction	OUTGOING (only outgoing transactions are screened)
amount, currency	The amount in the transaction currency (EUR, USD (US dollar) or GBP (British pound))
amountEur	The amount converted to EUR with the CBS exchange rate at the time of the transaction, without the FX markup. All thresholds use it
counterpartyName (optional)	The beneficiary; for cards, the merchant name
counterpartyIban (optional)	The beneficiary's IBAN; null for cards
counterpartyCountry	The counterparty's country (for an IBAN, its country code); for cards, the merchant's country
createdAt	When the transaction was created

Response 200:

{ "decision": "HOLD", "alertId": "uuid", "riskScore": 100 }

If decision is ALLOW and no rule matched, alertId is null and riskScore is 0.

Errors: 400 VALIDATION_ERROR when a required field is missing or invalid. A timeout or a 5xx response (500, 503) is handled by the caller as in F1, A4 (payments) or A6 (cards).

11.8.2 Back-office endpoints

Method and path	Purpose	Request	Success response	Errors
GET /api/v1/aml/alerts	List alerts for the queue	Query, all optional: status (NEW, IN_CASE or CLOSED; default NEW), ruleCode, minScore, maxScore, from, to, q (customer name, IBAN or alert ID), sort (riskScore or createdAt, "-" for descending; default -riskScore), page (default 0), size (default 50)	200, a page of alerts	400 VALIDATION_ERROR
POST /api/v1/aml/cases	Take an alert and create a case	alertId	201, the case, with a Location header	404 NOT_FOUND, 409 ALERT_ALREADY_TAKEN
GET /api/v1/aml/cases/{caseId}	Case details	none	200, the case details	404 NOT_FOUND
PATCH /api/v1/aml/cases/{caseId}	Release, or close as a false positive	status (only CLOSED_FALSE_POSITIVE), comment	200, the case	400 VALIDATION_ERROR, 404 NOT_FOUND, 409 CASE_CLOSED, 422 COMMENT_REQUIRED
POST /api/v1/aml/cases/{caseId}/fiu-reports	Report to FNTT and close the case	comment, reason, relatedTransactionIds (optional; ids from the case details)	201, fiuReportRef and reportedAt	404 NOT_FOUND, 409 CASE_CLOSED, 422 COMMENT_REQUIRED, 502 FIU_UNAVAILABLE

A page has the fields items, page, size and totalItems. An alert in the list has the fields id, riskScore, status, reason, source, customerId, customerName, ruleCodes, amount, currency, held (true if the transaction is ON_HOLD), createdAt and caseId.

Case details contain the AmlCase fields from section 11.7.2, plus:

- alerts: id, riskScore, reason, held and the matched rules with their weights;
- transaction: the transaction of the first alert (id, source, amount, currency, amountEur, counterpartyName, counterpartyIban, counterpartyCountry, status, createdAt);

- customer: id, name, type, kycRiskLevel, customerSince;
- earlierCases: id, status and closedAt of each earlier case of the Customer;
- history: the date and the total outgoing amountEur for each of the last 90 days (the chart);
- transactions: the Customer's transactions of the last 90 days (id, createdAt, direction, amount, currency, counterpartyName, status), for the Report to FNTT dialog;
- audit: time, employee name, action, oldValue and newValue of each AuditEntry of the case.

If a case is already closed, every action returns 409 CASE_CLOSED, and the screen shows "This case has already been closed".

11.8.3 Events

Events use the common envelope (section 3.2). If the same eventId is received twice, the receiver ignores the second one.

Events AML receives:

Event	From	Payload fields used	Used for
TRANSACTION_COMPLETED	CBS	transactionId, customerId, accountId, direction, amount, currency, amountEur, counterpartyName, counterpartyCountry	Updates AmlTransaction; checks unscreened transactions (F1, A5)
CARD_TRANSACTION_AUTHORIZED	Cards	cardAuthorizationId, customerId, amount, currency, amountEur, merchantName, merchantCountry	Updates AmlTransaction; checks unscreened card transactions (F1, A5)
AML_SCREENING_FAILED	CBS	transactionId, customerId	Alert for manual review (F1, A4)
SCREENING_REQUESTED	KYC	screeningId, customerId, partyRole (SUBJECT, REPRESENTATIVE or UBO (ultimate beneficial owner)), subjectType, name, birthDate (optional), personalCode or registrationNumber, countryCode	Watchlist match (AML-14). For partyRole = SUBJECT, also stores the name, type and code in AmlCustomer
KYC_APPROVED	KYC	customerId, riskLevel	Stores kycRiskLevel and kycStatus = APPROVED in AmlCustomer
KYC_REJECTED	KYC	customerId	Stores kycStatus = REJECTED in AmlCustomer (F1, A7)
CUSTOMER_ACTIVATED	CRM	customerId, and occurredAt from the envelope	Stores customerSince in AmlCustomer

Events AML publishes:

Event	Payload	Who reacts
TRANSACTION_RELEASED	transactionId, customerId, caseId	CBS executes the transaction
TRANSACTION_REJECTED	transactionId, customerId, userId (initiatedByUserId from the screening request), caseId	CBS sets REJECTED; the Notification system sends the neutral message to userId
SCREENING_COMPLETED	screeningId, customerId, result (CLEAR or HIT), matchDetails (for each match: listType, entry name, and whether it matched on the name only, the name and birth date, or the name and code)	KYC stores a VerificationCheck

TRANSACTION_RELEASED and TRANSACTION_REJECTED are published only for transactions that are ON_HOLD (F2, A4). TRANSACTION_REJECTED does not contain the reason, so no other module can show the customer that a report was made.

11.8.4 FNTT interface

This can be a mock during implementation.

POST {FNTT_URL}/reports with caseId, customer (name, personal or company code), transactions (id, amount, currency, date, counterparty) and reason. The response contains reference. The mock returns a random reference such as FNTT-2026-000123. If FNTT does not answer within 10 seconds or returns an error, F2 continues with A2.

12 Notification

12.1 Scope and ownership

The Notification system is a shared service used by every module. It receives events from the other modules, turns them into messages from templates, and sends them to customers by push, e-mail or SMS (short message service). Customers choose on which channels they receive each category of message. Mandatory messages, including all security messages, are always sent.

Module boundaries:

- The Notification system never decides on its own what to send or to whom. It only reacts to events. If a module needs a new message, it publishes a new event, and a template for that event is added here.
- Every event that has a template contains userId: the user to notify, chosen by the module that publishes the event (as in the CBS (core banking system) events, section 8.10.3). If several users must be notified (for example the approvers of a company payment), the module publishes one event per user.
- The Notification system sends messages only to customers. Messages for employees (for example a reconciliation discrepancy) are shown in the back-office of the module that owns them.
- One-time confirmation codes are generated by CBS. The Notification system only delivers them.
- The content of a message comes from the event payload. The Notification system does not read other modules' databases: it keeps its own copy of the contact details it needs (RecipientContact, section 12.7), filled from CRM events, and its own push tokens and language settings.
- The push service, e-mail provider and SMS provider are external systems. They are described as one interface in section 12.8.2, so they can be replaced with mocks during implementation.

12.2 Feature table

ID	Feature	Actor	Priority
NTF-01	The system sends a notification for every event that has a template in section 12.8.4.	System	Must
NTF-02	A notification is in the user's notification language (LT or EN). If no template exists in that language, the English one is used.	System	Must
NTF-03	A notification that is not mandatory is sent on every channel the customer enabled for its category, and never on a channel the customer turned off.	System	Must
NTF-04	Mandatory notifications are always sent, even if the customer turned off every channel. Every SECURITY template is mandatory; the template tables in section 12.8.4 mark which others are. One-time codes are sent only by SMS.	System	Must
NTF-05	If delivery of a mandatory notification fails, the system tries the next channel in the order push, SMS, e-mail, with at most 3 attempts in total.	System	Should
NTF-06	A notification never contains a full card number, a password, or the reason AML (anti-money laundering) rejected a payment. The stored copy of a one-time code message has the code masked.	System	Must
NTF-07	The customer turns push, e-mail and SMS on or off for each category except Security. The Marketing switches work only after the customer has given marketing consent in CRM.	Customer	Must
NTF-08	The mobile app registers the device's push token after login and removes it at sign-out.	Customer (mobile app)	Must
NTF-09	The customer chooses the language of notifications (LT or EN), separately from the language of the app.	Customer	Could

Won't (deliberately excluded from this release):

- A list (inbox) of past notifications in the app.
- Marketing messages: no event sends them yet. The Marketing category exists so that the customer's choice is stored.
- A screen for editing templates. They are loaded from sections 12.8.4 and 12.8.5.
- Messages to employees.

12.3 Business flows

12.3.1 F1: Send a notification

Actors: the module that publishes the event, the Notification system, the channel provider, the Customer.

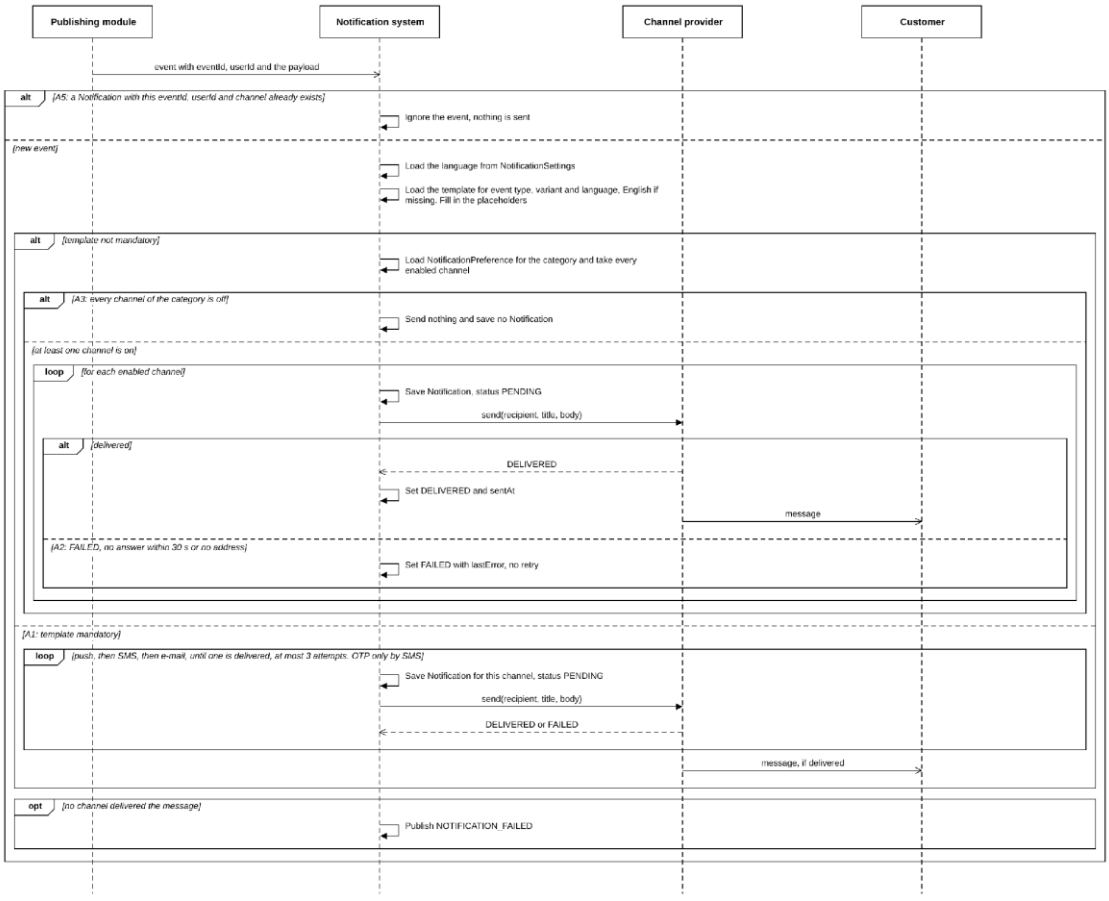
Preconditions: a module has published an event that has a template in section 12.8.4, with the userId of the user to notify.

Basic course (a notification that is not mandatory):

- The Notification system receives the event, for example TRANSACTION_COMPLETED.
- It reads userId from the payload and loads the user's notification language (LT or EN) from NotificationSettings.
- It loads the NotificationTemplate for the event type, variant and language, and fills in the placeholders from the payload.
- It loads the user's NotificationPreference for the template's category and takes every enabled channel. A missing preference row means the defaults in section 12.7.1.
- For each enabled channel, it saves a Notification with status PENDING and sends the message to that channel's provider: to every PushToken of the user, or to the e-mail address or phone number in RecipientContact.
- The provider returns DELIVERED. The Notification system sets that Notification to DELIVERED and records sentAt. A push message counts as delivered when at least one device accepted it.
- The Customer receives the message.

Alternate courses:

- A1. The template is mandatory. At step 4, the Notification system ignores the NotificationPreference and sends on one channel at a time: push first; if that fails, SMS; then e-mail. Each attempt is its own Notification row, with at most 3 attempts in total (NTF-05). OTP_REQUESTED is sent only by SMS, once; if that fails, the customer uses Resend in the one-time code dialog (section 5.2), which creates a new code and a new event.
- A2. Delivery fails: the provider returns FAILED, does not answer within 30 seconds, or the channel has no address (no push token, e-mail or phone). The Notification becomes FAILED, with lastError. A notification that is not mandatory is not retried and not moved to another channel; a mandatory one continues as in A1. When no channel delivered the message, the system publishes NOTIFICATION_FAILED.
- A3. The Customer turned off every channel for this category. The Notification system sends nothing and saves no Notification.
- A4. No NotificationTemplate exists in the user's language. The Notification system uses the English template.
- A5. The same event arrives again (the same eventId). A Notification with that eventId, userId and channel already exists, so nothing is sent again (NTF-NFR-02).



Notification F1: send a notification

12.3.2 F2: Register a push token

Actors: Customer (mobile app), Notification system, push service (Apple or Google).

Preconditions: the customer has logged in on the mobile app.

Basic course:

1. After login, the mobile app gets a push token for the device from the push service.
2. The app calls PUT /api/v1/push-tokens/{deviceId} with the platform and the token. deviceId is an id the app creates when it is installed.
3. The Notification system saves the PushToken, or replaces the token of that device.

Alternate courses:

- A1. The customer signs out. The app calls DELETE /api/v1/push-tokens/{deviceId}, and the Notification system deletes the token.
- A2. The push service answers that a token is no longer valid. The Notification system deletes that PushToken.

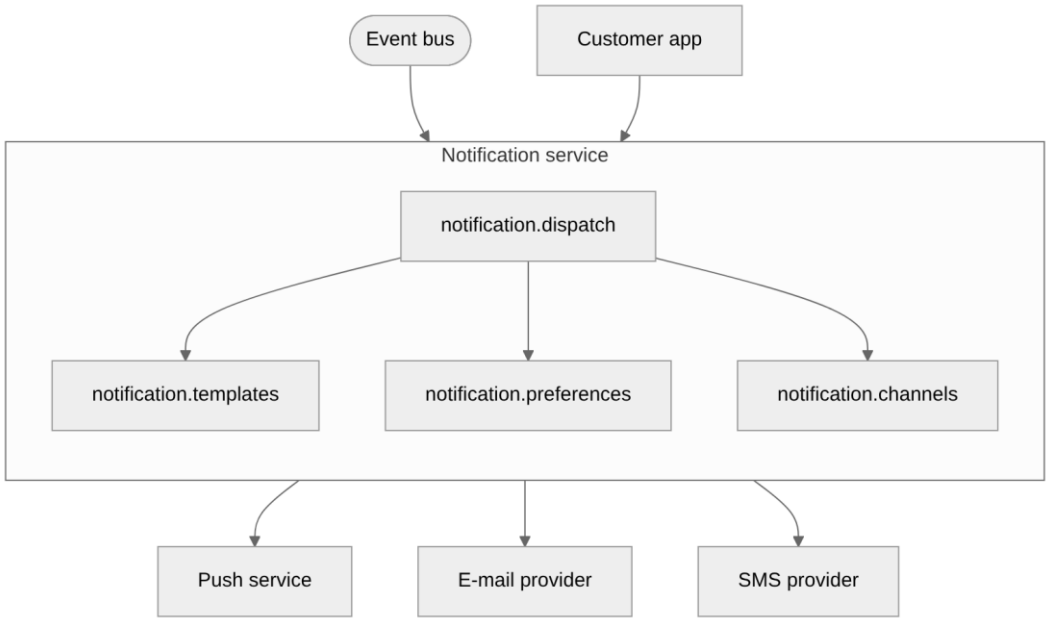
12.4 Non-functional requirements

These apply to this module. The requirements common to the whole system are in section 4.

ID	Requirement
NTF-NFR-01	95% of notifications are handed to the provider within 10 seconds after the event. For mandatory notifications, the limit is 5 seconds.
NTF-NFR-02	Every event is processed at least once. An event with an eventId that was already received does not create a second notification (eventId + userId + channel is unique).
NTF-NFR-03	A slow or unavailable provider does not slow down the module that published the event.
NTF-NFR-04	Notification rows are deleted 12 months after they are created. In the stored copy of an OTP_REQUESTED message, the code is replaced by six asterisks.
NTF-NFR-05	RecipientContact holds only what is needed to send messages: the e-mail address, the phone number and the marketing consent (GDPR (General Data Protection Regulation) data minimisation).

12.5 High-level architecture

The module consists of the packages below; the layering of the whole system is in section 3.3.



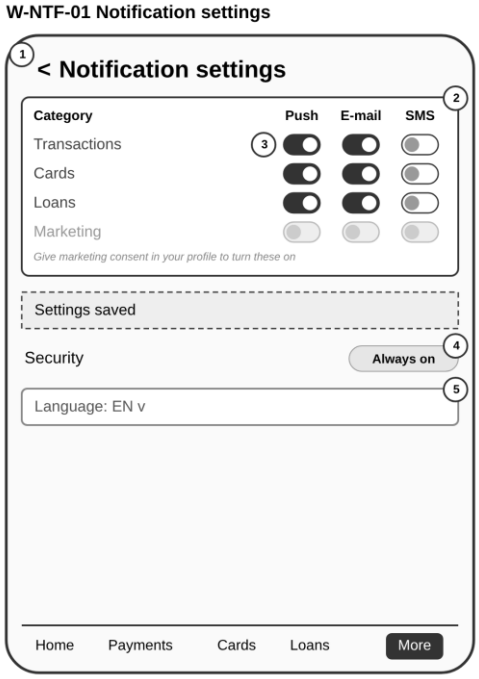
Notification packages

Package	Responsibility
notification.dispatch	Listens to the events that have a template, picks the template and the channels, and handles the attempts on another channel.
notification.templates	Stores templates and fills in placeholders.

Package	Responsibility
notification.preferences	Stores the channel preferences, the notification language, the contact copies (RecipientContact) and the push tokens. Listens to the CRM events and provides the customer endpoints.
notification.channels	One adapter per channel (push, e-mail, SMS) that implements the provider interface in section 12.8.2.

12.6 UI wireframes

There is one customer screen in the mobile app, reached from More (Profile).



W-NTF-01 Notification settings

12.6.1 W-NTF-01 Notification settings

#	Element	What happens when it's used
1	Back	Returns to Profile
2	One row per category: Transactions, Cards, Loans, Marketing	Each row has three switches, for push, e-mail and SMS. While the customer has not given marketing consent, the Marketing switches are disabled, with the hint "Give marketing consent in your profile to turn these on"
3	Channel switch	Saves immediately (PUT /api/v1/notification-preferences/{category}) and shows the message "Settings saved". If saving fails, the switch returns to its previous position and the message "Settings were not saved. Please try again." is shown
4	Security row	A label "Always on", not a switch. Tapping it shows the text "Security messages are always sent"
5	Language	Dropdown LT / EN. Changes the language of future notifications only, not of the app (PUT /api/v1/notification-settings)

All screen texts are message keys with LT and EN values (NFR-5).

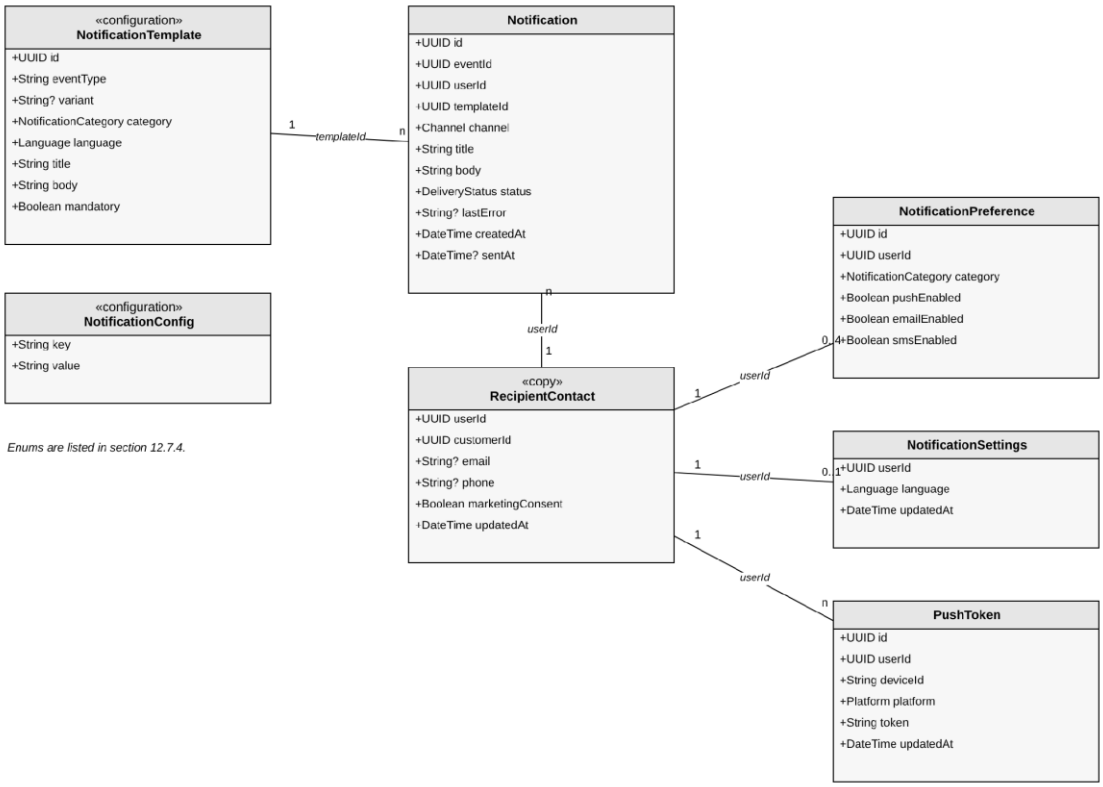
12.6.2 Screen transitions

From	Action	To
Profile	Notification settings	W-NTF-01
W-NTF-01	Back	Profile

12.7 Data model

Notification owns every entity below. User is a shared entity owned by KYC (know your customer) and CRM (section 3.4); Notification refers to it only by userId and keeps its own copy of the contact details, RecipientContact.

Notification is a shared entity owned by this module; other modules never write it. A "?" after a type marks a field that can be empty.



Notification data model

12.7.1 Configuration data

Loaded from the default data at deployment. There is no screen for editing it (section 12.2, Won't).

Entity	Fields
NotificationTemplate	id: UUID (universally unique identifier), eventType: String, variant: String? (for example INCOMING or OUTGOING), category: NotificationCategory, language: Language, title: String, body: String, mandatory: Boolean. eventType + variant + language is unique. Every SECURITY template is mandatory
NotificationConfig	key: String unique, value: String

Default configuration (NotificationConfig):

Key	Value	Used in
providerTimeoutSec	30	F1, A2
maxMandatoryAttempts	3	F1, A1 (NTF-05)
mandatoryChannelOrder	PUSH, SMS, EMAIL	F1, A1
defaultPreference	push on, e-mail on, SMS off	F1, step 4
defaultPreference.MARKETING	push off, e-mail off, SMS off	F1, step 4
retentionMonths	12	NTF-NFR-04

12.7.2 Business data

Entity	Fields
Notification	id: UUID, eventId: UUID, userId: UUID, templateId: UUID, channel: Channel, title: String, body: String (one-time codes masked), status: DeliveryStatus, lastError: String?, createdAt: DateTime, sentAt: DateTime?. eventId + userId + channel is unique
NotificationPreference	id: UUID, userId: UUID, category: NotificationCategory (not SECURITY), pushEnabled: Boolean, emailEnabled: Boolean, smsEnabled: Boolean. userId + category is unique. A missing row means the default preference
NotificationSettings	userId: UUID unique, language: Language, updatedAt: DateTime. Created from CUSTOMER_REGISTERED with the user's language
PushToken	id: UUID, userId: UUID, deviceId: String, platform: Platform, token: String, updatedAt: DateTime. userId + deviceId is unique

One template is used by many notifications. A user has at most one preference row per category (at most 4), one NotificationSettings row, and one push token per device.

12.7.3 Copies of other modules' data

Entity	Fields	Filled from
RecipientContact	userId: UUID unique, customerId: UUID, email: String?, phone: String? (E.164, for example +37060000000), marketingConsent: Boolean, updatedAt: DateTime	CUSTOMER_REGISTERED, CUSTOMER_PROFILE_UPDATED and CONSENT_CHANGED (section 12.8.3)

12.7.4 Enums

Enum	Values
NotificationCategory	TRANSACTIONS, CARDS, LOANS, SECURITY, MARKETING
Channel	PUSH, EMAIL, SMS
DeliveryStatus	PENDING, DELIVERED, FAILED
Language	LT, EN
Platform	IOS, ANDROID

12.8 API contract

All endpoints use JSON (JavaScript Object Notation). Times are in ISO 8601 (the international date and time format) UTC (Coordinated Universal Time), for example "2026-10-11T10:42:00Z". Errors use the common error body with code and message (section 3.2). Request fields are required unless marked optional.

Customer endpoints need a customer token and only return that customer's data. Without a valid token they return 401, as the CBS endpoints do (section 8.10.4).

12.8.1 Customer endpoints

Method and path	Purpose	Request body	Success response	Errors
GET /api/v1/notification-preferences	Get the channel settings	none	200, a list of rows with category, pushEnabled, emailEnabled, smsEnabled and editable (false for SECURITY, and for MARKETING without consent)	401
PUT /api/v1/notification-preferences/{category}	Change one category	pushEnabled, emailEnabled, smsEnabled	200, the saved row	400 VALIDATION_ERROR, 401, 404 NOT_FOUND for an unknown category, 422 CATEGORY_NOT_EDITABLE for SECURITY, or for MARKETING without consent
GET /api/v1/notification-settings	Get the notification language	none	200, language	401
PUT /api/v1/notification-settings	Change the notification language	language (LT or EN)	200, the saved settings	400 VALIDATION_ERROR, 401
PUT /api/v1/push-tokens/{deviceId}	Register or update this device's push token (F2)	platform (IOS or ANDROID), token	200, the saved row	400 VALIDATION_ERROR, 401
DELETE /api/v1/push-tokens/{deviceId}	Remove this device's push token at sign-out (F2)	none	204	401, 404 NOT_FOUND

12.8.2 Provider interface

This can be a mock during implementation.

Each channel adapter implements send(recipient, title, body) and returns DELIVERED or FAILED within 30 seconds. The recipient is a push token, an e-mail address or a phone number. An SMS carries only the body. The mock can be set up to fail a given share of messages, so the attempts in F1 can be tested.

12.8.3 Events

Events use the common envelope (section 3.2). If the same eventId is received twice, the receiver ignores the second one.

Events the Notification system receives:

Event	From	Payload fields used	Used for
CUSTOMER_REGISTERED	CRM	userId, customerId, email, phone, language	Creates RecipientContact and NotificationSettings
CUSTOMER_PROFILE_UPDATED	CRM	userId, customerId, email, phone	Updates RecipientContact
CONSENT_CHANGED	CRM	userId, purpose, granted	Updates RecipientContact.marketingConsent when purpose is MARKETING
Every event with a template in section 12.8.4	Several modules	userId and the placeholders of its template	F1

Event the Notification system publishes:

Event	Payload	Who reacts
NOTIFICATION_FAILED	notificationId, eventId, eventType, userId	Audit log

CUSTOMER_ACTIVATED has no template: the customer gets only the KYC_APPROVED message.
RECONCILIATION_DISCREPANCY is not a customer message and has no template (section 12.1).

12.8.4 Starting templates (EN)

Every placeholder is a field of the event payload. Amounts are shown with two decimals, and dates in Europe/Vilnius time. Enum values (for example newStatus) are shown through message keys with LT and EN texts.
TRANSACTION_COMPLETED has two templates; the direction in the payload decides which variant is used.

Event	Variant	Category	Mandatory	Title	Body
TRANSACTION_COMPLETED	INCOMING	TRANSACTIONS	no	Payment received	You received {amount} {currency} from {counterpartyName}.
TRANSACTION_COMPLETED	OUTGOING	TRANSACTIONS	no	Payment sent	{amount} {currency} was sent to {counterpartyName}.
CARD_TRANSACTION_AUTHORIZED	–	CARDS	no	Card payment	{amount} {currency} at {merchantName}, card ending {cardLast4}.
TRANSACTION_REJECTED	–	SECURITY	yes	Payment not executed	Your payment could not be executed. Please contact the bank.
CARD_BLOCKED	–	SECURITY	yes	Card blocked	Your card ending {cardLast4} has been blocked.
OTP_REQUESTED	–	SECURITY	yes	Confirmation code	Your code is {code}. Never share it with anyone.
NEW_DEVICE_LOGIN	–	SECURITY	yes	New login	New login from {deviceName} in {city}. If this wasn't you, block access in the app.
LOAN_INSTALLMENT_DUE	–	LOANS	no	Installment due	{amount} EUR is due on {dueDate}.

Event	Variant	Category	Mandatory	Title	Body
LOAN_INSTALLMENT_MISSED	–	LOANS	no	Installment overdue	Your installment of {amount} EUR is overdue.
KYC_APPROVED	–	SECURITY	yes	Account verified	Your identity has been verified. You can now use all services.
KYC_INFO_REQUESTED	–	SECURITY	yes	More information needed	Please upload the requested documents in the app.
TRANSACTION_FAILED	–	TRANSACTIONS	yes	Payment failed	Your payment of {amount} {currency} to {beneficiaryName} was not executed.
PAYMENT_PENDING_APPROVAL	–	TRANSACTIONS	yes	Approval needed	A payment of {amount} {currency} to {beneficiaryName} is waiting for your approval.
PAYMENT_REJECTED	–	TRANSACTIONS	yes	Payment not approved	Your payment of {amount} {currency} to {beneficiaryName} was rejected or has expired.
ACCOUNT_OPENED	–	TRANSACTIONS	no	Account opened	Your new {currency} account {iban} is ready.
ACCOUNT_STATUS_CHANGED	–	SECURITY	yes	Account status changed	Your account {iban} is now {newStatus}.
MANDATE_CHANGED	–	SECURITY	yes	Account access changed	Your access to account {iban} was changed.

TRANSACTION_REJECTED never says why the payment was rejected. This is required by AML law (section 11.3.2, A1).

12.8.5 Starting templates (LT)

The LT templates use the same events, variants, categories, mandatory flags and placeholders as section 12.8.4.

Event	Variant	Title	Body
TRANSACTION_COMPLETED	INCOMING	Gautas mokėjimas	Gavote {amount} {currency} iš {counterpartyName}.
TRANSACTION_COMPLETED	OUTGOING	Mokėjimas išsiųstas	Išsiuntėte {amount} {currency} gavėjui {counterpartyName}.
CARD_TRANSACTION_AUTHORIZED	–	Mokėjimas kortele	Atsiskaityta kortele, kurios numeris baigiasi {cardLast4}: {amount} {currency}, {merchantName}.
TRANSACTION_REJECTED	–	Mokėjimas neįvykdytas	Jūsų mokėjimo įvykdyti nepavyko. Susisiekite su banku.
CARD_BLOCKED	–	Kortelė užblokuota	Jūsų kortelė, kurios numeris baigiasi {cardLast4}, užblokuota.
OTP_REQUESTED	–	Patvirtinimo kodas	Jūsų kodas: {code}. Niekam jo nesakykite.
NEW_DEVICE_LOGIN	–	Naujas prisijungimas	Prisijungta iš {deviceName} ({city}). Jei tai ne jūs, užblokuokite prieigą programėlėje.
LOAN_INSTALLMENT_DUE	–	Artėja įmokos terminas	Įmoką {amount} EUR reikia sumokėti iki {dueDate}.
LOAN_INSTALLMENT_MISSED	–	Pradelsta įmoka	Jūsų {amount} EUR įmoka pradelsta.
KYC_APPROVED	–	Tapatybė patvirtinta	Jūsų tapatybė patvirtinta. Dabar galite naudotis visomis paslaugomis.
KYC_INFO_REQUESTED	–	Reikia papildomos informacijos	Įkelkite prašomus dokumentus programėlėje.
TRANSACTION_FAILED	–	Mokėjimas nepavyko	Jūsų mokėjimas {amount} {currency} gavėjui {beneficiaryName} neįvykdytas.

Event	Variant	Title	Body
PAYMENT_PENDING_APPROVAL	–	Laukia jūsų patvirtinimo	Mokėjimas {amount} {currency} gavėjui {beneficiaryName} laukia jūsų patvirtinimo.
PAYMENT_REJECTED	–	Mokėjimas nepatvirtintas	Jūsų mokėjimas {amount} {currency} gavėjui {beneficiaryName} atmestas arba nebegalioja.
ACCOUNT_OPENED	–	Sąskaita atidaryta	Jūsų nauja {currency} sąskaita {iban} paruošta.
ACCOUNT_STATUS_CHANGED	–	Pasikeitė sąskaitos būsena	Jūsų sąskaitos {iban} būsena: {newStatus}.
MANDATE_CHANGED	–	Pasikeitė prieiga prie sąskaitos	Pakeista jūsų prieiga prie sąskaitos {iban}.

13 Glossary

Term	Meaning
AML	Anti-money laundering: checks of transactions and customers for money laundering and terrorist financing.
AML alert, AML case	An alert is created when a transaction matches an AML rule. A case groups the alerts of one customer for one review by an officer.
Available balance	Account balance minus the sum of active holds. A payment is checked against it.
BIC	Bank identifier code (8 or 11 characters) of a bank in a SWIFT payment.
CBS	Core banking system: accounts, balances, holds and transactions.
Dual signature	A payment from a company account that must be approved by a second user who did not create it.
EOD	End of day: the nightly batch that posts the day’s transactions to the ledger and reconciles.
False positive	An alert that turns out to be legitimate activity after review.
FIU	Financial intelligence unit, the authority that receives suspicious-activity reports. In Lithuania it is FNTT.
FNTT	Financial Crime Investigation Service, the Lithuanian financial intelligence unit that receives suspicious-activity reports.
FX markup	Percentage the bank adds to the exchange rate when the payment currency differs from the account currency (1.0%).
GDPR	EU General Data Protection Regulation.
Hold	A reservation on an account (ACTIVE, CONSUMED, RELEASED) that reduces the available balance before money is deducted.
IBAN	International bank account number. Lithuanian IBANs are LT plus 18 digits.
Idempotency-Key	HTTP header that makes a repeated request return the first result instead of creating a second one.
IDV	Identity verification provider (document authenticity and liveness check).
KYC	Know your customer: the process that decides whether the bank may trust a customer’s identity.
Ledger	The general ledger of the bank: balanced debit and credit entries produced by Accounting.
Mandate	The permissions (VIEW, CREATE_PAYMENTS, APPROVE_PAYMENTS) and role a user has on a company account.
Mandatory notification	A notification sent whatever the customer's channel settings: every security message and the templates marked mandatory.
MoSCoW	Priority scale: Must, Should, Could, Won't. Won't means deliberately excluded from this release.
Outbox	Pattern in which an event is saved with the business change in one database transaction and published afterwards, so no event is lost.
OTP	One-time password: the six-digit code used for strong customer authentication.
p95	The value that 95% of measurements do not exceed (for example response time).
PEP	Politically exposed person, who needs enhanced checks.
PSD2	EU payment services directive that requires strong customer authentication.
Push token	An identifier from Apple or Google that lets the bank send push messages to one device.
Refresh token	A token the app uses to get a new access token while the session is still active (15 minutes of inactivity at most).
Risk score	The 0 to 100 score of a transaction in AML: the sum of the weights of the matched rules, multiplied by the customer risk factor.
RPO / RTO	Recovery point objective (data that may be lost) and recovery time objective (time to recover).
Sanctions list	A list of persons and companies the bank may not do business with.

Term	Meaning
SCA	Strong customer authentication: the one-time code at login and for payments above EUR 500.
SEPA	Single Euro Payments Area: euro transfers inside Europe.
Session	The server-side record of a login, ended by logout or by 15 minutes of inactivity.
Structuring	Splitting a large amount into several smaller transfers to stay below a reporting threshold. AML rule R2 looks for it.
SWIFT	International payment network for transfers that are not SEPA.
Tipping-off	Telling a customer that a suspicious-activity report was made. It is forbidden by AML law and never shown in the system.
UBO	Ultimate beneficial owner: a person who owns more than 25% of a company.
Watchlist	The sanctions and PEP entries that AML matches persons and companies against.